

Course Notes

Financial Time Series

0 Financial Time Series Course

These are notes to accompany the MSCF Financial Times Series Course. The course assumes students will have taken Financial Data Science 1 and 2 and be able to program in python. These notes are intended to provide adequate coverage for the material to be covered in the class, and in many cases go into greater depth than the material presented in class. Of course students may wish to refer to a true textbook on financial time series. There are many choices, a but set of excellent references is given in the following list:

- Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and G.M. Ljung, *Time Series Analysis: Forecasting and Control, 5th Edition*, John Wiley, 2015.
- Brockwell, P.J. and R. Davis, *Introduction to Time Series and Forecasting, 3rd Edition*, Springer, 2016.
- Chan, N.H, *Time Series: Applications to Finance with R and S-Plus*, Wiley, 2010.
- Hamilton, James, *Time Series Analysis*, Princeton University Press, 1994.
- Lai, T.L. and Xing, H., *Statistical Models and Methods for Financial Markets*, Springer, 2008.
- McKinney, W., *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython, 2nd Edition*, O'Reilly, 2018.
- Ruppert, D. and D.S. Matteson, *Statistics and Data Analysis for Financial Engineering, 2nd Edition*, Springer, 2015.
- Sargent, T.J. and J. Stachurski, *Quantitative Economics with Python*, 2020, on-line <https://python.quantecon.org>
- Tsay, R.S. *Analysis of Financial Time Series, 3rd Edition*, Wiley, 2019.
- Tsay, R.S. *Multivariate Time Series Analysis with R and Financial Applications*, Wiley, 2014.
- Tsay, R.S., *An Introduction to Analysis of Financial Data with R*, Wiley, 2013.
- Velu, R., Hardy, M. and D. Nehren, *Algorithmic Trading and Quantitative Strategies*, CRC Press, 2020.

Note, while most of these books with a computational component use the language R, three use python (McKinney, Sargent and Stachurski, and Velu, Hardy and Nehren).

The course will have many financial and economic data examples and exercises. These will require the use of python, and often special python packages. The reader can consult <http://pypi.org> which is a repository of

python packages. In addition to the familiar [numpy](#), [scipy](#), [pandas](#) and [matplotlib](#) some relevant time series packages include

- [statsmodels](#), a very large collection of statistical methodology including linear models (glm) and time series models, estimation, fitting and prediction.
- [quantecon](#), a large collection of procedures for quantitative economics and time series supporting the text book by Sargent and Stachurski, very useful for simulating ARMA processes and spectral analysis.
- [pmdarima](#), a useful set of procedures for ARIMA and SARIMA models as well as ARIMAX and SARIMAX models including fitting models and prediction. It claims to provide the functionality of the R package auto.arima.
- [arch](#), a package designed for ARCH/GARCH modeling and prediction.
- [johansen](#), a package for the Johansen cointegration test.
- [pyflux](#), a library for time series analysis and prediction including ARIMA(X) and GARCH models.
- [state-space](#), a package to support state space modeling methods.
- [pykalman](#), a package to support Kalman methodology (filter, prediction, smoothing).

1 Introduction

Time series analysis is a set of models and methodologies designed to handle data with a temporal (or sometimes spatial) dependence structure. This branch of statistics has been developed over many years and has great importance in a wide range of disciplines and application areas including signal processing in engineering, economics and finance.

The theory presented in most basic and intermediate level statistics courses (e.g. estimation, hypothesis testing, confidence intervals and linear models) is largely based on i.i.d. data; that is, the noise associated with each data point is assumed to have identical distributional properties, and those data are assumed to be generated independently from each other. Indeed, the common distribution is often assumed to be normal. When the dataset is a time series, say of economic indicators or security prices, then a dependency structure enters the picture. These series depend on many factors, whose distribution changes constantly, and any particular series will be impacted in large and small ways by other security price and economic series. Thus a single economic or financial series will not have an i.i.d. structure, and the components of a multivariate series will exhibit dependencies either directly or through common factors. The i.i.d. assumption will need to be replaced by a more general structure allowing for time dependencies, but this new structure must be one that is both general enough to provide adequate models for financial time series variables, while being simple enough to allow for the creation of usable methodology. The generalization will be from i.i.d. to (weak) stationarity, a concept to be defined shortly. There are a number of specific uses for time series analysis. Many MSCF students will encounter them in their summer internships and ultimately in their careers. The material also will be very useful in second year classes. Some uses include:

- **Prediction and Forecasting**

One of the major activities of financial economists is to predict the course of economic variables over some period. Asset managers make stock selections based on the predicted performance of those stocks over some time period. Risk managers are charged with predicting the likelihood of (large) losses from a certain trading position over some period. Indeed, for sell-side banks, the Basel regulations require the bank to post sufficient capital to be able to cover losses equal to a risk measure such as value-at-risk (VaR) or expected shortfall. Thus one of the major goals of time series methodology is to predict or forecast changes in the values of various portfolios. Of course, the future is very random and the predictions must be based on statistical models, so we need to think in terms of *predictive distributions* rather than point predictions.

- **Using time series methods to develop trading strategies**

Given that most all economic and financial time series exhibit non-stationary behavior, it would seem impossible to predict their future course with any degree of accuracy. Indeed, the efficient market hypothesis (originally articulated by E. Fama in 1970¹) and the popularity of the random walk model of stock prices popularized by B. Malkiel² would imply that prediction is equivalent to guessing. Beginning in 1993, there were a number of published quantitative studies showing that there were trading strategies (statistical arbitrage strategies) that made consistent profits over long periods of time. In 2003, the Nobel Prize in economics was awarded to C. Granger and R. Engle in part for developing

¹E. Fama, "Efficient capital markets: A review of theory and empirical work," *Journal of Finance*, **25**(2), 1970, 383-417.

²B. Malkiel, *A Random Walk Down Wall Street*, 12th Edition, Norton, 2019

the concept of *cointegration*. While individual series may be non-stationary, it is very possible to find a linear combination of a small number of series that behave in a stationary fashion. If one can find such a set of time series, then stationarity would entail that they exhibit *mean reversion*. This means that when a pair of stock price series move away from their mean values, they will return. This immediately gives us a potential trading strategy. When the stock prices move away from the mean relationship, one will be relatively high and one will be relatively low. By taking a long position in the low-value stock and a short position in the high-value stock, and closing both positions when they converge to equilibrium, a profit is guaranteed, so long as the cointegration relationship is maintained. The idea described here was presented in terms of stocks; however, it applies more generally to other asset classes and is generally known as *convergence trading*.

- **Volatility**

One of the most important concepts in finance is volatility and its connection to risk. The higher the volatility, the higher the return demanded by an investor to compensate for assuming the risk of holding the security. We know that the price of an option varies directly with the volatility of the underlying asset. Thus volatility modeling and estimation are important parts of many financial studies. There are also *volatility trading strategies*. Consider a call option on a particular stock and compute the implied volatility from the current price. Now compare that to an estimate of the current volatility of the underlying stock. If the two disagree, say the implied volatility is higher than the estimated volatility, then the option is overvalued relative to its value using the estimated volatility. Consequently, one would short the option as part of a convergence trade.

2 Stylized Facts

A 2001 paper by Rama Cont³ outlines 11 “stylized facts” about asset price behavior for a number of asset classes including equities and FX. We list them here with brief descriptions and a few examples.

1. Absence of autocorrelations

“Linear autocorrelations of asset returns are often insignificant, except for very small intraday time scales for which microstructure effects come into play.”

We will discuss the autocorrelation function (acf) later in these notes. It arises as the correlation of $\{X_t, t \geq 0\}$ with the same series shifted by a lag value, say k , $\{X_{t+k}, t \geq 0\}$. The stylized fact indicates that if we plot the empirical autocorrelation function for a typical stock, we should see at most two spikes, one with height 1 at 0 corresponding to the perfect correlation of the series with itself and possibly a spike at lag 1 indicating there is correlation between the prices at times t and $t+1$. For values of $k > 1$, the plot will be near 0.

2. Heavy Tails

“The (unconditional) distributions of returns seem to display a power law or Pareto-like tail, with a tail index which is finite, higher than two and less than five for most data sets studied. In particular this excludes stable laws with infinite variance and the normal distribution. However, the precise form of the tails is difficult to determine.”

A random variable, X , is said to obey a power law or have a Pareto-like tail if $P(X > x) \approx x^{-\alpha}$ where α is the tail index.

3. Gains/loss asymmetry

“One observes large drawdowns in stock prices and stock index values but not equally large upward movements (this property is not true for FX where there is symmetry).”

4. Aggregational Gaussianity

“As one increases the time scale over which returns are calculated, their distribution looks more like a normal distribution. The shape is not the same at different time scales.”

5. Intermittency

“Returns display, at any time scale, a high degree of variability. This is quantified by the presence of irregular bursts in time series of a wide variety of volatility estimators.”

6. Volatility Clustering

“Different measures of volatility display a positive autocorrelation over several days, which quantifies the fact that high-volatility events tend to cluster in time.”

7. Conditional heavy tails

“Even after correcting returns for volatility clustering (e.g. via GARCH-type models), the residual time series still exhibit heavy tails. However, the tails are less heavy than in the unconditional distribution of returns.”

³Rama Cont, “Empirical properties of asset returns, stylized facts and statistical issues,” *Quantitative Finance*, **1**, 2001, 223-236.

8. Slow decay of autocorrelation in absolute returns

“The autocorrelation function of absolute returns decays slowly as a function of the time lag, roughly as a power law with an exponent $\beta \in [0.2, 0.4]$. This is sometimes interpreted as a sign of long-range dependence.”

9. Leverage effect

“Most measures of volatility of an asset are negatively correlated with the returns of that asset.” This is explained on page 668 of Hamilton’s time series text⁴, “A number of researchers have found evidence of asymmetry in stock price behavior – negative surprises seem to increase volatility more than positive surprises.⁵ Since a lower stock price reduces the value of equity relative to corporate debt, a sharp decline in stock prices increases corporate leverage and could thus increase the risk of holding stocks. For this reason, the apparent finding ... is called the *leverage effect*.”

10. Volume/volatility correlation

“Trading volume is correlated with all measures of volatility.”

11. Asymmetry in time scales

“Coarse-grained measures of volatility predict fine-scale volatility better than the other way around.”

The stylized facts give the general behavior of financial time series but are not theorems. They can hold to a greater or lesser extent during various periods of time. Nevertheless, one should be aware of these stylized facts and consider using models that allow for them to hold. For example, the standard model for stock prices is geometric Brownian motion (GBM). Under the risk neutral measure, this model would predict that log-returns ($\log(S_{t+h}/S_t)$) would have a $N((\mu - .5\sigma^2)h, \sigma^2h)$ distribution. The stylized facts indicate that the tails must be much heavier for relatively small values of h . In addition, this model assumes constant volatility, yet the stylized facts call for volatility clustering. The introduction of white noise errors and GARCH volatility processes or stochastic volatility models will be helpful in developing models that conform more closely to the stylized facts listed above.

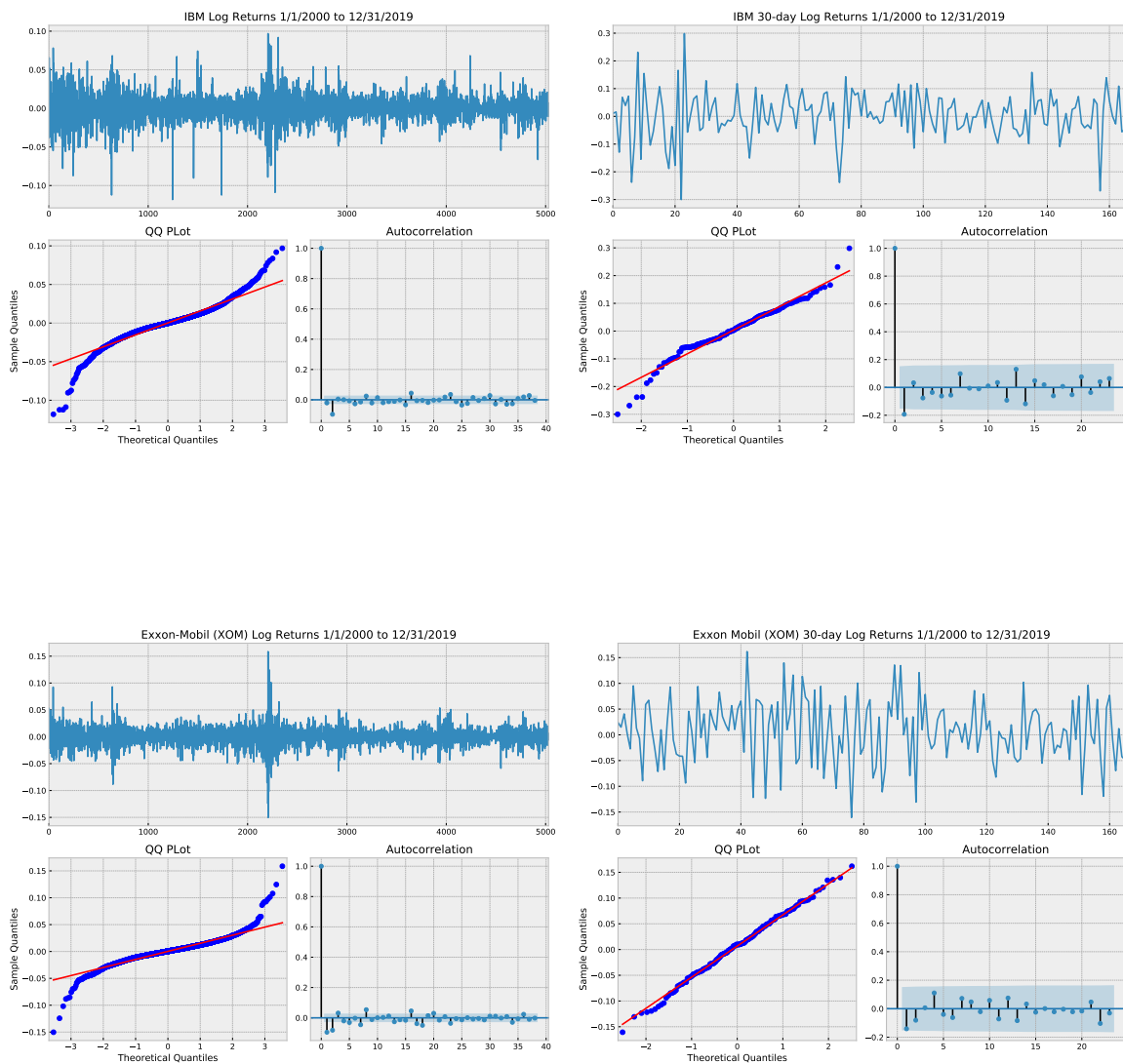
We present 3 representative figures to demonstrate the issues. Stock price data for IBM, Exxon-Mobil and the S&P 500 index over the 20-year period 1/1/2000 to 12/31/2019 were downloaded and the log-returns from the Adjusted Close were created. The one day log-return plot along with a Q-Q normal plot and the ACF plot are shown on the left-side figure in each of the three pairs. The corresponding right-side figure is from the same datasets; however, the log-returns are plotted and analyzed over 30-day non-overlapping periods (hence the size of the right-side data set is 1/30 that of the left-side).

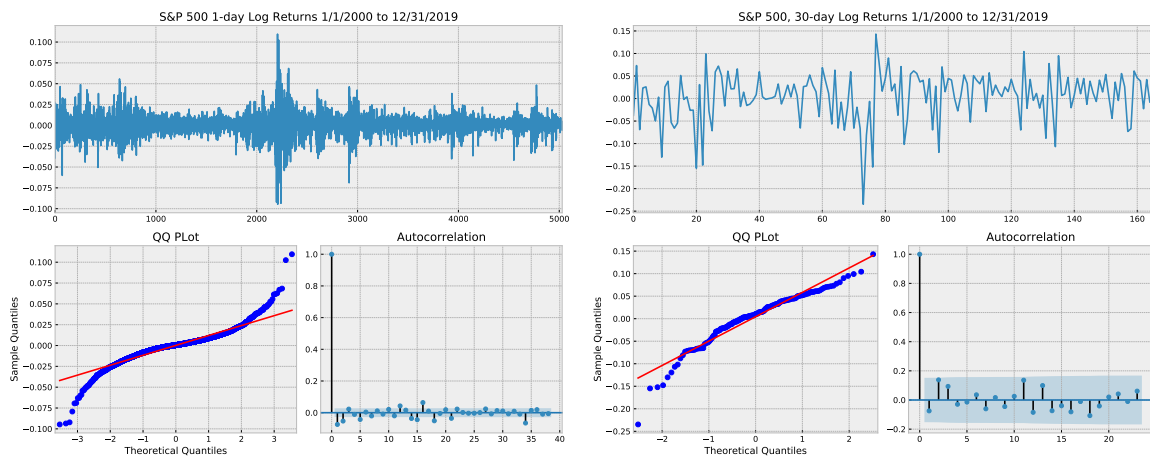
While the plots are very different, corresponding to the unique circumstances of each of these three, they also are very similar. Most striking is on the left-side daily Q-Q plots where the log-returns are very heavy tailed compared with the normal distribution. However, if one looks at the 30-day log returns, then the Q-Q plot is much closer to a normal distribution, although the tail still shows some significant outliers from extreme financial situations like the financial crisis of 2008-2009. The Q-Q plots show evidence of the “heavy tails” and “aggregational Gaussianity” stylized facts. The first stylized fact, absence of autocorrelation, is substantiated by the ACF plots in each of the three cases. There is very little autocorrelation in these time

⁴James D. Hamilton, *Time Series Analysis*, Princeton University Press, 1994.

⁵See Pagan and Schwert (1990), Engle and Ng (1991), and the studies cited in Bollerslev, Chou, and Kroner (1992, p.24),

series as shown in the ACF plots. Finally, the “volatility clustering” stylized fact is clearly evident in all of the plots. There are obvious time periods of huge volatility while there are also time periods of relatively low volatility. While these stylized facts are not theorems, at least four of them (absence of autocorrelation, heavy tails, aggregational Gaussianity, and volatility clustering) are very well demonstrated in these figures.





We give Python code to generate plots of the sort given above. Students will be able to modify it to include appropriate titles and the shape of the collection of plots.

```
import pandas as pd
import pandas_datareader as pdr
import numpy as np
import statsmodels as smm
import statsmodels.tsa.api as smt
import statsmodels.api as sm
import scipy.stats as scs
import matplotlib.pyplot as plt
```

```
def tsplot(y, lags=None, figsize = (10,8), style = 'bmh'):
    if not isinstance(y, pd.Series):
        y = pd.Series(y)
```

```
    with plt.style.context(style):
        fig = plt.figure(figsize=figsize)
        layout = (3,2)
        ts_ax = plt.subplot2grid(layout, (0,0), colspan = 2)
        acf_ax = plt.subplot2grid(layout, (1,0))
        pacf_ax = plt.subplot2grid(layout, (1,1))
        qq_ax = plt.subplot2grid(layout, (2,0))
        pp_ax = plt.subplot2grid(layout, (2,1))
```



```

y.plot(ax = ts_ax)
ts_ax.set_title('Time Series analysis Plots')
smt.graphics.plot_acf(y, lags = lags, ax=acf_ax, alpha = 0.05)
smt.graphics.plot_pacf(y, lags=lags, ax=pacf_ax, alpha = 0.05)
sm.qqplot(y, line = 's', ax=qq_ax)
qq_ax.set_title('QQ PLOT')
scs.probplot(y,sparams = (y.mean(), y.std()), plot=pp_ax)


plt.tight_layout()


return


start = '2000-01-01'
end = '2019-12-31'
# ibm data, but change to the whatever ticket is desired data = pdr.get_data_yahoo('ibm',start,end)
data_adjclose = data['Adj Close']
shf = 10
# select whatever shift is desired logret_shf = np.log(data_adjclose/data_adjclose.shift(shf)).dropna()
L = len(logret)
LL = int(L/shf)
logret_shf = np.zeros(LL)
for i in range(int(L/shf)):
    logret_shf[i] = logret_shf[i*shf]


tsplot(logret_shf)
plt.show()

```

3 Stationarity

Given that economic and financial time series exhibit significant dependencies over time, we need to develop a concept of time dependency for a time series, $\{X_t, t \geq 0\}$, that is more general than independence but is still one that is “workable,” i.e. a concept that will allow for statistical inference, including prediction and asymptotic properties that will allow for consistent estimation and testing of model parameters. The general concept is *stationarity*. There are two common versions of this concept: *strong stationarity* and *weak stationarity*, the latter also known by other names including *covariance stationarity*, *second-order stationarity* and *wide-sense stationarity* among others.

Strong stationarity is a very strong requirement on the distributional structure of a time series. For any k , take any subset of k time points, $0 \leq t_1 < t_2 < \dots < t_k$ (t_i are integers), consider the joint distributions of $(X_{t_1}, X_{t_2}, \dots, X_{t_k})$ and $(X_{t_1+j}, X_{t_2+j}, \dots, X_{t_k+j})$. These two k -dimensional joint distributions must be identical for any integer $j \geq 1$. By letting $k = 1$ we see that the marginal distributions of X_t and X_{t+j} must be identical for all t and j , hence the X_t variables are identically distributed. Letting $k = 2$, we see that all the two-dimensional joint distributions also must be identical, and so on, except that the specific distributions are left unstated. This is obviously true for an i.i.d. sequence, but we will need to allow for a much wider classes of processes.

To begin, consider a time series $\{X_t, t = 0, 1, \dots\}$ with common mean μ and variance σ^2 , but the marginal and joint distributions are unspecified. Then this process is called a *white noise process*, denoted $WN(\mu, \sigma^2)$, if the covariance function takes the form:

$$\sigma(s, t) = Cov(X_s, X_t) = \begin{cases} \sigma^2 & \text{if } s = t, \\ 0 & \text{if } s \neq t. \end{cases}$$

Note, while a white noise process allows for a non-zero mean μ as defined above, generally we set $\mu = 0$, so that when a white noise process is used as the error term for a statistical model, the mean term in the white noise process does not become confounded with the mean term in the statistical model.

The white noise process will be the “backbone” noise process for financial time series. This process requires that the random variables be uncorrelated with each other; however, it does not require any particular marginal or joint distribution structure, and so it is very general. There is, however, one important situation in which a weakly stationary process is also strongly stationary. This occurs where the joint distributions of all finite collections of the random variables have a joint normal distribution. The joint normal distribution is completely specified by its mean vector and covariance matrix, and the covariance matrix of a white noise process is given by $\sigma^2 \mathbf{I}$. A joint normal distribution with this covariance matrix is, in fact, a product of marginal normal distributions $N(\mu, \sigma^2)$. Consequently, a Gaussian (or normal) white noise process is a sequence of i.i.d. Gaussian random variables, which is strongly stationary.

Before we move on to weakly stationary processes, it is useful to note that there are two special classes that are white noise processes: i.i.d. sequences and Martingale Difference Sequences (MDS). I.i.d sequences are very familiar to statistics students and are the backbone of basic probability and statistics classes. Students are very familiar with the important limit theorem like the law of large numbers and the central limit theorem that hold for i.i.d. sequences. MDS will be discussed shortly but are singled out here, because they are a

special class of white noise processes for which, under mild assumptions, the law of large numbers and the central limit theorem still hold. That these theorems hold is important, because they allow for the asymptotic characterization of the behavior of estimates and hypothesis tests based on normal theory.

The strong stationarity property requires specification of the joint distribution structure of the time series, something that is often not known. For this reason, we move to a much weaker version of stationarity, one that does not involve the specific distributions at all, but only involves the first two moments of these distributions. Specifically, we define a time series, $\{X_t, t \geq 0\}$, to be *weakly stationary* if

1. $E(X_t) = \mu, |\mu| < \infty, \forall t,$
2. $Var(X_t) = \sigma^2 < \infty, \forall t,$
3. $Cov(X_s, X_t) = Cov(X_{s+j}, X_{t+j}), \forall s, t, j.$

If we let $j = -s$, then property 3 becomes, $Cov(X_s, X_t) = Cov(X_0, X_{t-s}) = \gamma(t-s)$. Consequently, the covariance function (also known as the autocovariance function) is a function only of the time difference or lag. Thus $\gamma(0) = \sigma^2$, the common variance of all variables in the time series. We can also define the correlation function (also called the *autocorrelation function*) by normalizing the covariance function,

$$\rho(t) = \frac{\gamma(t)}{\gamma(0)} = \frac{\gamma(t)}{\sigma^2}.$$

Weakly stationary processes include strong stationary processes, but are a much larger class, since they only require stationarity of the first and second moments and the covariance function and require nothing about the joint distribution structure. Again weakly stationary processes are also sometimes called covariance stationary, second-order stationary or wide-sense stationary

We will find that much can be done in assessing both the probabilistic and statistical properties of covariance stationary processes, especially in determining the asymptotic behavior of parameter estimates for a wide range of time series processes. Often this will involve making assumptions about the higher moments of the random variables comprising the process.

3.1 Asymptotics of stationary processes

MSCF students are very familiar with important asymptotic results for i.i.d. processes including laws of large numbers and the central limit theorem. These theorems are of special importance in statistical analyses including linear models. We use these theorems to determine the asymptotic behavior of parameter estimates in those models.

Statistical analyses associated with economic and financial models, for example asset price processes, require us to generalize standard linear models in two ways. If we consider the standard linear model:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where ϵ is a vector of i.i.d error terms with mean 0 and variance σ^2 , $\mathbf{X}$ is a constant matrix of independent variables, and $\mathbf{Y}$ is a vector of dependent variables. The least squares estimate of β is given by

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y},$$

and

$$\hat{\beta} - \beta \Rightarrow N(\mathbf{0}, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}),$$

as the sample size increases. If, in addition, ϵ has a normal distribution, then $\hat{\beta}$ is also the maximum likelihood estimator and has an exactly normal distribution.

When constructing linear models involving financial and economic variables, and time series in general, there are three important difference from standard linear models: $\mathbf{X}$ is often random, not constant, ϵ is not i.i.d, (usually strictly or weakly stationary), and $\mathbf{X}$ and ϵ may be correlated. A simple autoregressive process of order 1, AR(1), illustrates all three of these issues:

$$X_t = \phi X_{t-1} + \epsilon_t,$$

where $\{\epsilon_t, t \geq 0\}$ is weakly stationary with mean 0 and variance σ^2 .

Since on its face this is a simple linear model taking $\mathbf{Y} = (X_t, \dots, X_1)^T$ and $\mathbf{X} = (X_{t-1}, \dots, X_0)^T$, the least square estimator is still given by $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$. The question is whether the same asymptotic properties, e.g. consistency and asymptotic normality, apply. One problem is immediately obvious namely that the least squares estimator cannot be asymptotically normally distributed with covariance matrix $\sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$, because this is random, not constant.

To see some of the difficulties, let us consider the law of large numbers for weakly stationary processes with a simple example. Let X_1 be a random variable with Bernoulli(.5) distribution, and define $X_t = X_0$ for $t > 0$. Now consider $\{X_t, t \geq 0\}$. This sequence takes two possible values: $(1, 1, 1, \dots)$ and $(0, 0, \dots)$, and each variable has a marginal Bernoulli(.5) distribution with $E(X_t) = .5$. The sequence is strictly stationary and is weakly stationary with autocorrelation function $\rho(k) = 1$ for $k \geq 0$. Clearly $\bar{X}_t = X_0$, so $\bar{X}_t \rightarrow X_0$, a.s. However, for a law of large numbers we expect the average to converge to the common mean, .5, not a random variable, X_0 . The change from independence to total dependence has entirely changed the character of the limit.

One famous strong law of large numbers for strictly stationary sequences was given by Birkhoff, namely if $\{X_t, t \geq 0\}$ is strictly stationary with $E(X_t) = \mu$. Then there exists a random variable, X_∞ with $E(X_\infty) = \mu$, for which $\bar{X}_t \rightarrow X_\infty$, a.s.

There is a similarly theorem for weakly stationary processes; however, in general the convergence is in probability not a.s. and some additional conditions are needed to compensate for the loss of strict stationarity.

Suppose that $\{X_t, t \geq 1\}$ is a weakly stationary process with $E(X_t) = \mu$, $\text{Var}(X_t) = \sigma^2$, and $\sum_{i=1}^{\infty} |\gamma(i)| <$

∞ . Then $\bar{X}_t \rightarrow \mu$ in probability. The proof is a simple application of Chebyshev's inequality:

$$\begin{aligned}
P(|\bar{X}_t - \mu| > \epsilon) &\leq \frac{\text{Var}(\bar{X}_t)}{\epsilon^2} \\
&= \frac{(t\sigma^2 + 2 \sum_{i=1}^t (t-i)\gamma(i))}{(t\epsilon)^2} \\
&\leq \frac{(\sigma^2 + 2 \sum_{i=1}^t |\gamma(i)|)}{(t\epsilon)^2} \\
&\leq \frac{(\sigma^2 + 2 \sum_{i=1}^{\infty} |\gamma(i)|)}{(t\epsilon)^2} \\
&\rightarrow 0 \text{ as } t \rightarrow \infty, \forall \epsilon > 0.
\end{aligned}$$

Note that the same result holds with the slightly weaker condition $\frac{\sum_{i=1}^t \gamma(i)}{t} \rightarrow 0$ as $t \rightarrow \infty$.

It follows that there are reasonable conditions (like the absolute summability of the autocovariance function) that will ensure the weak law of large numbers for a weakly stationary sequence. This will typically ensure the consistency of the least squares estimators of model coefficients. However, we want to go further and have our estimate be asymptotically normally distributed, as they are in the i.i.d. case. For this to be the case, we need to introduce a special subset of the weakly stationary processes, *homoscedastic martingale difference sequences*.

We first introduce the more general class of processes, *martingale difference sequences*. This class places no restrictions on the second moments of the random variables nor on their covariance structure. We will eventually need to restrict this class to having constant variance (homoscedasticity), and then we will see that the homoscedastic martingale difference sequences are white noise processes. The benefit of restricting to this class is that with some further assumptions, these processes satisfy the central limit theorem, something that is important for implementing statistical inference for time series processes.

To begin, we enumerate some of the standard facts about conditional expectation. Consider a time series, $\{X_t, t \geq 0\}$. Define $\mathcal{F}_t$ as the σ -algebra generated by the random variables $(X_0, \dots, X_t)$. $\mathcal{F}_t$ intuitively means “all the information in the random variables $(X_0, \dots, X_t)$.” There are some important properties of conditional expectation, familiar to MSCF students from Stochastic Calculus 1. All random variables are assumed to have finite expected values.

- $\mathcal{F}_s \subseteq \mathcal{F}_t, s \leq t$.
- $E(X|\mathcal{F}_s)$ is an $\mathcal{F}_s$ measurable random variable.
- $E(Y) = E(E(Y|\mathcal{F}_t))$ for any t and Y satisfying $E(|Y|) < \infty$. A familiar version of this is $E(E(Y|X)) = E(Y)$ assuming $E(|X|) < \infty$ and $E(|Y|) < \infty$.
- $E(X|\mathcal{F}_t) = X$, a.s. if X is $\mathcal{F}_t$ measurable.
- $E(XY|\mathcal{F}_t) = YE(X|\mathcal{F}_t)$, a.s. if Y is $\mathcal{F}_t$ measurable.

- $E(E(X|\mathcal{F}_s)|\mathcal{F}_t) = E(E(X|\mathcal{F}_t)|\mathcal{F}_s) = E(X|\mathcal{F}_s)$ for $s \leq t$.

There is an additional property of conditional expectation that will be used in the section on prediction, that conditional expectation provides the minimal mean square error predictor. Specifically, suppose we have information, for example the values of a time series, $\{X_t, 1 \leq t \leq T\}$, and we wish to predict a future value, say X_{T+s} . Let $\mathcal{F}_T$ be the σ -algebra generated by the time series $\{X_t, 1 \leq t \leq T\}$. Then the predictor $E(X_{T+s}|\mathcal{F}_T)$ minimizes $E((Y - X_{T+s})^2)$ among all $\mathcal{F}_T$ -measurable random variables. The proof is given in the section of these notes on prediction. Of course, $E(X_{T+s}|\mathcal{F}_T)$ may be difficult to calculate. If the joint distributions are Gaussian, then $E(X_{T+s}|\mathcal{F}_T)$ is linear in the observations. In general, we will use linear estimation in all cases, and that will be optimal in the case of Gaussian random variables.

All MSCF students are very familiar with a *martingale sequence*, a pair $\{(X_t, \mathcal{F}_t), t \geq 1\}$ where X_t is $\mathcal{F}_t$ measurable, $\mathcal{F}_s \subset \mathcal{F}_t, \forall s \leq t$ and $E|X_t| < \infty, \forall t$ satisfying

$$E(X_{t+1}|\mathcal{F}_t) = X_t \text{ a.s. } t = 0, 1, \dots$$

We can also define a *martingale difference sequence* as a pair $\{(Y_t, \mathcal{F}_t), t \geq 0\}$ where Y_t is $\mathcal{F}_t$ measurable satisfying $E|Y_t| < \infty$ and

$$E(Y_{t+1}|\mathcal{F}_t) = 0 \text{ a.s. } t \geq 0.$$

Note, we can rewrite the martingale property as $E(X_{t+1} - X_t|\mathcal{F}_t) = 0, t \geq 0$. So if $\{(X_t, \mathcal{F}_t), t \geq 0\}$, then we let $Y_{t+1} = X_{t+1} - X_t$, it follows that $\{Y_t, t \geq 0\}$ is a martingale difference sequence (MDS).

There are three different types of weakly stationary processes, $\{X_t, t = 0, 1, \dots\}$, that are typically considered in time series analysis, each having constant mean and variance, $E(X_t) = \mu$, and $\text{Var}(X_t) = \sigma^2$:

1. Independent sequences in which the variables are mutually independent. Sometimes this is specialized to i.i.d. sequences, and even further to $\{X_t, t = 0, 1, \dots\}$ being i.i.d. with $N(\mu, \sigma^2)$ distribution.
2. Homoscedastic (constant variance) martingale difference sequences (MDS), $E(X_{t+1}|\mathcal{F}_t) = 0$ a.s.,
3. Weakly Stationary processes in which $\text{Cov}(X_s, X_t) = \gamma(t - s) = \gamma(s - t)$.

It should be clear that an independent sequence with $\mu = 0$ is also a MDS, since $E(X_{t+1}|\mathcal{F}_t) = E(X_{t+1}) = 0$.

The converse is not true, and a simple example suffices to show that not all MDS are independent sequences. Assume that $\{X_t, t = 0, 1, \dots\}$ is an independent sequence with $\mu = 0$. Let $Y_t = X_t X_{t-1}$. Let $\mathcal{F}_t$ be the σ -algebra generated by $\{X_0, \dots, X_t\}$. If we compute $E(Y_{t+1}|\mathcal{F}_t) = E(X_{t+1} X_t|\mathcal{F}_t) = X_t E(X_{t+1}|\mathcal{F}_t) = X_t \cdot 0 = 0$, because the X_t sequence is assumed to be an independent sequence with mean 0. It follows that the Y_t sequence is an MDS. However, the Y_t sequence is not an independent sequence. For an independent sequence, the property $E(g(Y_{t+1})|\mathcal{F}_t) = E(g(Y_{t+1}))$, a constant, holds for any function g . But for $g(x) = x^2$, we have $E(Y_{t+1}^2|\mathcal{F}_t) = E((X_{t+1} X_t)^2|\mathcal{F}_t) = X_t^2 E(X_{t+1}^2|\mathcal{F}_t) = X_t^2 \sigma^2$ which is not a constant, as

would be required for an independent sequence. Thus the set of MDS is larger than the set of mean 0 independent sequences.

Note that if $\{X_t, t = 0, 1, \dots\}$ is a MDS with constant variance, then for $s < t$, $Cov(X_s, X_t) = E(X_s X_t) - E(X_s)E(X_t) = E(E(X_s X_t | \mathcal{F}_{t-1})) - 0 \cdot 0 = E(X_s E(X_t | \mathcal{F}_{t-1})) = E(X_s \cdot 0) = 0$. Consequently, the X_t process is weakly stationary with covariance function

$$\gamma(t) = \begin{cases} \sigma^2 & \text{if } t = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Hence, a MDS is a white noise process.

The previous derivation shows that any $\{X_t, t \geq 0\}$ MDS process must have a particular covariance function, so any weakly stationary process having a different covariance function will not be an MDS. A simple example of this is an autoregressive process of order 1:

$$X_{t+1} = \phi X_t + \epsilon_t,$$

where $|\phi| < 1$, $\{\epsilon_t, t = 0, 1, \dots\}$ is an i.i.d. sequence with mean 0 and $Var(\epsilon_t) = \sigma^2$. It follows that $E(X_{t+1} | \mathcal{F}_t) = E(\phi X_t + \epsilon_{t+1} | \mathcal{F}_t) = \phi X_t + 0$, which is not, in general, equal to 0 as required for an MDS sequence.

We close by giving a theorem that provides conditions for an MDS sequence to satisfy a central limit theorem.

Theorem (MDS Central Limit Theorem)

Suppose that $\{(X_t, \mathcal{F}_t), t = 0, 1, \dots\}$ is a martingale difference sequence, and let $\bar{X}_T = \frac{1}{T} \sum_{t=1}^T X_t$. Suppose that

- $E(X_t^2) = \sigma_t^2 > 0$ with $\frac{1}{T} \sum_{t=1}^T \sigma_t^2 \rightarrow \sigma^2$,
- $E|X_t|^r < \infty$ for some $r > 2$ and all t , and
- $\frac{1}{T} \sum_{t=1}^T X_t^2 \rightarrow \sigma^2$.

Then $\sqrt{T} \bar{X}_T \Rightarrow N(0, \sigma^2)$.

Note that the above theorem allows for non-constant variance.

4 Standard Time Series Models

4.1 Linear and Causal Processes

As was mentioned earlier, in time series analysis, white noise processes form the backbone of time series models. Suppose we begin with a white noise process, $\{\epsilon_t, 0 \leq t < \infty\}$ or $\{\epsilon_t, -\infty < t < \infty\}$, in each case the individual terms have a common mean 0 and variance σ^2 . We can build a time series process $\{X_t, 0 \leq t\}$ or $\{X_t, -\infty < t < \infty\}$ from this noise sequence. If the X_t time series represents a financial or economic variable, then we would like a value, X_t , to depend only on the noise sequence up to time t , not on the future noise terms. If we define $\{\mathcal{F}_t, -\infty < t < \infty\}$, where $\mathcal{F}_t$ is the σ -algebra generated by $\{\epsilon_s, -\infty < s \leq t\}$, then the time series is said to be *causal* if X_t is $\mathcal{F}_t$ measurable. *Non-anticipative* is the common terminology in stochastic calculus.

There are two special classes of time series, *linear processes* and *causal processes*. A *linear process* $\{X_t, -\infty < t < \infty\}$ with respect to the noise process $\{\epsilon_t, -\infty < t < \infty\}$ is one in which there is a sequence of constants $\{\psi_j, -\infty < j < \infty\}$ for which the time series $\{X_t, -\infty < t < \infty\}$ has the representation

$$X_t = \sum_{j=-\infty}^{\infty} \psi_j \epsilon_{t-j},$$

for all t where $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$.

If we modify the above definition to

$$X_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j},$$

for $\sum_{j=0}^{\infty} |\psi_j| < \infty$, then the process is both linear and causal as X_t is $\mathcal{F}_t$ measurable where $\mathcal{F}_t$ is the σ -algebra generated by $\{\epsilon_s, s \leq t\}$.

4.2 Wold Decomposition

While the class of weakly stationary processes seems very large compared with the more familiar class of independent (or even i.i.d.) processes, there is a very important theorem that shows that white noise is a fundamental weakly stationary process and that essentially all other processes can be built from white noise. This result is known as [Wold's decomposition](#). At an intuitive level, Wold's decomposition shows that nearly all weakly stationary processes can be built from a white noise process.

We begin with a weakly stationary process, $\{X_t, -\infty < t < \infty\}$ with $E(X_t) = 0$, $Var(X_t) = \sigma^2$, and autocorrelation function $\rho(h)$. Note that we are assuming $-\infty < t < \infty$ for integer valued t so that the X process is fully weakly stationary, not just asymptotically stationary (defined later). We define $\mathcal{F}_t$ to be the σ -algebra generated by the random variables $\{X_s, s \leq t\}$. Note that $\mathcal{F}_s \subset \mathcal{F}_t$ for $s < t$. We also assume that $\cap_{s \leq t} \mathcal{F}_s = \{\phi, \Omega\}$. This latter assumption is similar to assuming that a probability measure is

absolutely continuous with respect to Lebesgue measure and does not have any discrete components. An example illustrating this is given later.

Given this setup, the Wold decomposition indicates that one can find a white noise process $\{\epsilon_t, -\infty < t < \infty\}$ and a set of coefficients $\{\psi_t, -\infty < t < \infty\}$ for which

$$X_t = \sum_{i=0}^{\infty} \psi_i \epsilon_{t-i}$$

That is, we can find a white noise process ϵ , so that the X process is a linear, causal process with respect to ϵ . The important remaining question is given the X process, how can we find the ϵ process and the coefficients $\{\psi_i\}$? The solution is

$$\epsilon_t = X_t - E(X_t | \mathcal{F}_{t-1}),$$

and

$$\psi_i = \gamma_X(i) / \sigma^2 = \rho_X(i).$$

The intuition behind $\{\epsilon_t\}$ being white noise is that the conditional expectation operator is a projection onto the space generated by the σ -algebra in this case, the space generated by the random variables $\{X_s, s \leq t-1\}$. The residual from the projection, $\epsilon_t = X_t - E(X_t | \mathcal{F}_{t-1})$ will be orthogonal (have covariance 0) to all random variables measurable with respect to $\mathcal{F}_{t-1}$. This will ensure that ϵ_t is uncorrelated with $\{\epsilon_s, \forall s < t\}$. We next show the actual calculation. Suppose h is an integer satisfying $h \geq 1$. Note $E(\epsilon_t) = E(X_t - E(X_t | \mathcal{F}_{t-1})) = E(X_t) - E(X_t) = 0$.

$$\begin{aligned} \gamma_\epsilon(h) &= Cov(\epsilon_t, \epsilon_{t+h}) \\ &= E(\epsilon_t \epsilon_{t+h}) \\ &= E((X_t - E(X_t | \mathcal{F}_{t-1}))(X_{t+h} - E(X_{t+h} | \mathcal{F}_{t+h-1}))) \\ &= E(X_t X_{t+h}) - E(X_t E(X_{t+h} | \mathcal{F}_{t+h-1})) \\ &\quad + E(E(X_t | \mathcal{F}_{t-1}) E(X_{t+h} | \mathcal{F}_{t+h-1})) - E(X_{t+h} E(X_t | \mathcal{F}_{t-1})) \\ &= E(E(X_t X_{t+h} | \mathcal{F}_{t+h-1})) - E(X_t E(X_{t+h} | \mathcal{F}_{t+h-1})) \\ &\quad + E(E(X_t | \mathcal{F}_{t-1}) E(X_{t+h} | \mathcal{F}_{t+h-1})) - E(E(X_{t+h} E(X_t | \mathcal{F}_{t-1}) | \mathcal{F}_{t+h-1})) \\ &= E(X_t E(X_{t+h} | \mathcal{F}_{t+h-1})) - E(X_t E(X_{t+h} | \mathcal{F}_{t+h-1})) \\ &\quad + E(E(X_t | \mathcal{F}_{t-1}) E(X_{t+h} | \mathcal{F}_{t+h-1})) - E(E(X_t | \mathcal{F}_{t-1}) E(X_{t+h} | \mathcal{F}_{t+h-1})) \\ &= 0 + 0 \\ &= 0. \end{aligned}$$

Thus the $\{\epsilon_i\}$ random variables form an orthogonal basis for the $\{X_t\}$ process, that is X_t can be represented as a linear combination of the ϵ sequence. Therefore,

$$X_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}.$$

We first note that $X_t = E(X_t|\mathcal{F}_{t-1}) + X_t - E(X_t|\mathcal{F}_{t-1}) = E(X_t|\mathcal{F}_{t-1}) + \epsilon_1 = E(X_t|\mathcal{F}_{t-1}) + \psi_1\epsilon_1$ where $\psi_1 = 1$. The remaining coefficients can be found using the idea that the ϵ_j defined above form a basis for all random variables that are measurable with respect to $\mathcal{F}_{t-1}$ including $E(X_t|\mathcal{F}_{t-1})$. Thus $E(X_t|\mathcal{F}_{t-1})$ can be *uniquely* represented as a linear combination of the white noise process $\{\epsilon_j, j \geq 0\}$ where the coefficients can be found as follows:

$$\begin{aligned} Cov(X_t, \epsilon_{t-j}) &= Cov\left(\sum_{j=1}^{\infty} \psi_j \epsilon_{t-j}, \epsilon_{t-j}\right) \\ &= Cov(\psi_j \epsilon_{t-j}, \epsilon_{t-j}) \\ &= \psi_j \sigma^2. \end{aligned}$$

We solve this to find $\psi_j = Cov(X_t, \epsilon_{t-j})/\sigma^2$.

The previous derivation of the Wold decomposition was done under two assumptions: $E(X_t) = 0$ and $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \{\phi, \Omega\}$. We can easily allow for a mean value $E(X_t) = \mu$ by simply adding μ to the representation,

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}$$

where $\psi_0 = 1$.

The above representation is especially important, because it illustrates the fundamental nature of the white noise process ($\{\epsilon_t, -\infty < t < \infty\}$ in the case above) in representing weakly stationary processes. Indeed assuming $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \{\phi, \Omega\}$, then the X_t process can be represented as a linear, causal process. We will see in the next section that such a process is a *moving average process of infinite order* with respect to the white noise process $\{\epsilon_t, -\infty < t < \infty\}$. This characterization of weakly stationary processes is critical for developing models that we will use in the rest of the course for financial and economic times series. Of course, such models have an infinite set of parameters, $\{\psi_j, 1 \leq j < \infty\}$, and we will only have a finite data set to estimate and perform statistical inferences about them. Consequently, we will need to limit the class of models to those having only a finite (and preferably small) number of parameters. In the next section, we will discuss such models, namely the autoregressive (AR(p)), moving average (MA(q)), and their combination autoregressive moving average models (ARMA(p,q)).

Before moving on to discussing the standard time series models, we consider a very simple example of a weakly stationary process for which the property, $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \{\phi, \Omega\}$ doesn't hold:

Example

Suppose U and V are uncorrelated random variables with $E(U) = E(V) = 0$ and $Var(U) = Var(V) = \sigma^2$. Define

$$X_t = U \cos(\theta t) + V \sin(\theta t).$$

Clearly $E(X_t) = 0$ and $Var(X_t) = \sigma^2(\cos^2(\theta t) + \sin^2(\theta t)) = \sigma^2$. The covariance function is given by

$$\begin{aligned}\gamma_X(h) &= Cov(X_t, X_{t+h}) \\ &= Cov(U \cos(\theta t) + V \sin(\theta t), U \cos(\theta(t+h)) + V \sin(\theta(t+h))) \\ &= \sigma^2(\cos(\theta t) \cos(\theta(t+h)) + \sin(\theta t) \sin(\theta(t+h))) \\ &= \sigma^2 \cos(\theta h)\end{aligned}$$

which is independent of t . Hence X_t is weakly stationary. The above calculation used the trigonometric identities $\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$, $\cos(-x) = \cos(x)$, and $\sin(-x) = -\sin(x)$. Now, for this time series, $\mathcal{F}_t = \sigma(U, V)$, $\forall t$. Consequently, $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \sigma(U, V)$, not $\{\phi, \Omega\}$ as required for the analysis given above.

We can see that the Wold decomposition given above is not appropriate for this time series. In fact, conditional on $\mathcal{F}_{t-1}$, both U and V are known, thus X_t is also known, i.e. $E(X_t | \mathcal{F}_{t-1}) = X_t$ a.s. Although the X_t process defined in this example is weakly stationary, it cannot be decomposed as described before. In the time series literature, such a weakly stationary process is called *deterministic*, very confusing terminology since it is, in fact, stochastic, as U and V are random variables. The idea is that if we know two values of X_t for any two times, t_1 and t_2 , no matter how far in the past, and as long as $\theta t_1 \neq \theta t_2 + 2\pi k$ for some k , then X_{t_1} and X_{t_2} determine U and V and X_t is known $\forall t \geq \max(t_1, t_2)$. Such a process is totally predictable at all future times, which justifies the name *deterministic*. Actually, *conditionally deterministic* would be a better description, as the process is not unconditionally deterministic. Using the Wold decomposition presented by Brockwell and Davis (2nd edition, page 187), we summarize the full Wold Decomposition theorem:

Theorem (Wold Decomposition)

Given a weakly stationary process $\{X_t, -\infty < t < \infty\}$ with $E(X_t) = \mu$ and $Var(X_t) = \sigma^2 > 0$,

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j} + Y_t,$$

where

- $\psi_0 = 1$ and $\sum_{j=0}^{\infty} \psi_j^2 < \infty$,
- $\{\epsilon_t\}$ is $WN(0, \sigma^2)$,
- The process is linear and causal in $\{\epsilon_j\}$,
- Y_t is measurable with respect to $\cap_{j=0}^{\infty} \mathcal{F}_{t-j}$ and $E(Y_t | \cap_{j=0}^{\infty} \mathcal{F}_{t-j}) = 0$,
- Y_t is “deterministic.”

4.3 Some Simple Time Series Models

In the preceeding section, ignoring unimportant special cases we learned that weakly stationary processes can be represented as a moving average process of infinite order with respect to a white noise process, i.e.

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j},$$

where $\{\epsilon_j\}$ is $WN(0, \sigma^2)$. Thus, given a stationary X_t process, we want to make inferences about the coefficients ψ_j . Of course, the data set used to make such inferences will be finite, and there are infinitely many parameters in this model. Clearly this is an impossible situation, and we must constrain the statistical model so there are only finitely many parameters (indeed only a few) to estimate. This leads to several subsets of models with a smaller number of parameters. One approach is to simply truncate the infinite series,

$$X_t = \mu + \sum_{j=0}^q \psi_j \epsilon_{t-j}.$$

Such models are called *moving average process of order q*, MA(q).

A second approach is to introduce a small number of parameters, and allow ψ_j to depend upon them. The most important version of this subset of models is the *autoregressive models of order p*, AR(p). A typical example would be the AR(p) process, where the parameters are $\{\phi_j, 1 \leq j \leq p\}$ is

$$X_t = \mu + \phi_j X_{t-j} + \epsilon_t.$$

We will see by iterating backward in time, we can identify the ψ_j , although the $\{\phi_j, 1 \leq j \leq p\}$ are of greatest interest.

Of course, the most important class is the combination of these two, the *autoregressive moving average processes of order (p,q)*, ARMA(p,q). This is the general weakly stationary class, and is, by far, the most important class of time series models for financial and economic time series.

The three most basic classes of models encountered in time series analysis are autoregressive processes (AR), moving average processes (MA) and the combination of those two, the autoregressive moving average (ARMA) processes. We consider each in the following, and later will discuss the ARIMA(p,d,q) processes (AutoRegressive Integrated Moving Average processes), a generalization of ARMA processes in which a third parameter, d , indicates the number of differences needed to transform the process into an ARMA(p,q) process.

4.3.1 Autoregressive Models

We begin with a white noise process, $\{\epsilon_t, -\infty < t < \infty\}$. From this white noise process, we define an AR(p) autoregressive time series process by

$$X_t = \mu + \phi_1(X_{t-1} - \mu) + \cdots + \phi_p(X_{t-p} - \mu) + \epsilon_t$$

for constants $\mu, \phi_1, \dots, \phi_p$.

If we define $Y_t = X_t - \mu$, then the Y_t process satisfies

$$Y_t = \sum_{j=1}^p \phi_j Y_{t-j} + \epsilon_t,$$

which removes the parameter μ from consideration.

To study the properties of AR(p) processes, it is easiest to first consider the often used special case $p = 1$, $Y_t = X_t - \mu$ and to remove the subscript on ϕ_1 since there is only a single autoregressive parameter. Thus

$$Y_t = \phi Y_{t-1} + \epsilon_t,$$

where $\{\epsilon_t, -\infty < t < \infty\}$ is $\text{WN}(0, \sigma^2)$.

Iterating backward, we find

$$\begin{aligned} Y_t &= \phi Y_{t-1} + \epsilon_t \\ &= \phi(\phi Y_{t-2} + \epsilon_{t-1}) + \epsilon_t \\ &= \phi^2 Y_{t-2} + \phi \epsilon_{t-1} + \epsilon_t \\ &= \dots \\ &= \phi^t Y_0 + \sum_{j=0}^{t-1} \phi^j \epsilon_{t-j} \end{aligned}$$

If we iterate back to $-\infty$, then we must assume $|\phi| < 1$ so that the first term converges to 0, else the first term and the terms in the sum of white noise terms will explode if $|\phi| > 1$ or form a random walk if $|\phi| = 1$, which we will see is not stationary when we calculate the variance.

Thus, a necessary condition for weak stationarity of the Y_t series is $|\phi| < 1$, in which case we can represent $Y_t = \sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}$, and $X_t = \mu + \sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}$.

To check that the Y_t process (and therefore also the X_t process) is weakly stationary, we need to find the covariance function of the Y_t process, and show that it depends only on the difference in time indices.

First, consider $\text{Var}(Y_t) = \text{Var}(\sum_{j=0}^{\infty} \phi^j \epsilon_j)$. The variance of a sum of random variables is the sum of the individual variances plus a correction term involving the covariances terms of the form $\text{Cov}(\epsilon_i, \epsilon_j)$. For $i \neq j$, all of these are 0, because the $\{\epsilon_t\}$ process is white noise. Consequently, for $s < t$, $\text{Var}(Y_t) =$

$\sum_{j=0}^{\infty} \phi^{2j} \sigma^2 = \frac{\sigma^2}{1-\phi^2}$ again assuming $|\phi| < 1$. Next we consider for $s < t$:

$$\begin{aligned}
Cov(Y_s, Y_t) &= Cov\left(\sum_{i=0}^{\infty} \phi^i \epsilon_{s-i}, \sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}\right) \\
&= Cov\left(\sum_{i=0}^{\infty} \phi^i \epsilon_{s-i}, \sum_{j=0}^{t-s-1} \phi^j \epsilon_{t-j} + \sum_{j=t-s}^{\infty} \phi^j \epsilon_{t-j}\right) \\
&= 0 + Cov\left(\sum_{i=0}^{\infty} \phi^i \epsilon_{s-i}, \phi^{t-s} \sum_{j=0}^{\infty} \phi^j \epsilon_{s-j}\right) \\
&= \phi^{(t-s)} Var(Y_s) \\
&= \frac{\sigma^2 \phi^{(t-s)}}{1 - \phi^2}.
\end{aligned}$$

Thus the covariance function depends only upon $t - s$, not the specific values of s and t . Therefore, the AR(1) process is weakly stationary assuming that $|\phi| < 1$. In this case, dividing $\gamma(h)$ by the variance, $\gamma(0)$, we find the autocorrelation function to be give by:

$$\rho(h) = \phi^{|h|}.$$

We did the above calculations for the AR(1) process Y_t assuming $E(Y_t) = 0$. We start with a different version of the AR(1) model in which we don't assume that the mean is 0, i.e.

$$X_t = \phi_0 + \phi_1 X_{t-1} + \epsilon_t.$$

Assume the mean is a constant given by μ and $|\phi_1| < 1$. Taking expectations of both sides of the model, μ must satisfy $\mu = E(X_t) = \phi_0 + \phi_1 E(X_{t-1}) + 0$, or $\mu = \phi_0 + \phi_1 \mu$. This gives $\mu = \phi_0 / (1 - \phi_1)$.

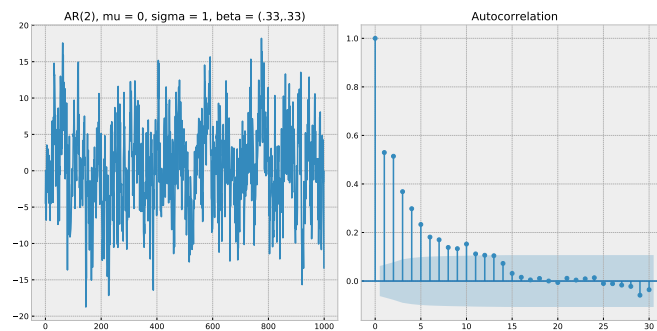
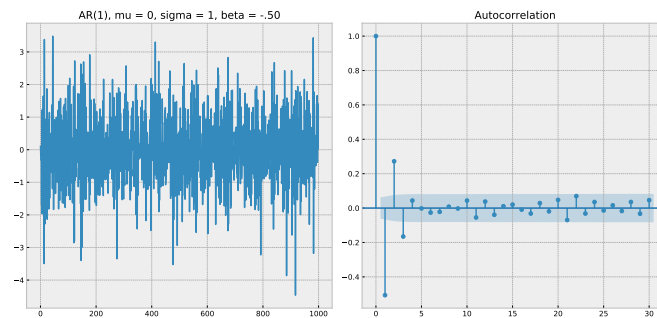
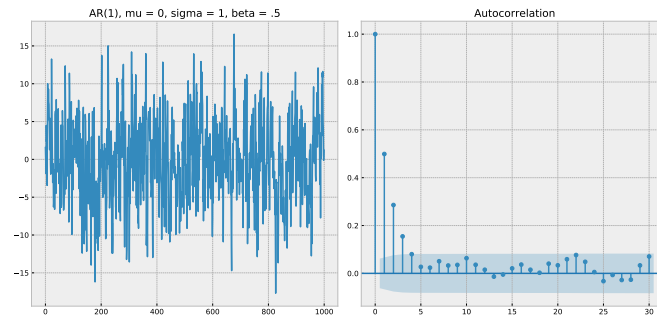
To summarize some results for the AR(1) process given by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \epsilon_t, \quad t \geq 0, \quad |\phi_1| < 1,$$

where $\{\epsilon_t\}$ is $WN(0, \sigma_\epsilon^2)$

- $E(X_t) = \phi_0 / (1 - \phi_1)$
- $Cov(X_t, X_{t+h}) = \gamma(h) = \frac{\phi_1^{|h|}}{1 - \phi_1^2} \sigma_\epsilon^2$
- $Var(X_t) = \sigma_\epsilon^2 / (1 - \phi_1^2)$
- $\rho(h) = \phi_1^{|h|}$

In the figures below we show some plots of simulated AR(1) processes with $\phi_1 = .5$ and $-.5$ respectively. One can observe the differences (primarily the oscillatory behavior associated with the negative autoregressive parameter), and the exponential decline in the ACF as the lag increases. The figure below is for an AR(2) process with $\phi_1 = \phi_2 = .33$. The ACF shows that it is not an AR(1) process, but it is hard to deduce otherwise. The PACF, to be discussed later, will be a very useful tool to deduce the order of an AR process.



The code to simulate and display the sample data and the acf plots is given below. Note that the data is generated by simulation rather than calling a package. Many python time series packages have commands to simulate data from ARMA processes; however, there is an advantage to using standard simulation methodology, especially since it allows one to change the distribution from a normal to a heavy tailed distribution.

```
import numpy as np
import matplotlib.pyplot as plt

def tsplot(y, lags=None, figsize = (12,6), style = 'bmh'):
    if not isinstance(y, pd.Series):
        y = pd.Series(y)

    with plt.style.context(style):
        fig = plt.figure(figsize=figsize)
        layout = (1,2)
        ts_ax = plt.subplot2grid(layout, (0,0))
        acf_ax = plt.subplot2grid(layout, (0,1))

        y.plot(ax = ts_ax)
        ts_ax.set_title('AR(2), mu = 0, sigma = 1, beta = (.33,-.33)')

        plt.tight_layout()
    return

n = 1000
sigma = 5
mu = 0
beta1 = .33
beta2 = -.33
z = np.random.randn(n)
x = np.zeros(n)
x[0] = sigma*z[0]
for i in range(n-2):
    x[i+2] = mu + beta1*x[i+1] + beta2*x[i] + sigma*z[i+2]

tsplot(x)
plt.show()
```

We next move to the general autoregressive process model, $AR(p)$, $p \geq 1$. This model is given by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t,$$

where the ϵ_t process is $WN(0, \sigma^2)$. For this time series to be weakly stationary, we need it to have constant mean, μ . Assuming for the moment that the process is weakly stationary, we take expectations across the

equation, and since the mean, $E(X_t) = \mu$, is constant we find $\mu = \phi_0 + \mu(\phi_1 + \dots + \phi_p)$. Solving, we find $\mu = \phi_0 / (1 - \sum_{i=1}^p \phi_i)$. For convenience in the remainder of this section we set $\phi_0 = 0$, hence $\mu = 0$. Equivalently, we focus on $Y_t = X_t - \mu$.

To study the behavior of an AR(p) process and to find conditions under which it is weakly stationary, it is convenient to reframe the model in matrix notation. To do this, we create equations for $Y_t, Y_{t-1}, \dots, Y_{t-p+1}$ as follows:

$$\begin{aligned} Y_t &= \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \epsilon_t \\ Y_{t-1} &= Y_{t-1} \\ &\vdots \\ &\vdots \\ &\vdots \\ Y_{t-p+1} &= Y_{t-p+1} \end{aligned}$$

If we define $\mathbf{Y}_t = (Y_t, \dots, Y_{t-p+1})^T$, $\boldsymbol{\epsilon}_t = (\epsilon_t, 0, \dots, 0)^T$ and

$$\mathbf{M} = \begin{bmatrix} \phi_1 & \phi_2 & \dots & \phi_{p-1} & \phi_p \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{bmatrix},$$

then the resulting AR(p) model now becomes in matrix form

$$\mathbf{Y}_t = \mathbf{M}\mathbf{Y}_{t-1} + \boldsymbol{\epsilon}_t.$$

Now, the stationarity of this time series will depend on the eigenvalues of the matrix $\mathbf{M}$. Assume that the eigenvalues are distinct, and we can represent $\mathbf{M} = \mathbf{R}\mathbf{D}\mathbf{R}^{-1}$, where $\mathbf{R}$ contains the eigenvectors of $\mathbf{M}$ and $\mathbf{D}$ is the diagonal matrix of eigenvalues. Iterating backward,

$$\begin{aligned} \mathbf{Y}_t &= \mathbf{M}(\mathbf{M}\mathbf{Y}_{t-1} + \boldsymbol{\epsilon}_{t-1}) + \boldsymbol{\epsilon}_t \\ &= \mathbf{M}^2\mathbf{Y}_{t-2} + \mathbf{M}\boldsymbol{\epsilon}_{t-1} + \boldsymbol{\epsilon}_t \\ &\dots \\ &= \mathbf{M}^t\mathbf{Y}_0 + \sum_{i=0}^{t-1} \mathbf{M}^i\boldsymbol{\epsilon}_{t-i}, \end{aligned}$$

we find Y_t can be expressed as the sum of terms containing powers of $\mathbf{M}$. Now $\mathbf{M}^i = \mathbf{R}\mathbf{D}^i\mathbf{R}^{-1}$. Since $\mathbf{D}$ is diagonal, $\mathbf{D}^i$ is also a diagonal matrix with each term raised to the i th power. For $\mathbf{M}^i$ to not explode and for the moments to converge, we need the eigenvalues of $\mathbf{M}$ to have modulus less than 1. Put differently, weak stationarity of the AR(p) process is equivalent to the eigenvalues of the $\mathbf{M}$ matrix to all lie *inside the*

unit circle (have modulus less than 1). Later we will see that if any of the eigenvalues lie on the unit circle (“has a unit root”), then the AR(p) process will *not* be stationary. Recall that the eigenvalues, λ , are the p roots of the determinantal equation:

$$\det(\mathbf{M} - \lambda\mathbf{I}) = 0.$$

The matrix $\mathbf{M}$ has a simple form, and it is possible to explicitly write out the above equation for AR(p) processes as a function of the parameters $\phi_1, \dots, \phi_p$. This means that the eigenvalues are the roots of the polynomial equation

$$\lambda^p - \phi_1\lambda^{p-1} - \dots - \phi_{p-1}\lambda - \phi_p = 0.$$

Generally, if $p \geq 3$, the eigenvalues are found numerically as the roots of a pth-degree polynomial, some pairs of which may be complex numbers. In the special cases $p = 1$ or 2 , the eigenvalue can be expressed in closed form. For $p = 1$, $\lambda_1 = \phi_1$ from which the requirement $|\phi_1| < 1$ is derived. For $p = 2$, the two eigenvalues are the roots of

$$(\phi_1 - \lambda)(-\lambda) - \phi_2(1) = \lambda^2 - \phi_1\lambda - \phi_2 = 0.$$

The solutions are

$$\lambda_1 = \frac{\phi_1 + \sqrt{\phi_1^2 + \phi_2}}{2},$$

$$\lambda_2 = \frac{\phi_1 - \sqrt{\phi_1^2 + \phi_2}}{2},$$

and we need $|\lambda_i| < 1$, $i = 1, 2$ for stationarity. The values of ϕ_1 and ϕ_2 for which the AR(2) process is weakly stationary lie in a triangular region defined by the constraints:

- $\phi_1 + \phi_2 < 1$,
- $\phi_1 - \phi_2 > -1$,
- $|\phi_2| < 1$

Now, by the above matrix arguments, AR(p) processes are weakly stationary if the eigenvalues of the $\mathbf{M}$ matrix all have modulus strictly less than 1. Note, since the eigenvalues are roots of a polynomial equations, some may be complex, hence we refer to their modulus (or length).

There is a second equivalent specification of the conditions needed for an AR(p) process to be stationary. If we define $z = 1/\lambda$ in the determinantal equation above, then it becomes (after multiplying through by z^p).

$$1 - \phi_1 z - \dots - \phi_{p-1} z^{p-1} - \phi_p z^p = 0.$$

This is essentially the same polynomial equation with λ replaced with $1/z$, but it is called the **autoregressive polynomial**, and its roots are the reciprocal of the determinantal equation. Thus if a root of the determinantal equation has modulus less than 1, the corresponding roots of the autoregressive polynomial must have modulus greater than 1. Thus the requirement that an AR(p) process be weakly stationary can be framed in two equivalent ways:

1. All the roots of the autoregressive polynomial have modulus > 1 ,
2. All the roots of the determinantal equation corresponding to the eigenvalues of $\mathbf{M}$ have modulus < 1 .

This, of course, can be a cause of significant confusion for students.

4.3.2 The Backshift or Lag Operator

In some cases, time series notation can be simplified by introducing the **backshift operator** (B) (some authors prefer to define this as the **lag operator** and use the symbol $\mathcal{L}$). The backshift operator is applied to the entire time series and $BX_t = X_{t-1}$, that is, the backshifted time series is shifted one time unit. Consider the AR(1) process,

$$X_t = \phi X_{t-1} + \epsilon_t, \quad -\infty < t < \infty.$$

This can be written as

$$X_t = \phi BX_t + \epsilon_t.$$

More generally, we can write the AR(p) process as

$$\begin{aligned} X_t &= \phi_1 BX_t + \cdots + \phi_p B^p X_t + \epsilon_t \\ &= (\phi_1 B + \cdots + \phi_p B^p) X_t + \epsilon_t. \end{aligned}$$

This equation is often recast in the form

$$X_t - \phi_1 BX_t - \cdots - \phi_p B^p X_t = \epsilon_t,$$

or

$$\phi(B)X_t = \epsilon_t,$$

where

$$\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p.$$

Replacing the backshift operator by z , we find the autoregressive polynomial equation

$$\phi_1 z + \cdots + \phi_p z^p = 1.$$

Setting $z = 1/\lambda$ the polynomial equation becomes

$$\lambda^p = \phi_1 \lambda^{p-1} + \cdots + \phi_p,$$

which is exactly the determinantal equation characterizing the eigenvalues discussed earlier. The above is intended to indicate the context from which the autoregressive polynomial equation arises.

4.4 Moving Average Processes

We have seen by iteration that the AR(1) process can be, assuming $|\phi_1| < 1$, written as

$$\begin{aligned}
 X_t &= \phi_0 + \phi_1 X_{t-1} + \epsilon_t \\
 &= \phi_0 + \phi_1(\phi_0 + \phi_1 X_{t-2} + \epsilon_{t-1}) + \epsilon_t \\
 &= \phi_0(1 + \phi_1 + \phi_1^2) + \phi_1^3 X_{t-3} + \sum_{i=0}^2 \phi_1^i \epsilon_{t-i} \\
 &= \dots \\
 &\rightarrow \frac{\phi_0}{1 - \phi_1} + \sum_{i=0}^{\infty} \phi_1^i \epsilon_{t-i}.
 \end{aligned}$$

Thus

$$X_t = \phi_0/(1 - \phi_1) + \sum_{i=1}^{\infty} \phi_1^i \epsilon_{t-i} + \epsilon_t.$$

which is an example of a **moving average process of infinite order**. In general, we define a **moving average process of order q** , MA(q), by

$$X_t = \mu + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \epsilon_t,$$

where $\{\epsilon_t\}$ is WN($0, \sigma^2$). Some properties of a MA(q) process are easily computed:

- Taking expectations of both sides, we find $E(X_t) = \mu + E(\epsilon_t) + \sum_{i=1}^q \theta_i E(\epsilon_{t+1-i}) = \mu + 0 + 0 = \mu$.
- $Var(X_t) = Var(\mu + \epsilon_t + \sum_{i=1}^q \theta_i \epsilon_{t-i}) = \sigma_\epsilon^2(1 + \sum_{i=1}^q \theta_i^2)$.
- For $m > 0$, $Cov(X_t, X_{t+m}) = Cov(\epsilon_t + \sum_{i=0}^q \theta_i \epsilon_{t-i}, \epsilon_{t+m} + \sum_{j=0}^q \theta_j \epsilon_{t+m-j}) = \sigma_\epsilon^2 \sum_{k=0}^{q-m} \theta_k \theta_{k+m}$, where $\theta_0 = 0$ by definition. Note, the final summation is empty when $m > q$, thus the autocovariance function becomes 0 for $m > q$. This provides a good way to detect the order of a moving average process: plotting the empirical autocovariance function and noting the argument at which it drops within the noise limits of 0. The same is true for the autocorrelation function, which is simply the autocovariance divided by the variance.
- For $q = 1$, $\gamma(1) = \theta_1 \sigma^2$ and $\gamma(m) = 0$ for $m > 1$. In addition, the autocorrelation function behaves similarly: $\rho(1) = \theta_1/(1 + \theta_1^2)$ and $\rho(m) = 0$ for $m > 1$.

4.5 ACF, PACF and Bartlett's Equation

For a given weakly stationary time series, $\{X_t, t \geq 0\}$, the mean function is constant, i.e. $\mu(t) = E(X_t) = \mu$, the autocorrelation function at lag m is defined by $\gamma(m) = Cov(X_t, X_{t+m})$, which is assumed to be

constant with respect to t . The autocorrelation function at lag m is defined as $\rho(m) = \frac{\gamma(m)}{\gamma(0)}$, which is the correlation of X_t and X_{t+m} for any t .

The mean, autocovariance and autocorrelation functions have corresponding sample versions calculated from the observed time series. The sample mean, autocovariance function and sample autocorrelation function are given by

$$\begin{aligned}\bar{X} &= \frac{1}{n} \sum_{i=1}^n X_i, \\ \hat{\gamma}(m) &= \frac{1}{n} \sum_{i=1}^{n-|m|} (X_{t+|h|} - \bar{X})(X_t - \bar{X}), \quad -n < m < n, \\ \hat{\rho}(m) &= \frac{\hat{\gamma}(m)}{\hat{\gamma}(0)}, \quad -n < m < n,\end{aligned}$$

respectively. The sample autocorrelation function is especially useful as a diagnostic tool to identify (or reject) a particular model for the time series. For example, some of the standard time series models have easily identifiable autocovariance functions. We next consider a few examples of this.

Example: Suppose that $\{X_t, 1 \leq t \leq n\}$ is just a collection of i.i.d. random variables with cdf F , mean μ and variance σ^2 . To study the asymptotic behavior of the autocovariance and autocorrelation functions, it is without loss of generality that we can assume $\mu = 0$ and $\sigma = 1$, because

$$\sum_{i=1}^{n-|m|} (X_{t+|h|} - \bar{X})(X_t - \bar{X}) = \sum_{i=1}^{n-|m|} ((X_{t+|h|} - \mu) - (\bar{X} - \mu))((X_t - \mu) - (\bar{X} - \mu)),$$

the distribution of which does not depend upon μ . Similarly,

$$\hat{\rho}(m) = \frac{\hat{\gamma}(m)}{\hat{\gamma}(0)} = \frac{\hat{\gamma}(m)/\sigma}{\hat{\gamma}(0)/\sigma},$$

both numerator and denominator of which do not depend upon σ .

With this simplification, and for n sufficiently large, $\hat{\gamma}(0) \rightarrow 1$, almost surely.

Assuming that n is sufficiently large, we expect $\frac{\bar{X}-\mu}{\sqrt{n}} \approx N(0, \sigma^2)$. Moreover, by the law of large numbers, $\hat{\gamma}(0) = \frac{1}{n} \sum_{i=1}^n X_i^2 \rightarrow E(X^2)$.

The asymptotic behavior of the sample autocorrelation function has been extensively studied, beginning with the work of Bartlett (1946)⁶. Bartlett found that when $\rho(k) = 0$, then $\sqrt{n}\hat{\rho}(k) \rightarrow N(0, 1)$. Thus it is standard to find graphs of the sample ACF plotted using standard time series software having two approximately horizontal lines plotted at $\pm 1.96/\sqrt{n}$. If $\hat{\rho}(k)/\sqrt{n}$ lies inside these two lines, the inference that $\rho(k) = 0$ is made at the 95% confidence level. For the technical details underlying this asymptotic analysis, the reader is referred to the paper by Bartlett.

⁶Bartlett, M., "On the theoretical specification and sampling properties of autocorrelated time series," *Journal of the American Statistical Association, Supplement*, **8**, 1946, 27-41, 85-94 (Corr.), 1948, **10**, 200.

A paper by Romano and Thombs (1996)⁷ extends the work of Bartlett. The beginning of the paper, copied almost verbatim below describes the asymptotic behavior of the sample correlation function for time series in which the errors (innovations) are pure white noise (i.i.d.) leading to Bartlett's formula, and then goes on to indicate how that asymptotic description can be incorrect when the errors are only weak white noise (i.e. mutually uncorrelated but not independent).

1. INTRODUCTION

Let $X_1, \dots, X_n$ denote a realization of a real-valued stationary time series. Unless stated otherwise, we assume that the time series is stationary in the strict sense. This article focuses on inference for the autocorrelation structure of the series. Define the autocovariance and autocorrelation for lag k by $R(k) = \text{Cov}(X_t, X_{t+k})$ and $\rho(k) = \text{Corr}(X_t, X_{t+k})$.

The usual estimates of $R(k)$ and $\rho(k)$ are

$$\hat{R}_n(k) = \frac{1}{n} \sum_{i=1}^{n-k} (X_i - \bar{X}_n)(X_{i+k} - \bar{X}_n)$$

and $\hat{\rho}(k) = \hat{R}_n(k)/\hat{R}_n(0)$, where $\bar{X}_n = \sum_{i=1}^n X_i/n$ is the sample mean. For convenience, we also consider

$$\tilde{R}_n(k) = \frac{1}{n} \sum_{i=1}^{n-k} (X_i - \mu)(X_{i+k} - \mu),$$

where $\mu = E(X_i)$.

Inference for $\hat{\rho}(k)$ where the process $\{X_n\}$ is linear with i.i.d. innovations is well known, and the approach developed by Box and Jenkins (1976) has been quite useful in practice. In particular, if $\{X_t\}$ is the stationary process

$$X_t = \mu + \sum_{j=-\infty}^{\infty} \psi_j \epsilon_{t-j},$$

where $\{\epsilon_t\}$ are independent with mean zero satisfying $E(\epsilon_t^4) < \infty$, then the distribution of the first k sample autocorrelations are asymptotically normal with mean $(\rho(1), \rho(2), \dots, \rho(k))$ and covariance matrix $\mathbf{W}/\sqrt{n}$, where the (i, j) th element of $\mathbf{W}$ is given by

$$\begin{aligned} w_{ij} &= \sum_{j=-\infty}^{\infty} (\rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j)) \\ &+ 2\rho(i)\rho(j)\rho^2(k) - 2\rho(i)\rho(k)\rho(k+j) \\ &- 2\rho(j)\rho(k)\rho(k+i). \end{aligned}$$

These results are used extensively in time series modeling. Perhaps the most widely used application of Bartlett's (1946) formula is when the process $\{X_t\}$ is i.i.d. In this case, $\rho(k) = 0$ for all $k \neq 0$, leading to the standard result, $\hat{\rho}(k) \approx N(0, 1/n)$. a common diagnostic check on

⁷Romano, J.P. and L.A. Thombs, "Inference for autocorrelations under weak assumptions", *Journal of the American Statistical Association*, **91**(434), 1996, 590-600.

a fitted model is to apply this result to the estimated autocorrelation function of the residuals. Typically, $\pm 2/n^{1/2}$ limits are superimposed on a graph of $\hat{\rho}(k)$ versus k . Indeed, it will be seen that the asymptotic distribution of $\hat{\rho}(k)$ generally cannot be approximated by the normal distribution with mean zero and variance $1/n$ if we only assume that the underlying process is uncorrelated. In particular, there exist uncorrelated processes for which the asymptotic variance of $\hat{\rho}(k)$ is any nonnegative value.

Let us consider Bartlett's formula in another setting. For the MA(1) process, $X_t = \epsilon + \theta\epsilon_{t-1}$, the limiting variance of $\sqrt{n}\hat{\rho}(1)$ reduces to $1 - 3\rho^2(1) + 4\rho^4(1)$, and for lags greater than 1, $n\text{Var}(\hat{\rho}_n(k)) \rightarrow 1 + 2\rho^2(1)$. To check whether an observed series follows a MA(1) model, one may use these results on the autocorrelation function of the data. But if the innovations $\{\epsilon_t\}$ were only uncorrelated (instead of i.i.d.), the inference is incorrect. The MA(1) model may be "rejected" when in fact it is the true model. On the other hand, the standard inference that assumes i.i.d. may incorrectly "accept" a model as appropriate. More generally, inference for autoregressive moving average (ARMA) parameter estimates based on formulas derived under the assumption of i.i.d. errors are invalid if the errors are only assumed to be uncorrelated.

4.6 Ljung-Box Test

When one generates an ACF plot, one typically looks to see the set of integers k for which $\hat{\rho}(k)$ lies within the $\pm 2/\sqrt{n}$ limits. The largest k whose $\hat{\rho}(k)$ lies outside the region is then taken to be an upper bound on the parameter k in an AR(k) process or an MA(k) process. In the case of the MA(k) process, the autocorrelation function becomes 0 for lags larger than k . In the case of an AR(1), the autocorrelation function decreases exponentially; however, if the autoregressive parameter is relatively large, the sample autocorrelation function will be positive for lags much larger than 1, hence we will overestimate the order of the process.

While finding the lag at which the sample autocorrelation becomes insignificant does give some insight into the nature of the time series, there is a second issue. By looking at the ACF plot to find the lag at which it becomes insignificant, we are making a set of hypothesis tests, one for each k , and using the results of all of them to make the final inference. Thus we are using simultaneous hypothesis tests, without correcting for that fact. The idea for creating a simultaneous hypothesis test that will control the experimentwise type 1 error rate was first done by Box and Pierce (1970)⁸. This was later improved upon by Ljung and Box (1978)⁹. To describe these tests it is easiest to directly quote the text by Ruey Tsay¹⁰, pp32-33 given next.

Portmanteau Test

Financial applications often require to test jointly that several autocorrelations of X_t are zero.

⁸Distribution of residual autocorrelations in autoregressive-integrated moving average time series models," *Journal of the American Statistical Association*, **65(332)**, 1970, 1509-1526.

⁹Ljung, G.M. and G.E.P. Box, "On a measure of lack of fit in time series models," *Biometrika*, **65(2)**, 1978, 297-303

¹⁰R.G. Tsay, *Analysis of Financial Time Series*, 3rd Ed., Wiley, 2010.

Box and Pierce (1970) propose the Portmanteau statistics

$$Q^*(m) = T \sum_{l=1}^m \hat{\rho}^2(l)$$

as a test statistic for the null hypothesis $H_0 : \rho_1 = \dots = \rho_m = 0$ against the alternative hypothesis $H_a : \rho_i \neq 0$ for some $i \in \{1, \dots, m\}$. Under the assumption that $\{X_t\}$ is an i.i.d. sequence with certain moment conditions, $Q^*(m)$ is asymptotically a chi-squared random variable with m degrees of freedom.

Ljung and Box (1978) modify the $Q^*(m)$ statistic as below to increase the power of the test in finite samples,

$$Q(m) = T(T+2) \sum_{l=1}^m \frac{\hat{\rho}_l^2}{T-l}.$$

The decision rule is to reject H_0 if $Q(m) > \chi_{\alpha}^2$, where χ_{α}^2 denotes the $100(1-\alpha)th$ percentile of a chi-squared distribution with m degrees of freedom. Most software packages will provide the p value of $Q(m)$. The decision rule is then to reject H_0 if the p value is less than or equal to α , the significance level.

In practice, the choice of m may affect the performance of the $Q(m)$ statistic. Several values of m are often used. Simulation studies suggest that the choice of $m \approx \log(T)$ provides better power performance. This general rule needs modification in analysis of seasonal time series for which autocorrelations with lags at multiples of the seasonality are more important.

The Ljung-Box test can be performed in python using the statsmodels package,
statsmodels.stats.diagnostic.acorr_ljungbox

4.7 Partial Correlation Coefficients

In multivariate statistical analysis, one tries to characterize the dependencies among a set of variables. For example, consider (X, Y, Z) . The familiar pairwise correlation coefficient, e.g. $\rho(X, Y)$, gives the linear relationship of X and Y without any consideration of Z . For example, one could start with the joint distribution, $f_{X,Y,Z}(x, y, z)$, integrate out Z leaving $f_{X,Y}(x, y)$ and then compute $\rho(X, Y)$. However, we might want to first eliminate any effects that Z may have on X and Y , and then compute the correlation. This is known as the *partial correlation coefficient*.

The partial correlation of $\rho(X, Y|Z)$ can be found by regressing X on Z and Y on Z separately, then computing the correlation coefficient of the residuals from each of those regressions. Note that Z could be a set of random variables, i.e. $\mathbf{Z} = (Z_1, \dots, Z_m)$. One would again separately regress X and Y on $\mathbf{Z}$ and then find the sample correlation of the two residuals from the regressions.

The concept of partial correlation can be useful in determining the order of a time series model, especially in the AR process case. Suppose that the data came from a true autoregressive process with order p . We fit

an autoregressive model but want to estimate the order. If the true model has order p , then

$$X_t = \mu + \beta_1 X_{t-1} + \cdots + \beta_p X_{t-p} + \epsilon_t.$$

The model indicates that the correlation of X_{t-k} and X_t will be 0 conditional on $X_{t-1}, \dots, X_{t-p}$ for $k > p$ whereas these will be correlated if $k < p$. How can we use this to devise a method to determine the order of an autoregressive process?

Suppose the true order of the AR process is p for some $p \geq 1$. The k th partial autocorrelation, $\phi_{k,k}$, is estimated by $\hat{\phi}_{k,k}$ in the regression

$$X_t = \phi_{0,k} + \phi_{1,k} X_{t-1} + \cdots + \phi_{k,k} X_{t-k} + \epsilon_{k,t}.$$

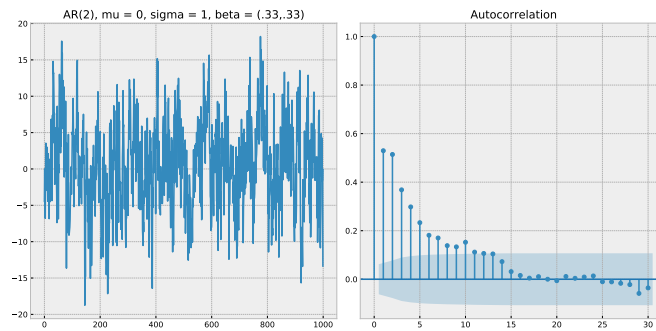
The coefficients are doubly indexed to distinguish the different statistical models. Thus $\phi_{1,k}$ is the coefficient of X_{t-1} in the AR(k) model, which can be different from the coefficient of X_{t-1} , $\phi_{1,k+1}$, in the AR($k+1$) model. Note that we contemplate running a series of regressions of the type above as we increase $k = 2, 3, \dots$. The $\epsilon_{k,t}$ have the same structure as the errors in the actual autoregressive process but are different, as are the regression coefficients. If X_t is an AR(p) process, then $\phi_{k,k} = 0$ for $k > p$. Testing that $\phi_{k,k} = 0$ can be used to determine the value of p .

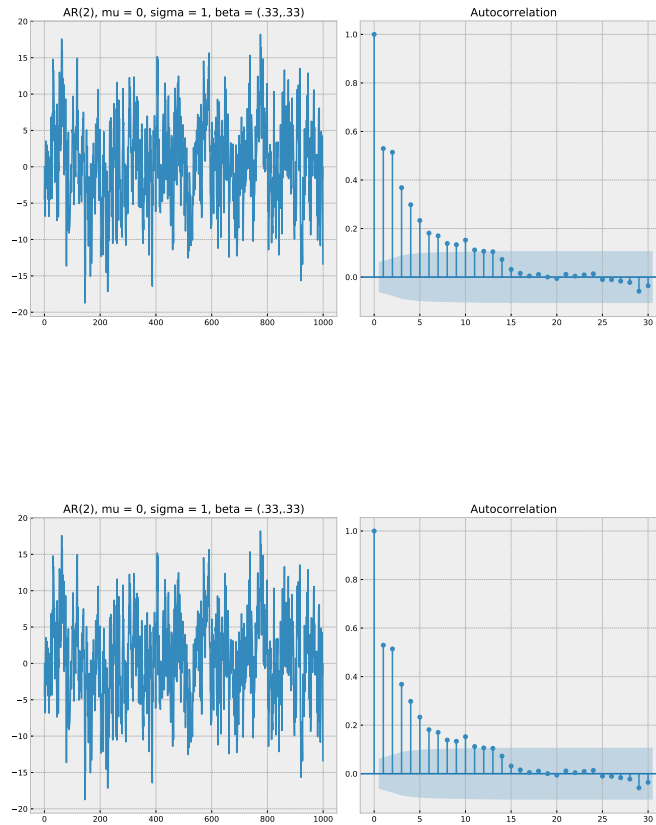
Thus, given time series data from an AR(p) process, $\{X_t, 1 \leq t \leq T\}$, one can consider a series of regressions of the form

$$X_t = \phi_{0,k} + \phi_{1,k} X_{t-1} + \cdots + \phi_{k,k} X_{t-k} + \epsilon_{k,t},$$

each time testing whether $\phi_{k,k} = 0$. The smallest k for which the test allows us to conclude $\phi_{k,k} = 0$ will then give us our estimated order for the process. Note that we are making multiple hypothesis tests, each with its own probability of type 1 error. Thus the overall (experimentwise) type i error is greater than the α -level used for each test.

The Box-Ljung test controls the overall α -level in the multiple testing case. The figure that follows show plots of an AR(3) process, its ACF and PACF. It is hard to deduce the form of the process from a plot of the data, and the ACF plot does not clearly indicate that 3 lags are needed. However, the PACF clearly show that 3 should be chosen as the order of the process.





4.8 Formal Notation for Basic Time Series Models

We start with a MA(q) sequence X_t with mean μ and let $Y_t = X_t - \mu$, to eliminate the mean. This can be written as

$$Y_t = \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t.$$

This can be concisely symbolized as

$$Y_t = \theta(B)\epsilon_t,$$

where

$$\theta(B) = (1 + \theta_1 B + \cdots + \theta_q B^q).$$

There is a similar representation for AR(p) processes. Again using Y_t the process satisfies

$$Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \epsilon_t.$$

This can be represented as

$$\phi(B)Y_t = \epsilon_t,$$

where

$$\phi(B) = (1 - \phi_1 B - \cdots - \phi_p B^p).$$

For an ARMA(p,q) model given by

$$Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p} = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q},$$

we can express this using the backshift notation as

$$\theta(B)Y_t = \phi(B)\epsilon_t.$$

A process X_t is an ARMA(p,q) process if

- X_t is weakly stationary,
- $\phi(B)X_t = \theta(B)\epsilon_t$, $\forall t$ where ϵ_t is $\text{WN}(0, \sigma^2)$
- A process X_t is an ARMA(p,q) process with mean μ if Y_t is an ARMA(p,q) process where $Y_t = X_t - \mu$.

Suppose W_t is an ARMA(p,q) process, so $\phi(B)W_t = \theta(B)\epsilon_t$, $\forall t$ where ϵ_t is $\text{WN}(0, \sigma^2)$. Now suppose $W_t = (1 - B)^d Y_t$, then Y_t is an ARIMA(p,d,q) process. This can be written in the notation

$$\phi(B)(1 - B)^d Y_t = \theta(B)\epsilon_t.$$

4.9 Parameter Estimation

We begin with the estimation of autoregressive models. If we have assumptions about the distribution of ϵ_t , then we can use standard maximum likelihood methods. On the other hand, if the error process is only assumed to be $\text{WN}(0, \sigma^2)$ with no specific distributional assumption, then we need to fall back on moment methods. We want to assume we are working with a weakly stationary model. The standard autoregressive model for Y_t with $E(Y_t) = 0$ can be written in vector form as

$$Y_t = \mathbf{Y}_{t-1}^T \boldsymbol{\phi} + \epsilon_t,$$

where $\mathbf{Y}_{t-1} = (Y_{t-1}, \dots, Y_{t-p})^T$ and $\boldsymbol{\phi} = (\phi_1, \dots, \phi_p)^T$.

Putting this together with data $(Y_1, \dots, Y_n)$, we can find the least squares solution for $\boldsymbol{\phi}$:

$$\hat{\boldsymbol{\phi}} = \left(\sum_{t=p}^n \mathbf{Y}_{t-1} \mathbf{Y}_{t-1}^T \right)^{-1} \left(\sum_{t=p+1}^n \mathbf{Y}_{t-1} Y_t \right).$$

If the ϵ_t sequence is iid $N(0, \sigma^2)$, then the estimator is also the MLE. In the simple case that $p = 1$, the formula simplifies to

$$\hat{\phi} = \sum_{t=1}^{n-1} Y_t Y_{t+1} / \sum_{t=1}^{n-1} Y_t^2.$$

The least squares estimators do not make any assumptions about the error distribution, yet given that the errors are white noise, we can work out the asymptotic distribution of those estimators. In particular they are asymptotically normal:

$$\sqrt{n}(\hat{\phi} - \phi) \Rightarrow N_p(\mathbf{0}, \sigma_\epsilon^2 \mathbf{\Gamma}_p^{-1}),$$

where

$$\mathbf{\Gamma}_p = E(\mathbf{Y}^T \mathbf{Y}),$$

with $\mathbf{Y} = (Y_1, \dots, Y_p)$.

The (ij) element of this matrix is $E(Y_i Y_j) = \gamma(i - j)$.

For $p=1$, using the AR(1) model, we have $\mathbf{\Gamma}_1 = (\gamma(0)) = \sigma_\epsilon^2 / (1 - \phi_1^2)$. This gives

$$\sqrt{n}(\hat{\phi}_1 - \phi_1) \Rightarrow N(0, \sigma_\epsilon^2 / \sigma_\epsilon^2 / (1 - \phi_1^2)) = N(0, (1 - \phi_1^2)).$$

Approximate confidence intervals can be generated and hypothesis tests conducted about the parameter ϕ_1 or the parameters $\phi = (\phi_1, \dots, \phi_p)$ using methods based on the normal distribution. We next turn to the familiar Yule-Walker (YW) equations approach, which is based on using the method of moments.

4.10 Yule Walker Equations

We begin with a standard autoregression process characterized by $\phi(B)Y_t = \epsilon_t$. We next multiply both sides by Y_{t-k} and take expectations. This leads to a set of linear equations for the autocovariance values, and if one normalizes by dividing by γ_0 , we get linear equations for the autocorrelations. Specifically, we get for $1 \leq k \leq p$:

$$\begin{aligned}\gamma(k) &= \phi_1 \gamma(k-1) + \dots + \phi_p \gamma(k-p), \\ \rho(k) &= \phi_1 \rho(k-1) + \dots + \phi_p \rho(k-p).\end{aligned}$$

If we let $\boldsymbol{\rho} = (\rho(1), \dots, \rho(p))^T$ and $\mathbf{R}$ be a $p \times p$ matrix whose ij element is $\rho(i - j)$, then

$$\boldsymbol{\rho} = \mathbf{R}\boldsymbol{\phi},$$

the Yule-Walker (YW) equations. We want to use these equations, plugging in estimates of $\boldsymbol{\rho}$ to generate an estimate of $\boldsymbol{\phi}$. Replacing $\rho(k)$ by $\hat{\rho}(k) = r_k$, $1 \leq k \leq p$ (of course $\rho(0) = \hat{\rho}(0) = 1$) we have the YW estimates for $\boldsymbol{\phi}$:

$$\hat{\boldsymbol{\phi}} = \mathbf{R}^{-1} \mathbf{r}.$$

If appropriate, one would prefer to use MLE methods for parameter estimation. However, as we will see, the MLEs often must be solved numerically by a search algorithm. This sort of algorithm requires a starting value, and the estimates from the YW equation approach can often provide a good starting value. We will soon turn to forecasting/prediction. For this problem, we have data, from which we can fit a model and estimate the model parameters, then we will use that model to forecast future values.

For example, if one fits an AR(P) model to data $Y_1, \dots, Y_t$ and then wants to predict a future value, say Y_{t+k} , the minimum mean square error predictor would be given by $E(Y_{t+k}|\mathcal{F}_t)$. This will often be given by extrapolating the model from time t to time $t+k$ using the estimated coefficients. This will be demonstrated later. If one is going to engage in forecasting on a daily (or even shorter) basis, updating the data each time, it would be computationally much more efficient to have a recursive or on-line method to take the old results and update them with the new data.

As a simple example of this, consider a stream of numbers, $X_1, X_2, \dots$. Suppose we want to compute the mean as each new number arrives, so we have the running mean, $\bar{X}_n = \sum_{i=1}^n X_i/n$. Each time, rather than going back to the beginning and adding all the numbers up (having to store them all), one can note that

$$\bar{X}_n = \bar{X}_{n-1} + \frac{X_n - \bar{X}_{n-1}}{n}.$$

Thus, one only needs to know the current value of $n-1$ and $\bar{X}_{n-1}$ and the new number, X_n in order to compute $\bar{X}_n$.

There is a similar on-line algorithm to update the sample variance knowing only its current value, the sample size and the new value. There is a recursive algorithm of this sort for the solution to the YW equations given by the Durbin Levinson algorithm, which is presented later in this section. This algorithm is especially useful if one has a current AR(p) model estimated using YW, adds new data and updates the estimated model.

In estimating the parameters from a MA(2) model, the problem is far more difficult and must be done numerically. This, of course, carries over to ARMA models which have a MA component. In such cases, the estimation is done using likelihood methods, and standard packages based this on i.i.d. white noise errors.

4.11 Likelihood Methods

We begin with the simplest model, AR(1) with a single parameter with iid Normal $WN(0, \sigma_\epsilon^2)$ with three unknown parameters, $\mu, \phi_1, \sigma_\epsilon^2$. Suppose the AR(1) data is $(Y_1, Y_2, \dots, Y_n)$. We must make an assumption concerning the distribution of Y_1 , the first data point that initiates the process. We might assume that it has the stationary distribution, $N(\mu, \sigma_\epsilon^2/(1 - \phi_1^2))$, or that it is unknown. Conditional on the starting value, Y_1 , the sequence is a Markov process, and the conditional distribution of $Y_{i+1}|\mathcal{F}_i$ has a $N(\mu + \phi_1(Y_i - \mu), \sigma_\epsilon^2)$ distribution. Consequently, we can easily write out the likelihood function for the data:

The MLE for σ_ϵ^2 is found by the usual methods and is given by

$$\hat{\sigma}_\epsilon^2 = \frac{1}{n-1} \sum_{i=2}^n (Y_i - (\hat{\mu} + \hat{\phi}_1(Y_{i-1} - \hat{\mu})))^2,$$

the residuals from comparing the data with the fitted values for the data using the model.

The extension of likelihood methods for AR(1) to AR(p) processes is in some sense straightforward, but it will again rely on numerical solutions. For AR(p) models and a data set $(Y_1, \dots, Y_n)$, the model is no longer a Markov process.

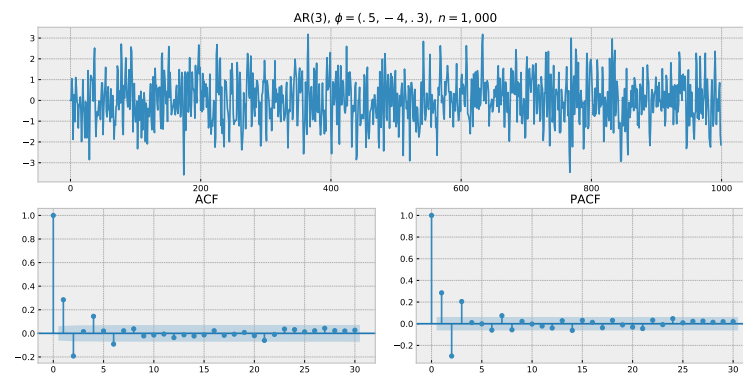
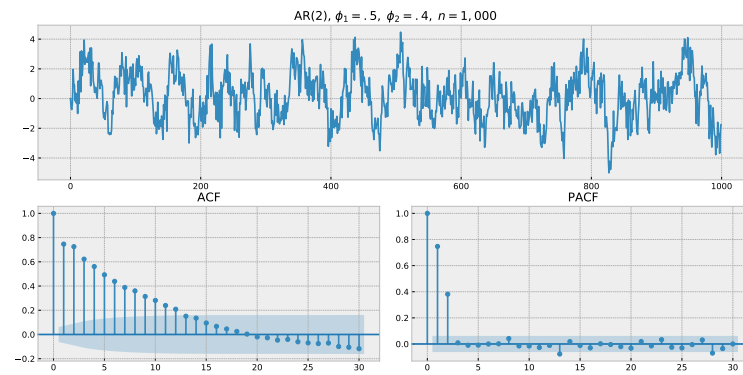
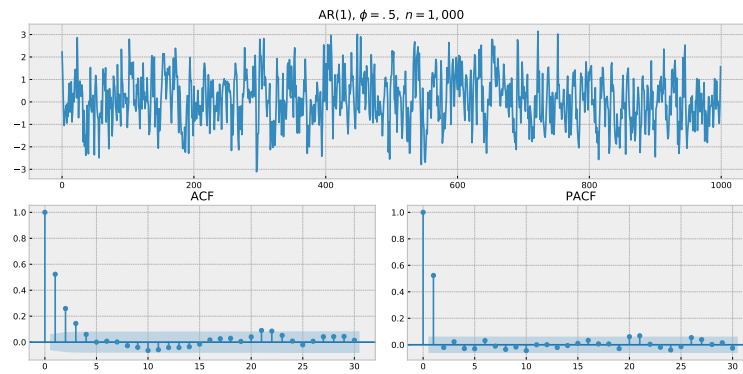
The next value in the time series, Y_{n+1} will depend upon $Y_n, Y_{n-1}, \dots, Y_{n+1-p}$, but not on any data further in the past. Also there will be an initial part of the data set, $Y_1, \dots, Y_p$ for which we must determine their joint distribution. One obvious choice is the p-dimensional stationary distribution of the AR(p) process.

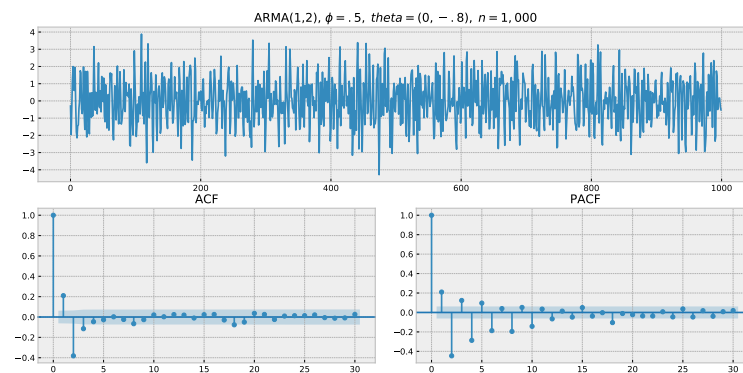
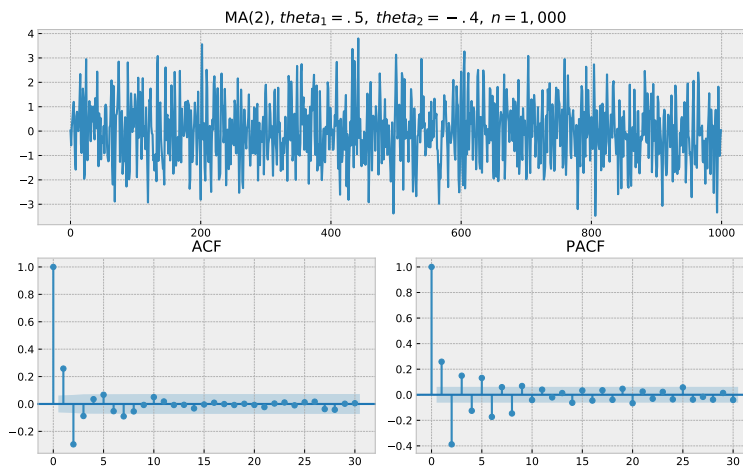
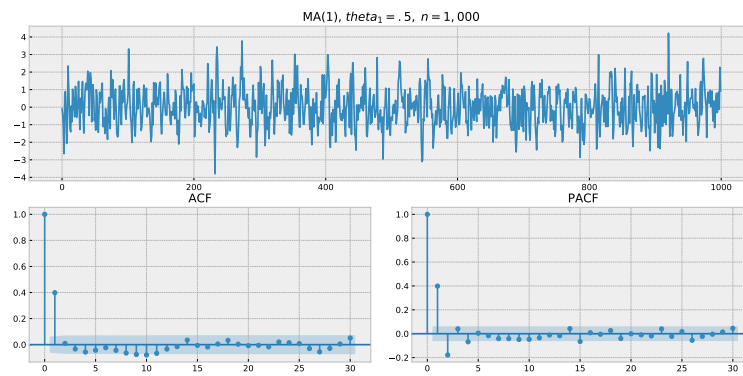
This again calls for numerical methods to find the maximum likelihood, and it is common to use conditional least squares. Nevertheless, the MLE of σ_ϵ^2 is still given by the sum of squared residuals divided by the appropriate denominator.

Having given this description of the MLE approaches used for the simple AR(1) and AR(p) models, it should be clear that using likelihood methods for MA(1) and ARMA(p,q) models will be more complicated and will certainly require numerical methods.

4.12 Examples

We give a series of examples showing simulated time series (using the quantecon python package) that were generated using the exact model. The associated ACF and PACF graphs are also included. A python program to create such graphs for general ARMA processes is given after these figures. One of the points of these graphs is to show that the PACF is a reliable tool for determining the order of an AR process (assuming the data are actually generated by an AR process). In all three cases, the PACF falls (or nearly so) into the noise region for all lags larger than the order of the process. Nevertheless, the PACF is no longer relevant for MA or ARMA processes. We can see through these examples that the PACF can have significant values at lags larger than the largest order of the process.





The code follows:

```
import numpy as np
from quantecon import ARMA
import matplotlib.pyplot as plt
import pandas as pd
import statsmodels as smm
import statsmodels.formula.api as smf
import statsmodels.tsa.api as smt
import statsmodels.api as sm

def tsplot(y, lags=None, figsize = (10,8), style = 'bmh'):
    if not isinstance(y, pd.Series):
        y = pd.Series(y)

    with plt.style.context(style):
        fig = plt.figure(figsize=figsize)
        layout = (2,2)
        ts_ax = plt.subplot2grid(layout, (0,0), colspan = 2)
        acf_ax = plt.subplot2grid(layout, (1,0))
        pacf_ax= plt.subplot2grid(layout, (1,1))

        y.plot(ax = ts_ax)
        ts_ax.set_title('ARMA(1,2),  $\phi = .5$ ,  $\theta = (0, -.8)$ ,  $n = 1,000$ ')
        smt.graphics.plot_acf(y, lags = lags, ax=acf_ax, alpha = 0.05)
        smt.graphics.plot_pacf(y, lags=lags, ax=pacf_ax, alpha = 0.05)
        acf_ax.set_title('ACF')
        pacf_ax.set_title('PACF')
        plt.tight_layout()

    return

n = 1000
maparams = np.array([.50,-.50]) # MA coefficients
ma = np.r_[1, maparams] # add zero-lag

arparams = np.array([.30, .20, .10]) #Ar coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
ar_process = ArmaProcess(ar)

lp = ARMA(.5, (0,-.8))
X = lp.simulation(ts_length = n)
```

```

tsplot(X)
plt.show()

arparams = np.array([.60, -.40]) # AR coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
maparams = np.array([.50, -.50]) # MA coefficients
ma = np.r_[1, maparams] # add zero-lag
arma_process = ArmaProcess(ar, ma)

```

4.13 Fitting an ARMA Model

We can use either the statsmodels or the pmdarima packages to fit an ARMA (and ARIMA) model to data. In this case we use pmdarima. The examples shown are for Microsoft (MSFT) and Goldman Sachs (GS) respectively. We use the data gathering is done using the python program before, except the dates are January 1, 2020 to December 31, 2020. The additional code needed to fit an ARIMA model are:

```

import pmdarima as pm
from pmdarima import auto_arima

auto = pm.auto_arima(logret_shf, seasonal=False, stepwise=True,
                    suppress_warnings = True, error_action = "ignore",
                    max_p=6, max_order=None, trace=True)

```

With “trace = True” we see the sequence of models that are tested and the resulting value of the information criterion (in this case AIC). The ARIMA model and its seasonal version are discussed in the next section; however, for reference considering the ARIMA(p,d,q) process, the p and q parameters correspond to an ARMA(p,q) process and the d parameter corresponds to the number of differences needed to result in a weakly stationary process. The addition parameters (0,0,0)(0) refer to seasonal factors, which are not included here.

For MSFT:

Performing stepwise search to minimize aic

```

ARIMA(2,0,2)(0,0,0)[0]      : AIC = -1131.419, Time = 0.34sec
ARIMA(0,0,0)(0,0,0)[0]      : AIC=-1089.938, Time=0.03 sec
ARIMA(1,0,0)(0,0,0)[0]      : AIC = -1133.408, Time = 0.03sec
ARIMA(0,0,1)(0,0,0)[0]      : AIC=-1124.028, Time=0.08 sec
ARIMA(2,0,0)(0,0,0)[0]      : AIC = -1131.548, Time = 0.03sec
ARIMA(1,0,1)(0,0,0)[0]      : AIC=-1131.495, Time=0.14 sec
ARIMA(2,0,1)(0,0,0)[0]      : AIC = -1130.474, Time = 0.19sec
ARIMA(1,0,0)(0,0,0)[0]intercept : AIC=-1132.821, Time=0.08 sec

```

Best model: ARIMA(1,0,0)(0,0,0)[0]
Total fit time: 0.938 seconds

For GS:

Performing stepwise search to minimize aic ARIMA(2,0,2)(0,0,0)[0] : $AIC = -1131.419, Time = 0.38sec$

ARIMA(0, 0, 0)(0, 0, 0)[0]	: AIC=-1089.938, Time=0.03 sec
ARIMA(1,0,0)(0,0,0)[0]	: $AIC = -1133.408, Time = 0.05sec$
ARIMA(0, 0, 1)(0, 0, 0)[0]	: AIC=-1124.028, Time=0.08 sec
ARIMA(2,0,0)(0,0,0)[0]	: $AIC = -1131.548, Time = 0.03sec$
ARIMA(1, 0, 1)(0, 0, 0)[0]	: AIC=-1131.495, Time=0.13 sec
ARIMA(2,0,1)(0,0,0)[0]	: $AIC = -1130.474, Time = 0.20sec$
ARIMA(1, 0, 0)(0, 0, 0)[0]intercept	: AIC=-1132.821, Time=0.06 sec

Best model: ARIMA(1,0,0)(0,0,0)[0]
Total fit time: 0.953 seconds

5 Transforming a Non-Stationary Time Series to a Stationary Time Series

As we have discussed before, most financial and economic time series are non-stationary, and they also have dependencies with other time series. The methodology being developed in the MSCF Financial Time Series course are applicable to weakly stationary time series, thus to use the methods we must first transform the non-stationary series to become weakly stationary. Two important sources of non-stationarity are *trend* and *seasonal components*. This can be represented as

$$X_t = Y_t + m_t + s_t + \epsilon_t,$$

where $\{X_t, t \geq 1\}$ is the original time series, $\{m_t, t \geq 1\}$ is a trend process, $\{s_t, t \geq 1\}$ is a seasonal component, and $\{\epsilon_t, t \geq 1\}$ is a stationary time series with $E(\epsilon_t) = 0, t \geq 1$ and $\{Y_t, t \geq 1\}$ is the centered time series with $E(Y_t) = 0$. In the case of the seasonal component, $\{s_t, t \geq 1\}$, it will have a period, d , and this will lead to the restrictions $s_{t+d} = s_t$ and $\sum_{i=1}^d s_i = 0$.

5.1 No Seasonal Component

We begin with the case of no seasonal component, so

$$X_t = m_t + Y_t.$$

There are two common approaches to estimation of the trend: $\hat{m}_t$. least squares and moving average smoothing. These apply when the trend is modelled parametrically.

1. Least Squares:

One first decides on a parametric model for the trend and then fits the estimated trend using that parametric model. For example:

(a) Linear trend: $m_t = \beta_0 + \beta_1 t$,

(b) Quadratic trend: $m_t = \beta_0 + \beta_1 t + \beta_2 t^2$

Estimate the β coefficients using least squares to minimize $\sum (x_t - m_t)^2$

2. Moving Average Smoother

Choose a non-negative integer q . For $t = q + 1, \dots, n - q - 1$

$$\hat{m}_t = \frac{1}{2q + 1} \sum_{i=-q}^q x_{t+i}.$$

This is the uniform averaging of consecutive values of the time series centered at the current value. Of course other smoothing functions are possible that do not uniformly average adjacent values. For example, we might wish to give more weight to nearby values.

Note that the moving average smoother preserves a linear trend. Assume that $m_t = \beta_0 + \beta_1 t$. Then applying the moving average smoother:

$$\frac{1}{2q+1} \sum_{j=-q}^q m_{t+j} = \beta_0 + \beta_1 t + \frac{1}{2q+1} \beta_1 \sum_{j=-q}^q j = \beta_0 + \beta_1 t.$$

5.2 Seasonal Component Present

We next consider the case in which there is a seasonal component with period d present. Both the least squares and the moving average approaches described above can be used with some minor modification.

Least Squares

First, we define indicator functions,

$$I_j(t) = \begin{cases} 1 & \text{if } t \bmod d = j, \\ 0 & \text{otherwise,} \end{cases}$$

for $1 \leq j \leq d$. Second, we consider a model for the trend, for example a linear trend, $m_t = \beta_0 + \beta_1 t$. Thus our model becomes

$$m_t = \beta_0 + \beta_1 t + \sum_{j=1}^d \omega_j I_j(t),$$

and we estimate the $d+2$ parameters β_0, β_1 and $\{\omega_i, i = 1, \dots, d\}$ using least squares by minimizing $\sum (m_t - \hat{m}_t)^2$. Now the seasonal effects parameters represent the differential impact associated with the particular steps in the d -step cycle. Hence, we require that these effects summed over the cycle sum to 0. Consequently, we require that $\sum_{i=1}^d \omega_i = 0$. This constraint can be dealt with by estimating the $d+2$ parameters, then computing $\bar{\omega} = \frac{1}{d} \sum_{i=1}^d \hat{\omega}_i$. We then define $\beta_{i+1} = \omega_i - \bar{\omega}$. Since $\sum_{i=1}^d I_i(t) = 1, \forall t$, we have

$$\begin{aligned} m_t &= \beta_0 + \beta_1 t + \sum_{i=1}^d \omega_i I_i(t) \\ &= \beta_0 + \beta_1 t + \sum_{i=1}^d (\beta_{i+1} + \bar{\omega}) I_i(t) \\ &= \beta_0 + \beta_1 t + \bar{\omega} + \sum_{i=1}^d \beta_{i+1} I_i(t), \end{aligned}$$

and the seasonal component is given by $\sum_{j=1}^d \beta_{j+1} I_j(t)$. The same ideas can be applied for more complicated trend models.

The most ideal circumstances for forecasting are when a time series is weakly stationary and the variables have a normal distribution. With raw financial data, this is rarely the case. We have discussed the necessity

to detect and remove time trends and seasonality, but only when those are additive. It is possible that trends and seasonality can enter multiplicatively, say with models of the form $X_t = m_t s_t Y_t + \epsilon_t$. In such a case, a logarithmic transformation ($\tilde{X}_t = \log(X_t)$) will convert the multiplicative form to a linear form.

6 Differencing to Obtain Stationarity

Differencing is an alternative approach to explicitly modeling the trend. Recall the backshift operator, B , where $BX_t = X_{t-1}$. Similarly, we define the differencing operator, “del” = ∇ by $\nabla X_t = X_t - X_{t-1}$. Note that $\nabla X_t = (1 - B)X_t$. One can define and apply polynomials in ∇ and B , for example

$$\nabla^3 X_t = (1 - B)^3 X_t = (1 - 3B + 3B^2 - B^3)X_t = X_t - 3X_{t-1} + 3X_{t-2} - X_{t-3}.$$

Appropriate differencing can remove polynomial trends. We show this for the linear and quadratic cases. Suppose that $m_t = \beta_0 + \beta_1 t$. That is $X_t = \beta_0 + \beta_1 t + \epsilon_t$, where the ϵ_t process is assumed to be weakly stationary. It follows that

$$\begin{aligned} \nabla X_t &= X_t - X_{t-1} \\ &= \beta_0 + \beta_1 t + \epsilon_t - \beta_0 - \beta_1(t-1) - \epsilon_{t-1} \\ &= \beta_1 + \epsilon_t - \epsilon_{t-1} \\ &= \beta_1 + \text{a weakly stationary process.} \end{aligned}$$

Similarly, suppose instead there is a quadratic term, i.e. $X_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \epsilon_t$. Applying differencing twice, $(1 - B)^2 = 1 - 2B + B^2$, to the time series, we find

$$\begin{aligned} (1 - B)^2 X_t &= X_t - 2X_{t-1} + X_{t-2} \\ &= \beta_0 + \beta_1 t + \beta_2 t^2 + \epsilon_t \\ &\quad - 2(\beta_0 + \beta_1(t-1) + \beta_2(t-1)^2) + \epsilon_{t-1} \\ &\quad + \beta_0 + \beta_1(t-2) + \beta_2(t-2)^2 + \epsilon_{t-2} \\ &= 2\beta_2 + (\epsilon_t - 2\epsilon_{t-1} + \epsilon_{t-2}) \\ &= 2\beta_2 + \text{a weakly stationary process.} \end{aligned}$$

This can be generalized to polynomial trends in the obvious way. One can also use differencing to remove both a seasonal and a linear trend. Suppose that $X_t = \beta_0 + \beta_1 t + s_t + \epsilon_t$, and the seasonal term has period d , i.e. $s_t = s_{t-d}$. In this case the differencing should be done over d time periods rather than simply one time period using ∇ . Instead we define a “ d -lagged difference operator,”

$$\nabla_d = X_t - X_{t-d} = (1 - B^d) \neq (1 - B)^d = \nabla^d.$$

Care should be taken not to confuse ∇_d with ∇^d . Applying the ∇_d operator to the time series, we find

$$\begin{aligned} \nabla_d X_t &= \nabla_d m_t + \nabla_d s_t + \nabla_d \epsilon_t \\ &= m_t - m_{t-d} + s_t - s_{t-d} + \epsilon_t - \epsilon_{t-d} \\ &= \beta_1(t - (t-d)) + 0 + \epsilon_t - \epsilon_{t-d} \\ &= \beta_1 d + \text{a weakly stationary process.} \end{aligned}$$

Thus this differencing operator has transformed the original time series into a weakly stationary process.

Definition

A univariate time series $\{X_t\}$ is *integrated of order k* (denoted by $\mathcal{I}(k)$) if $\nabla^{k-1} X_t$ is non-stationary (hence $\nabla^{k-i} X_t$ is non-stationary for $i = 1, \dots, k-1$); however $\nabla^k X_t$ is stationary.

6.1 Testing for Stationarity: Dickey Fuller Test

We begin with the AR(1) process, $X_t = \beta X_{t-1} + \epsilon_t$, with ϵ_t assumed to be a stationary process. The $\{X_t\}$ process is stationary if and only if $|\beta| < 1$. This leads to the hypothesis test:

$H_0 : |\beta| = 1$ versus $H_1 : |\beta| < 1$.

There are two approaches to test this hypothesis:

1. Regress X_t on X_{t-1} and set up a “t-statistic,”
2. Subtract X_{t-1} from both sides, so $\nabla X_t = (\beta - 1)X_{t-1} + \epsilon_t$, regress ∇X_t on X_{t-1} , and set up a “t-statistic”.

Both approaches give identical estimates of β and identical “t-statistics”. To test for the unit root, we need the distribution of the test statistics under $H_0 : \beta = 1$. In this case, the time series behaves like a random walk, or for long series, when properly scaled like a Wiener process. This gives rise to the Dickey-Fuller test described next. The estimator of β is given by

$$\begin{aligned}\hat{\beta} &= \frac{\sum_{t=1}^T X_{t-1} X_t}{\sum_{t=1}^T X_{t-1}^2} \\ \hat{\beta} - 1 &= \frac{\sum_{t=1}^T X_{t-1} (X_t - X_{t-1})}{\sum_{t=1}^T X_{t-1}^2} \\ s_T^2 &= \frac{1}{T-1} \sum_{t=1}^T (X_t - \hat{\beta} X_{t-1})^2 \\ t_T &= \frac{\hat{\beta} - 1}{s_T / \sqrt{\sum_{t=1}^T X_{t-1}^2}}\end{aligned}$$

Under H_0 , $\beta = 1$ and the time series is a random walk. For large samples, both numerator and denominator can be approximated using central limit theorem ideas. Specifically:

$$\begin{aligned}Y_t &= \frac{X_t}{\sigma\sqrt{T}} \Rightarrow W\left(\frac{t}{T}\right) = W(s), \quad s = \frac{t}{T}, \quad 0 \leq s \leq 1. \\ T(\hat{\beta} - 1) &= \frac{\sum_{t=1}^T Y_{t-1} (Y_t - Y_{t-1})}{\sum_{t=1}^T Y_{t-1}^2 \frac{1}{T}} \Rightarrow \frac{\int_0^1 W(s) dW(s)}{\int_0^1 W^2(s) ds}. \\ t_T &= \frac{\hat{\beta} - 1}{s_T / \sqrt{\sum_{t=1}^T X_{t-1}^2}} \\ t_T &\Rightarrow \frac{\int_0^1 W(s) dW(s)}{\sqrt{\int_0^1 W^2(s) ds}} = \frac{\frac{1}{2}(W^2(1) - 1)}{\sqrt{\int_0^1 W^2(s) ds}}.\end{aligned}$$

The limiting probability distribution is different from the t-distribution, hence the statistics mentioned earlier in this section were referred to as “t-statistics,” because they do not have a t-distribution. For example $P(t_T < 0) = P(W^2(1) < 1) = P(-1 < Z < 1) = .68$, so the distribution is not symmetric, unlike the t-distribution. This distribution is tabled, and used in statistical packages. Remember that it is derived under certain specific assumptions. The analysis is correct for $T \rightarrow \infty$ as long as the ϵ_t has finite second moments. However, in smaller samples and heavy tailed distributions, the distribution will not be correct. One can use Monte Carlo methods to simulate the distribution of t_T under any specific model.

6.2 Box-Cox Transformation

Once the trend and seasonal components have been eliminated, there are still two common ways in which further transformation can be beneficial. One possible problem is *heteroscedascity*, or unequal variance. This is especially a problem when the variability of the data grows with the level of the data. For example, if we consider a single stock price or stock index price process, the variability can be expected to grow with the price, so at high values the variability is higher than at low values.

A second issue is the lack of symmetry. Ideally, we would like the marginal distributions of the data to be symmetric. Although we expect the tail behavior over short time intervals to be heavier than one would observe with the normal distribution, we would like to transform the data to make it more symmetric. This would make it more likely in large data sets that the normal distribution could be reasonably used.

Interestingly, there is a family of transformations, the Box-Cox family, that can be used to address either or possibly both of these issues. The Box-Cox family of data transformations (of X into $X^{(\alpha)}$) is defined by

$$X^{(\alpha)} = \begin{cases} (X^\alpha - 1)/\alpha & \alpha \neq 0 \\ \log(X) & \alpha = 0 \end{cases}$$

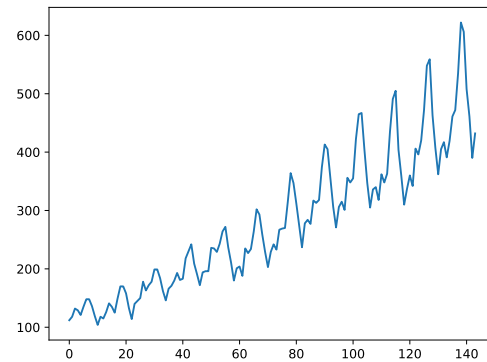
The α divisor is included to make the family continuous as $\alpha \rightarrow 0$, since

$$\lim_{\alpha \rightarrow 0} \frac{X^\alpha - 1}{\alpha} = \log(X).$$

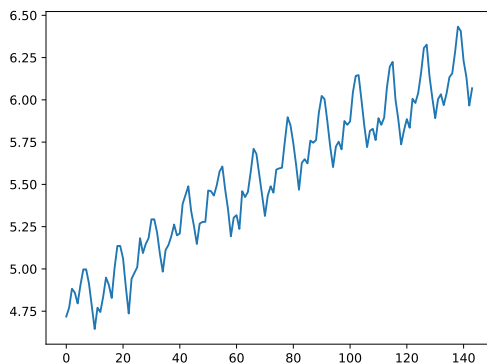
This transformation with proper choice of α can help to make the data more symmetric with more constant variance if the variance increases as the conditional mean increases. There is a fairly extensive discussion about the choice of α in section 4.6 (pp 67-71) in the text by Ruppert and Matteson.

A simple example (drawn from the Box and Jenkins or Ruppert and Matteson texts and Kaggle) provides a data set (Monthly International Airline Passengers, 1949-1969) that exhibits the need for a Box-Cox transformation, trend and seasonal adjustments. The first set of figures shows the raw data on the left. These data have obvious trend and seasonal effects (with the period being $d = 12$ to account for the monthly data patterns). Less obvious is the change in variability from the early years to the later years. It is clear that the range in the oscillations is quite small initially and grows much larger in the later years. A variance stabilization transformation is needed. A log transformation was chosen, and the results are pictured on the right. The result was successful in that the variability is relatively constant over the entire data set. One must note that the transformations we have discussed are designed to achieve stationarity, allowing us, for

example, to predict future behavior using time series techniques. However, once such predictions have been made using the transformed series, we must invert those transforms to return to the original time series.

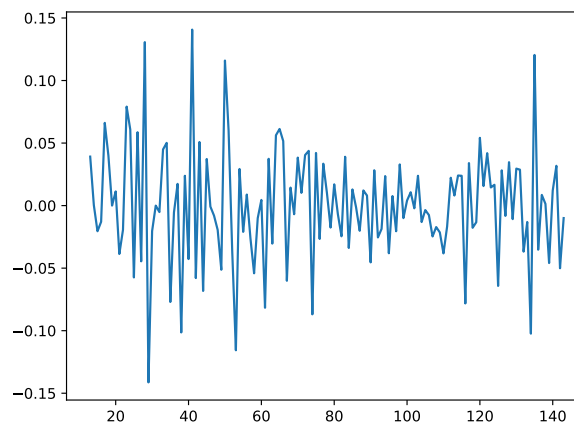
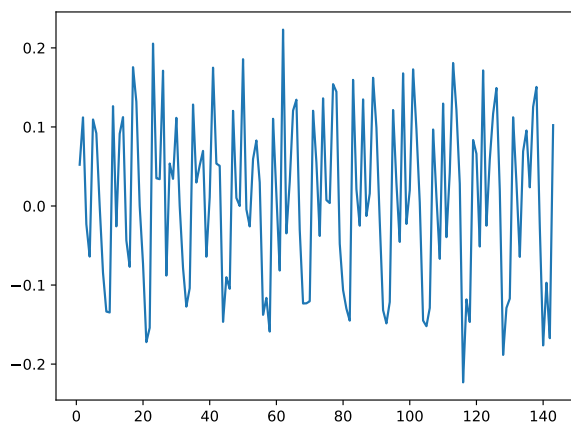


Difference(Log(Airline Passengers))



Having made the log transform, we next deal with the trend and seasonality. As one can see, the trend is nearly linear, and the seasonal periodicity is monthly. Consequently, a 12 month difference(∇_{12}) should address both of these simultaneously. The results are on the next pair of figures. The first shows simple differencing (∇), and the second shows 12-month differencing. The final result on the right is a time series that passes the weak stationary tests, and allows for the prediction of future values.

Difference(Log(Airline Passengers))



6.3 ARIMA, ARIMAX, SARIMA and SARIMAX Models

We have discussed AR, MA and ARMA processes. A generalization of ARMA processes are the ARIMA processes, in which the added letter “I” stands for “integrated,” thus Auto Regressive Integrated Moving Average process. These processes have three parameters: $\text{ARIMA}(p,d,q)$ where the p and q parameters are associated with the AR and MA components respectively. The d parameter refers to the number of differences of the process that must be taken so that the process becomes a weakly stationary $\text{ARMA}(p,q)$ process. That is, the $\text{ARIMA}(p,d,q)$ process is the d -fold integral of an $\text{ARMA}(p,q)$ process and is, therefore not weakly stationary before this differencing operation. Many of the relevant python time series packages such as statsmodels, pmdarima and arch, are built to handle the ARIMA processes.

An even broader class of processes, SARIMA, generalizes ARIMA processes where the “S” refers to “seasonal.” In a similar fashion, ARIMAX and SARIMAX processes are ARIMA and seasonal ARIMA process with an additional exogenous process included. While we do not pursue the possibility of including exogenous processes into the model description (i.e. both ARIMAX and SARIMAX processes) in these notes, this can be an important addition to a financial or economic time series model. For example, in modelling the returns of a stock price process we might want to include an interest rate process in the stock price process, especially if the stock is interest-rate sensitive.

The SARIMA process has additional parameters beyond the standard $\text{ARIMA}(p,d,q)$. The seasonal components have their own P,D,Q parameter along with the length of the periodic cycle, m . Thus a SARIMA process is parametrized as $\text{ARIMA}(p,d,q)(P,D,Q,m)$ or $\text{ARIMA}(p,d,q)(P,D,Q)m$, depending upon the particular author or package.

We now return to the international air passenger data set and fit with a SARIMA using the pmdarima package. The code to do this is given next, and then we show the resulting model fit. Later when we discuss prediction, we will use the prediction capabilities of pmdarima to split the data into a training set and a testing set where we know the actual data for comparison purposes. We can also use the entire data set to

predict future values, but there is no way to evaluate those predictions.

First, we consider the code. Note we first apply a log transform and then fit the model. The code contains both the fitting and the prediction.

```
import pmdarima as pm
from pmdarima.model_selection import train_test_split
import numpy as np
import matplotlib.pyplot as plt
# load data, this is an available data set in pmdarima
y = pm.datasets.load_airpassengers()
y = np.log(y) # log transformation
# split data into training and test sets
train, test = train_test_split(y, train_size=100)

# fit the model using the whole data set
fit1 = pm.auto_arma(y, m=12, trace=True, suppress_warnings=True)

# fit the model using the training set
model = pm.auto_arma(train, seasonal=True, m=12)

# create forecasts
forecasts = model.predict(test.shape[0]) # predict N steps into the future

# the following three lines transform the results back to the original scale
# y = np.exp(y)
# train = np.exp(train)
# forecasts = np.exp(forecasts)

# Visualize the forecasts (blue=train, red=forecasts)
x = np.arange(y.shape[0])
plt.plot(y, c = 'blue') # plot whole data set
plt.plot(x[100:], forecasts, c='red') #superimpose the predicted values
plt.show()
```

Here are the results for fitting a SARIMA model:

```
ARIMA(2,0,2)(1,1,1)[12] intercept : AIC=-476.679, Time=2.73 sec
ARIMA(0,0,0)(0,1,0)[12] intercept : AIC=-358.005, Time=0.05 sec
ARIMA(1,0,0)(1,1,0)[12] intercept : AIC=-472.836, Time=1.22 sec
ARIMA(0,0,1)(0,1,1)[12] intercept : AIC=-423.080, Time=0.80 sec
ARIMA(0,0,0)(0,1,0)[12] : AIC=-152.756, Time=0.02 sec
ARIMA(2,0,2)(0,1,1)[12] intercept : AIC=-485.488, Time=2.27 sec
ARIMA(2,0,2)(0,1,0)[12] intercept : AIC=-455.359, Time=0.70 sec
```

ARIMA(2,0,2)(0,1,2)[12] intercept : AIC=-483.586, Time=4.86 sec
 ARIMA(2,0,2)(1,1,0)[12] intercept : AIC=-479.076, Time=1.98 sec
 ARIMA(2,0,2)(1,1,2)[12] intercept : AIC=-483.248, Time=6.17 sec
 ARIMA(1,0,2)(0,1,1)[12] intercept : AIC=-486.609, Time=1.94 sec
 ARIMA(1,0,2)(0,1,0)[12] intercept : AIC=-455.568, Time=0.48 sec
 ARIMA(1,0,2)(1,1,1)[12] intercept : AIC=-477.516, Time=2.17 sec
 ARIMA(1,0,2)(0,1,2)[12] intercept : AIC=-484.619, Time=4.31 sec
 ARIMA(1,0,2)(1,1,0)[12] intercept : AIC=-480.652, Time=1.88 sec
 ARIMA(1,0,2)(1,1,2)[12] intercept : AIC=-483.150, Time=5.11 sec
 ARIMA(0,0,2)(0,1,1)[12] intercept : AIC=-455.593, Time=2.31 sec
 ARIMA(1,0,1)(0,1,1)[12] intercept : AIC=-489.188, Time=1.17 sec
 ARIMA(1,0,1)(0,1,0)[12] intercept : AIC=-457.177, Time=0.59 sec
 ARIMA(1,0,1)(1,1,1)[12] intercept : AIC=-478.580, Time=1.22 sec
 ARIMA(1,0,1)(0,1,2)[12] intercept : AIC=-487.386, Time=3.67 sec
 ARIMA(1,0,1)(1,1,0)[12] intercept : AIC=-482.645, Time=1.44 sec
 ARIMA(1,0,1)(1,1,2)[12] intercept : AIC=-485.197, Time=5.11 sec
 ARIMA(1,0,0)(0,1,1)[12] intercept : AIC=-482.026, Time=1.00 sec
 ARIMA(2,0,1)(0,1,1)[12] intercept : AIC=-486.762, Time=1.38 sec
 ARIMA(0,0,0)(0,1,1)[12] intercept : AIC=-367.061, Time=0.73 sec
 ARIMA(2,0,0)(0,1,1)[12] intercept : AIC=-489.295, Time=2.25 sec
 ARIMA(2,0,0)(0,1,0)[12] intercept : AIC=-458.262, Time=0.23 sec
 ARIMA(2,0,0)(1,1,1)[12] intercept : AIC=-483.841, Time=2.17 sec
 ARIMA(2,0,0)(0,1,2)[12] intercept : AIC=-487.396, Time=4.03 sec
 ARIMA(2,0,0)(1,1,0)[12] intercept : AIC=-482.539, Time=1.67 sec
 ARIMA(2,0,0)(1,1,2)[12] intercept : AIC=-485.903, Time=2.61 sec
 ARIMA(3,0,0)(0,1,1)[12] intercept : AIC=-487.305, Time=2.05 sec
 ARIMA(3,0,1)(0,1,1)[12] intercept : AIC=-485.296, Time=2.17 sec
 ARIMA(2,0,0)(0,1,1)[12] : AIC=-481.418, Time=1.20 sec

Best model: ARIMA(2,0,0)(0,1,1)[12] intercept

Total fit time: 73.760 seconds

Thus the final model, based on the entire data set is an AR(2) process with a seasonal component of $P = 2$, $D = 0$, $Q = 1$ with period 12 months.

7 Prediction/Forecasting

One of the most important inferences that is made in time series analysis is *prediction* or *forecasting*. For example, given the current and past values of a particular stock (and perhaps other stocks in the same industry), we want to predict a future value. The word *predict* generally connotes a single value, our best guess. Just as a simulation-based estimated is of minimal value without attaching a measure of accuracy (standard error), a single predicted value is also of little value. Rather, we need to think in terms of a *predictive distribution* or a *prediction interval*. Generally, this will be based on the normal distribution.

Most raw financial time series are not stationary. The general approach to prediction is to first transform the time series so that it is approximately stationary. The typical steps made are as follows:

- Plot the data checking and cleaning it looking for outliers, missing values, etc.
- After cleaning the data, plot it and look for time trends or seasonality, and apply transformations to remove these.
- Fit a stationary time series model
- Examine the residuals to be sure that they conform to the assumptions of the time series model.
- Generate the predictive distribution of the stationary model at some appropriate time in the future.
- Invert the transformation (i.e. undo the steps taken to reduce to the stationary model) and apply it to the predictive distribution to obtain a predictive distribution for the original series at the appropriate time.

7.1 Point Prediction

While we ultimately want to generate a prediction interval, we begin by developing a point prediction of a future value of a stationary time series.

Suppose $\{X_t, t = 0, 1, \dots\}$ is a weakly stationary process with mean μ and covariance function $\gamma_X(t)$. Suppose the current time is t , the information available is denoted by $\mathcal{F}_t$, and we wish to forecast (predict) a future value, say X_{t+k} . There are many ways to predict or forecast future values; however, we will consider *linear predictors*, and we will denote the best linear predictor of X_{t+k} by $\hat{X}_{t+k}$. Here, “best” will be defined as the forecast that minimizes the mean square error given the information up to time t , $E((X_{t+k} - \hat{X}_{t+k})^2 | \mathcal{F}_t)$.

There is a general approach to identifying the minimum mean square error predictor given the information in $\mathcal{F}_t$, namely $E(X_{t+k} | \mathcal{F}_t)$, a function of $\mathcal{F}_t$, the L^2 projection of X_{t+k} onto the space of measurable functions of $\mathcal{F}_t$. This is easy to show using the properties of conditional expectation. Let $\hat{X}_{t+k}$ be any L^2 measurable

function of $\mathcal{F}_t$. We will show that $E(X_{t+k}|\mathcal{F}_t)$ is the optimal choice to minimize mean square error. In the following calculation $\hat{X}_{t+k}$ is any $\mathcal{F}_t$ -measurable predictor.

$$\begin{aligned}
E((X_{t+k} - \hat{X}_{t+k})^2) &= E(E((X_{t+k} - \hat{X}_{t+k})^2|\mathcal{F}_t)) \\
&= E(E((X_{t+k} - E(X_{t+k}|\mathcal{F}_t) + E(X_{t+k}|\mathcal{F}_t) - \hat{X}_{t+k})^2|\mathcal{F}_t)) \\
&= E(E((X_{t+k} - E(X_{t+k}|\mathcal{F}_t))^2|\mathcal{F}_t)) + E(E(E((X_{t+k}|\mathcal{F}_t) - \hat{X}_{t+k})^2|\mathcal{F}_t)) \\
&\quad + 2E(E((X_{t+k} - E(X_{t+k}|\mathcal{F}_t))(E(X_{t+k}|\mathcal{F}_t) - \hat{X}_{t+k})|\mathcal{F}_t)) \\
&= E(E((X_{t+k} - E(X_{t+k}|\mathcal{F}_t))^2|\mathcal{F}_t)) + E(E(E((X_{t+k}|\mathcal{F}_t) - \hat{X}_{t+k})^2|\mathcal{F}_t)) \\
&\quad + 2E((E(X_{t+k}|\mathcal{F}_t) - \hat{X}_{t+k})E(X_{t+k} - E(X_{t+k}|\mathcal{F}_t)|\mathcal{F}_t)) \\
&= E(E((X_{t+k} - E(X_{t+k}|\mathcal{F}_t))^2|\mathcal{F}_t)) + E(E(E((X_{t+k}|\mathcal{F}_t) - \hat{X}_{t+k})^2|\mathcal{F}_t)) + 0 \\
&= 0
\end{aligned}$$

The steps in the above series of equalities follow because $\hat{X}_{t+k}$ is $\mathcal{F}_t$ measurable as is $E(X_{t+k}|\mathcal{F}_t)$, hence the second factor in the cross-product term factors outside the inner conditional expectation. The remaining term inside the conditional expectation is 0, hence the cross-product term vanishes. It is clear that both remaining terms are non-negative for every choice of $\hat{X}_{t+k}$. The first is independent of the choice of $\hat{X}_{t+k}$; however, the second can be minimized (=0) by choosing $\hat{X}_{t+k} = E(X_{t+k}|\mathcal{F}_t)$. Thus $E(X_{t+k}|\mathcal{F}_t)$ is the minimum mean square error estimator of X_{t+k} among all $\mathcal{F}_t$ measurable predictors. The resulting minimum mean square error is $E((X_{t+k} - E(X_{t+k}|\mathcal{F}_t))^2)$. Some manipulations using the properties of conditional expectation give the minimum mean square error to be $E(\text{Var}(X_{t+k}|\mathcal{F}_t))$ which can be expressed as $\text{Var}(X_{t+k}) - \text{Var}(E(X_{t+k}|\mathcal{F}_t)) = E(\text{Var}(X_{t+k}|\mathcal{F}_t))$. This gives us the optimal forecast ($E(X_{t+k}|\mathcal{F}_t)$), the resulting minimum mean square error ($E(\text{Var}(X_{t+k}|\mathcal{F}_t))$) and the reduction in mean square error from being given the information contained in $\mathcal{F}_t$, namely $\text{Var}(E(X_{t+k}|\mathcal{F}_t))$.

Later in this section we will calculate the best point forecast for a number of weakly stationary time series models including $\text{WN}(0, \sigma^2)$, $\text{AR}(p)$, $\text{MA}(q)$, and $\text{ARMA}(p, q)$.

7.2 Linear Forecasts

While we have characterized the minimum mean square error forecast of a future value of the time series, say X_{n+k} , given $\mathcal{F}_n$ by $E(X_{n+k}|\mathcal{F}_n)$. Unfortunately, this may be difficult to compute. For this reason, we focus on linear forecasts. These are widely used and there are three important benefits associated with the use of linear forecasts:

1. When the joint distributions of the random variables is jointly normal, then the optimal forecast is a linear function and the two methods agree.
2. The linear forecasts are relatively easily computed and can be computed recursively.
3. The linear forecasts only depend upon the first and second moments of the random variables and their correlation structure, whereas, excluding the case of joint normality, the optimal forecast will depend on the details of the joint distribution.

We now consider the special case of linear forecasts. That is, we wish to forecast X_{n+k} given $\{X_1, \dots, X_n\}$, considering all forecasts of the form $a_0 + \sum_{i=1}^n a_i X_{n-i}$. Here, for notational simplicity, we assume the earliest observation we make is denoted by X_1 rather than X_0 , and the current observation is X_n rather than X_t .

Since we are using linear predictors and have information $\mathcal{F}_n$, the best linear predictor will be of the form

$$\hat{X}_{n+k} = a_0 + a_1 X_n + \dots + a_n X_1,$$

with the coefficients $\{a_i, 0 \leq i \leq n\}$ chosen to minimize $E((X_{n+k} - \hat{X}_{n+k})^2 | \mathcal{F}_n)$. The coefficients will depend on k , but for simplicity we do not doubly subscript them until we discuss iterative calculations.

The optimal choices for the coefficients $a_0, \dots, a_n$ can be found by differentiating the mean squared error with respect to each of the coefficients, setting those equations to 0 and solving simultaneously. Since the minimization is quadratic, it is easiest to interchange the order of differentiation and expectation (which can be justified in this case), i.e. solving

$$\frac{\partial}{\partial a_i} E((X_{n+k} - \hat{X}_{n+k})^2 | \mathcal{F}_n) = E\left(\frac{\partial}{\partial a_i} (X_{n+k} - \hat{X}_{n+k})^2 | \mathcal{F}_n\right) = 0,$$

for $0 \leq i \leq n$. This gives a set of linear equations, which can be solved to find the optimal coefficients.

$$\text{Let } S(\mathbf{a}) = (X_{n+k} - (a_0 - \sum_{j=1}^n a_j X_{n+1-j}))^2.$$

For a_0 we have

$$\begin{aligned} 0 = \frac{\partial}{\partial a_0} E(S(\mathbf{a})) &= E\left(\frac{\partial}{\partial a_0} (X_{n+k} - a_0 - \sum_{j=1}^n a_j X_{n+1-j})^2\right) \\ &= -2E(X_{n+k} - a_0 - a_1 X_n - \dots - a_n X_1) \\ &= -2(\mu - a_0 - \mu(a_1 + \dots + a_n)) \\ &= -2(-a_0 + \mu(1 - (a_1 + \dots + a_n))). \end{aligned}$$

This is easily solved to find

$$\hat{a}_0 = \mu(1 - \sum_{j=1}^n a_j).$$

Differentiation with respect to a_i , $1 \leq i \leq n$ gives

$$0 = \frac{\partial}{\partial a_i} = -2E((X_{n+k} - a_0 - \sum_{j=1}^n a_j X_{n+1-j})X_{n+1-i}).$$

Equivalently,

$$E(X_{n+k}X_{n+1-i} - a_0X_{n+1-i} - a_1X_nX_{n+1-i} - \dots - a_nX_1X_{n+1-i}) = 0,$$

or

$$\gamma(k+i-1) + \mu^2 - a_0\mu - a_1(\gamma(i-1) + \mu^2) - \cdots - a_n(\gamma(n-i) + \mu^2) = 0.$$

This can be rewritten as

$$\gamma(k+i-1) = a_0\mu - \mu^2 + \sum_{j=1}^n a_j\mu^2 + \sum_{j=1}^n a_j\gamma(j-i).$$

Note $a_0\mu - \mu^2 + \sum_{j=1}^n a_j\mu^2 = 0$. This results in a set of linear equations relating the a_i coefficients in terms of the $\gamma(h)$ coefficients:

$$\mathbf{\Gamma}_n \mathbf{a}_n = \boldsymbol{\gamma}_n(k),$$

where $\mathbf{\Gamma}_n = (\gamma(i-j))_{i,j=1,\dots,n}$, $\boldsymbol{\gamma}_n(k) = (\gamma(k), \dots, \gamma(k+n-1))$, and $\mathbf{a}_n = (a_1, \dots, a_n)$. These are the **prediction equations**. Here $\mathbf{\Gamma}_n$ is the same matrix that arose in the general Yule-Walker equations, and we can find the coefficients of the best predictor as

$$\hat{\mathbf{a}}_n = \mathbf{\Gamma}_n^{-1} \boldsymbol{\gamma}_n(k).$$

The best linear predictor is

$$\hat{X}_{n+k} = \mu + \hat{a}_1(X_n - \mu) + \cdots + \hat{a}_n(X_1 - \mu),$$

and its mean square prediction error is

$$E((X_{n+k} - \hat{X}_{n+k})^2) = \gamma(0) - \mathbf{a}_n^T \boldsymbol{\gamma}_n(k).$$

Thus linear prediction for a weakly stationary time series model is straightforward:

- Solve the prediction equations using the autocovariance function of the process
- Write down the expression for the best linear predictor
- Evaluate its mean squared error.

We consider some simple examples

Predicting a white noise process, $\text{WN}(\mu, \sigma_\epsilon^2)$

Here $\gamma(h) = 1$ if $h = 0$ and 0 if $h \neq 0$. Thus

$$\mathbf{\Gamma}_n = \sigma_\epsilon^2 \mathbf{I}_n,$$

and the prediction equations become

$$\mathbf{a} = \boldsymbol{\gamma}_n = \mathbf{0}$$

Thus the best linear predictor of a white noise process with mean μ at a future point in time, ϵ_{n+k} , is given by $\hat{\mu} + 0 = \hat{\mu}$. The mean square error is $E((\epsilon_{n+k} - \mu)^2) = \sigma_\epsilon^2$. There is no information in the data that can improve the minimum mean square error prediction.

Predicting an AR(1) process

We next predict an AR(1) process, beginning with predicting X_{n+1} from $X_1, \dots, X_n$, the one-step-ahead predictor. Recall for an AR(1) process,

$$\gamma(h) = \frac{\sigma_\epsilon^2 \phi_1^{|h|}}{1 - \phi_1^2}$$

We need to find $\mathbf{a}_n$ to satisfy the prediction equations: $\mathbf{\Gamma}_n \mathbf{a}_n = \boldsymbol{\gamma}_n(1)$. Both $\mathbf{\Gamma}_n$ and $\boldsymbol{\gamma}_n(1)$ have a common factor $\sigma_\epsilon^2/(1 - \phi_1^2)$ that we omit in writing the equations. The first row of $\mathbf{\Gamma}$ is $(1, \phi_1, \dots, \phi_1^{n-1})$. Multiplying this by ϕ_1 gives exactly $\boldsymbol{\gamma}_n(1)$. Consequently, if we set

$$\hat{\mathbf{a}}_n = (\phi_1, 0, \dots, 0),$$

then we have a solution to the prediction equations. Consequently, the best linear one-step-ahead predictor of X_{n+1} given $\mathcal{F}_n$ is $\hat{X}_{n+1} = \hat{\mu} + \phi_1 X_n$. The minimum mean square error is

$$\begin{aligned} E((X_{n+1} - \hat{X}_{n+1})^2) &= \gamma(0) - \hat{\mathbf{a}}_n \boldsymbol{\gamma}_n(1) \\ &= \frac{\sigma_\epsilon^2}{1 - \phi_1^2} (1 - \phi_1^2) \\ &= \sigma_\epsilon^2 \end{aligned}$$

This is a reduction from $\sigma_\epsilon^2/(1 - \phi_1^2)$. Note we assume the parameter(s) of the model, ϕ_1 in this case, are known.

Now we can predict further in time, e.g. $\hat{X}_{n+k}$ for $k > 1$. The analysis is essentially identical, as all that changes is $\boldsymbol{\gamma}_n(k) = \phi_1^k \boldsymbol{\gamma}_n(1)$. The prediction equations are easily solved, we simply multiply $\mathbf{a}_n$ found before by ϕ_1^k yielding $(\phi_1^k, 0, \dots, 0)$. The minimum mean square error for this k-step-ahead predictor is

$$\sigma_\epsilon^2 \frac{1 - \phi_1^{2k}}{1 - \phi_1^2}$$

Note how the error increases as k increases, with the limit as $k \rightarrow \infty$ being the unconditional variance $\sigma_\epsilon^2/(1 - \phi_1^2)$, an intuitive result.

Now the best linear predictor we derived for a weakly stationary time series (coming from solving the prediction equations) used the parameters of the time series as though we knew them. Obviously, in general we will not know those parameters (unless we generated the time series by simulation). We saw this problem in MSCF Simulation for Option Pricing course where we needed to use, say, a standard deviation of the data in defining the optimal control variable. In that case we used the data to estimate the standard deviation, resulting in a standard error (estimated standard deviation of an estimate).

This is the same situation. We have an optimal estimator (predictor or forecast) involving unknown parameters, so we substitute those for the unknown true values. However, the minimum mean squared error we computed does not reflect the added variability coming from the uncertainty in the estimated values, including σ_ϵ . In addition to the additional uncertainty that arises from the estimation of the model parameter estimate, we should recognize that creating a “point” forecast is only part of the task.

We need to accompany the point forecast with a measure of uncertainty. Indeed, we really need to develop a [predictive distribution](#). The predictive distribution is the probability distribution of our optimal forecast. We

generally use the mean of that distribution if we must give a specific value, but the predictive distribution allows for the creation of confidence intervals for a prediction (“prediction intervals”). The generation of a predictive distribution can be done in theory for special favorable assumptions, or it can be done by Monte Carlo methods for more general assumptions. Most statistical software packages create prediction intervals based on the normal distribution. In fact, the distribution when X_{t+k} represents the log-returns, a heavier tailed distribution is more appropriate, such as a t_4 distribution, for example.

Generally, with enough data, the model parameter estimates will be asymptotically normally distributed (e.g. MLE or moment estimators are asymptotically normal by the central limit theorem). If n is sufficiently large, the error from the parameter estimates likely dominates the error from the model and from the noise process in the model. In most cases, statistical software like statsmodels is used and will automatically compute forecasts and their limits. But for more careful work (for example, a concern for heavy tails), Monte Carlo simulation should be applied.

For example, given an AR(1) process, parameters ϕ_0, ϕ_1 and σ_ϵ^2 , a model for $\{\epsilon_t, -\infty < t < \infty\}$, and data $(X_1, \dots, X_t)$ one can simulate many paths for the process using the AR(1) model, the parameter estimates and the assumptions about the noise process distribution. This would result in a set of values for X_{n+k} , and the distribution of these values gives the predictive distribution of the process. From that distribution, one can find a point prediction (usually the mean) and develop confidence limits using that distribution.

A new aspect arises when forecasting a MA(q) time series, therefore also in forecasting an ARMA(p,q) or an ARIMA(p,d,q) time series model. To see the issue, consider a MA(1) model:

$$Y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1}.$$

If we wish to forecast one-step-ahead, $\hat{Y}_{t+1}$, then we note that $Y_{t+1} = \mu + \epsilon_{t+1} + \theta_1 \epsilon_t$ and the forecast, $\hat{Y}_{t+1}$ must be based only on $\mathcal{F}_t$.

Clearly, our best estimate of ϵ_{t+1} is 0, and we will have estimates of μ and θ_1 and plug them into the equation. But what do we do with ϵ_t ? We replace ϵ_t by the residual, $\hat{\epsilon}_t$. Thus the forecast of Y_{t+1} is

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\theta}_1 \hat{\epsilon}_t.$$

For a MA(1) model, the autocorrelation is 0 for lags greater than 1. To forecast Y_{t+2} we note that the model gives

$$Y_{t+2} = \mu + \theta_1 \epsilon_{t+1} + \epsilon_{t+2}.$$

Our best estimates of the two error terms, ϵ_{t+1} and ϵ_{t+2} is 0. This gives

$$\hat{Y}_{t+k} = \hat{\mu}, \quad k \geq 2.$$

7.3 Iterative Calculations

We have seen that forecasting future values of the time series using linear forecasts can be done by simply solving a set of linear equations involving the mean, variance and covariance function of the stationary

process governing the time series. It is common to track the evolution of the time series and update the one-day-ahead forecast as new prices for the time series become available. This would be especially true in risk management where the calculation of risk measures like VaR for a future time period, say 10 days, is often done by taking the one-day-ahead forecast and multiplying by $\sqrt{10}$. To do this, we need to repeatedly solve a different set of linear equations each day. It would be more convenient to be able to use the results up to time n and update them taking into account the new data. We will also want to update our estimates of the mean, variance and covariance function as more data is gathered. To understand the use of iterative calculations, we digress briefly and consider two simple examples.

Suppose we are given $(X_1, \dots, X_{n-1})$. Let $\bar{X}_{n-1} = \frac{\sum_{i=1}^{n-1} X_i}{n-1}$ and $s_{n-1}^2 = \frac{\sum_{i=1}^{n-1} (X_i - \bar{X}_{n-1})^2}{n-2}$. Given these definitions of the sample mean and the sample variance, the question arises whether given a new data point, X_n , can we compute $\bar{X}_n$ and s_n^2 using just X_n , $\bar{X}_{n-1}$ and s_{n-1}^2 ? The answer is yes, and the formulas are given by

- $\bar{X}_n = \frac{n-1}{n} \bar{X}_{n-1} + \frac{X_n}{n},$
- $s_n^2 = \frac{n-2}{n-1} s_{n-1}^2 + \frac{(X_n - \bar{X}_{n-1})^2}{n}.$

Thus $\bar{X}_n$ and s_n^2 can be computed using only $\bar{X}_{n-1}$ and s_{n-1}^2 and the new value X_n .

For linear prediction, we can find the optimal coefficients, $(a_0, \dots, a_n)$ for forecasting X_{n+m} by solving the prediction equations $\Gamma_n(m) \alpha_n = \gamma_n(m)$ and setting $a_0 = \mu(1 - \sum_{i=1}^n a_i)$. First, it is convenient to set $\mu = 0$. This can be done without loss of generality, because we can introduce the mean zero series, $Y_n = X_n - \mu$, forecast the $\{Y_n, n \geq 0\}$ sequence, then add $\hat{\mu}$.

While linear prediction has the advantage of minimizing the mean squared error of prediction, there is a second advantage to using linear prediction. It is possible to utilize the coefficients determined given $\mathcal{F}_n$ to determine the optimal coefficients for the problem given $\mathcal{F}_{n+1}$. That is, first we forecast given $(X_1, \dots, X_n)$, then given new information, X_{n+1} , we can forecast a future value given $(X_1, \dots, X_{n+1})$ utilizing $\mathbf{a}_n$ without having to re-solve the prediction equations. To do this, we need a different set of forecast coefficients for each level of information. Thus the forecasting coefficients will need to be doubly indexed, $\mathbf{a}_n = (a_{1n}, \dots, a_{nn})$ given $\mathcal{F}_n$. For simplicity, we consider only one-step-ahead prediction. That is given $\mathcal{F}_n$, we predict X_{n+1} or more generally X_{n+k} . Then we observe X_{n+1} and seek to predict X_{n+2} , and so forth. We wish to find the coefficients $\mathbf{a}_{n+1} = (a_{1,n+1}, \dots, a_{n+1,n+1})$ in terms of $\mathbf{a}_n$. Before presenting the recursive expressions associated with the Durbin-Levinson algorithm, it is important to note that, $\alpha_0 = 0$ (because we set $\mu = 0$) and α_n are the partial autocorrelation coefficients discussed earlier. This should be clear as those coefficients provide the best linear predictor of X_{n+1} based on $(X_1, \dots, X_n)$. Consequently, this linear predictor can be used to remove the effect of those variables on X_{n+1} . When we discussed the use of the partial correlation, it focused on a_{nn} . We tested $a_{nn} = 0$ to determine if the order of the time series was $n - 1$.

7.3.1 Durbin Levinson Iteration

The calculation of the optimal prediction coefficients, $\mathbf{a}_n$, only requires the solution of a system of linear equations, the prediction equations. However, as the size of the system grows with more data, the solution requires a matrix inversion or its equivalent, which becomes increasingly time and space consuming. Thus, an on-line recursive algorithm to update $\mathbf{a}_n$ to $\mathbf{a}_{n+1}$ is an important step toward efficient prediction.

We return to the model with the mean included:

$$\hat{X}_{n+1} = \mu + a_{1,n}(X_k - \mu) + \cdots + a_{n,n}(X_1 - \mu),$$

and we define p_n to be the mean squared prediction error,

$$p_{n+1} = E((X_{n+1} - \hat{X}_{n+1})^2).$$

The Durbin-Levinson algorithm is a recursive update algorithm. The algorithm is initialized by

$$\hat{X}_1 = \mu, p_0 = E((X_1 - \hat{X}_1)^2) = \gamma_X(0).$$

Then for $n \geq 1$,

$$a_{n,n} = \frac{1}{p_{n-1}}(\gamma_X(n) - \sum_{j=1}^{n-1} a_{j,n-1}\gamma_X(n-j)),$$

and

$$\begin{pmatrix} a_{1,n} \\ \vdots \\ a_{n-1,n} \end{pmatrix} = \begin{pmatrix} a_{1,n-1} \\ \vdots \\ a_{n-1,n-1} \end{pmatrix} - a_{n,n} \begin{pmatrix} a_{n-1,n-1} \\ \vdots \\ a_{1,n-1} \end{pmatrix}$$

and

$$p_n = p_{n-1}(1 - a_{n,n}^2).$$

The reader is referred to Brockwell and Davis (3rd edition, pp61-62) for a proof that this recursion gives the desired optimal forecasting coefficients for the one-step-ahead linear prediction problem.

7.4 AIC, BIC and Examples

To make predictions using time series techniques, one needs a weakly stationary model and a method to fit one of the time series models we have been discussing. We have discussed methods to transform a series so it is weakly stationary, for example removing trend and seasonal components, applying variance stabilization using the Box-Cox family, and differencing. Once weak stationarity has been achieved, it remains to fit a time series model from the ARMA or ARIMA family. Because adding parameters to a

model will usually increase (and never decrease) the fit of the model; however, adding parameters without obtaining a substantial gain in fit usually results in overfitting. By overfitting we mean that more parameters than necessary have been added to the model, and the model is being overly tuned to the particular data set. When a model is overfit, its predictions made with new data are generally worse than could be achieved with a less overparametrized model.

In statistics there are some goodness-of-fit criteria that penalize the goodness of fit measure for using too many parameters. The two most commonly applied measures are AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion). We quote next the several paragraphs on 109-110 of the text by Rupert and Matteson explaining these criteria.

An important practical problem is choosing between two or more statistical models that might be appropriate for a data set. The maximized value of the log-likelihood, denoted here by $\log(L(\hat{\theta}_{ML}))$, can be increased simply by adding parameters to the model. The additional parameters do not necessarily mean that the model is a better description of the data-generating mechanism, because the additional model complexity due to added parameters may simply be fitting random noise in the data, a problem that is called *overfitting*. Therefore, models should be compared both by fit to the data and by model complexity. To find a parsimonious model one needs a good tradeoff between maximizing fit and minimizing model complexity.

AIC (Akaike's information criterion) and BIC (Bayesian information criterion) are two means for achieving a good tradeoff between fit and complexity. They differ slightly and BIC seeks a somewhat simpler model than AIC. They are defined by

$$AIC = -2 \log(L(\hat{\theta}_{ML})) + 2p,$$

$$BIC = -2 \log(L(\hat{\theta}_{ML})) + \log(n)p,$$

where p equals the number of parameters in the model and n is the sample size. For both criteria "smaller is better," since small values tend to maximize $L(\hat{\theta}_{ML})$ (minimize $-\log(L(\hat{\theta}_{ML}))$) and minimize p , which measures model complexity. The terms $2p$ and $\log(n)p$ are called "complexity penalties" since they penalize larger models.

The term *deviance* is often used for minus twice the log-likelihood, so $AIC = \text{deviance} + 2p$ and $BIC = \text{deviance} + \log(n)p$. Deviance quantifies model fit, with smaller values implying better fit.

Generally, from a group of candidate models, one selects the model that minimizes whichever criterion, AIC or BIC, is being used. However, any model that is within 2 or 3 of the minimum value is a good candidate and might be selected instead, for example, because it is simpler or more convenient to use than the model achieving the absolute minimum. Since $\log(n) > 2$ provided, as is typical, that $n > 8$, BIC penalizes model complexity more than AIC, and for this reason BIC tends to select simpler models than AIC. However, it is common for both criteria to select the same, or nearly the same, model. Of course, if several candidate models all have the same value of p , then AIC, BIC, and $-2 \log(L(\hat{\theta}_{ML}))$ are minimized by the same model.

While the above is a very good explanation of using these criteria for model choice, one should also recognize that it may be that the choice of a third model, one that minimizes neither AIC nor BIC, should be

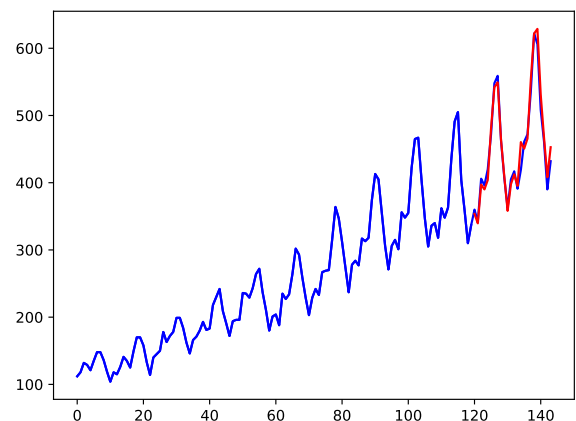
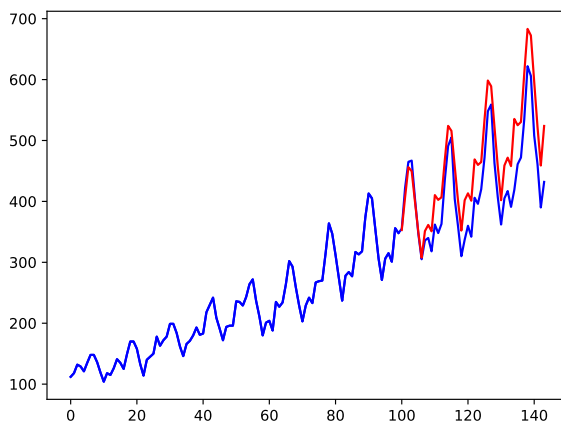
selected if, for example, it gives good predictions with possibly fewer parameters. Finally, there is a small sample correction for the AIC, namely

$$AICC = AIC + \frac{2p(p+1)}{n-p-1}.$$

Insights into the derivation of the AIC and AICC are given on page 112 of the text by Ruppert and Matteson and are not reproduced here.

7.5 Forecasting Example

We return to the international airline and use the prediction capabilities of `pmdarima` to illustrate forecasting. In this case, we divide the data set, which has 144 data points, into a training set of size 100 and 120 and then predict the final 44 and 24 values respectively. The results are given below. The data was first log-transformed, then a model was fit on the training data and predictions were made. The data and predictions were then inverse-transformed back to the original scale and the results are plotted below. As one can see the point predictions for the size 100 training set are reasonable, while the results for the 120 training set are nearly perfect.



8 ARCH/GARCH Models

8.1 Introduction

We mentioned earlier that most raw (untransformed) financial time series are not stationary. There are some standard methods to transform the raw data to make it more stationary, for example

- Removing trends and seasonality
- Differencing
- Box-Cox (power-log transformation, variance stabilization - especially when the variance increases as the mean increases)

After these transformations, the resulting time series may have non-constant volatility and volatility clusters.

Suppose we start with a time series $\{X_t, t \geq 1\}$ and define $\mathcal{F}_t = \sigma(X_s, s \leq t)$, the σ -algebra defined by the process up to time t . In the models considered thus far (e.g. AR, MA, ARMA and ARIMA), we assumed $Var(X_t|\mathcal{F}_{t-1}) = \sigma_\epsilon^2$, a constant. The constant volatility assumption is referred to as *conditional homoskedasticity*. We now consider the alternative situation in which this conditional variance is not constant, that is *conditional heteroskedasticity*.

From Rama Cont's stylized facts and a discussion of volatility on page 111 of Ruey Tsay's (2011) text, volatility is not directly observable, but some characteristics are commonly seen in stock price series:

- Volatility clustering
- Volatility evolves continuously with time, usually not with jumps,
- Volatility does not diverge to infinity, in fact it is approximately stationary,
- Volatility increases more following big price drops than it does after big price increases - also known as the leverage effect.

We represent a time series, $\{X_t\}$, as having a mean component and a volatility component:

$$X_t = \mu_t + a_t,$$

where $\mu_t = E(X_t|\mathcal{F}_{t-1})$ and $\sigma_t^2 = Var(a_t|\mathcal{F}_{t-1}) = Var(X_t|\mathcal{F}_{t-1})$. For most models we have studied so far (e.g. AR and ARMA), μ_t is non-constant and σ_t is constant. For example, consider an AR(1) model

$$X_t = \phi_0 + \phi_1 X_{t-1} + \epsilon_t.$$

$$E(X_t|\mathcal{F}_{t-1}) = E(\phi_0 + \phi_1 X_{t-1} + \epsilon_t|\mathcal{F}_{t-1}) = \phi_0 + \phi_1 X_{t-1},$$

and

$$Var(X_t|\mathcal{F}_{t-1}) = Var(\phi_0 + \phi_1 X_{t-1} + \epsilon_t|\mathcal{F}_{t-1}) = \sigma_\epsilon^2.$$

The AR(1) model is for the mean component of the X_t time series. We now consider models for the volatility component.

With conditional heteroskedastic (CH) models, the opposite of the mean models is true, namely the mean component, μ_t , is constant and the volatility component, σ_t , is non-constant.

There are two main approaches to CH models:

- Deterministic Models: $Var(X_t|\mathcal{F}_{t-1})$ is a deterministic function. ARCH/GARCH models fall into this category.
- Stochastic Models: $Var(X_t|\mathcal{F}_{t-1})$ is stochastic, conditional on $\mathcal{F}_{t-1}$. Stochastic volatility (SV) models fall into this category.

In this section on ARCH/GARCH models, we draw some text and an example from an introductory 2001 paper by Robert Engle explaining ARCH/GARCH modeling¹¹. We weave in a few quotes from this paper, and in a later section we present the major example given in this paper that addresses forecast of VaR (value-at-risk) using an ARCH/GARCH approach. The first quote (with very minor modifications) from page 158 gives some motivation for this approach and echoes some of the stylized facts of Rama Cont and the ideas discussed earlier in this section.

...sometimes the natural question facing the applied econometrician is the accuracy of the predictions of the model. In this case, the key issue is the variance of the error terms and what makes them large. This question often arises in financial applications where the dependent variable is the return on an asset or portfolio and the variance of the return represents the risk level of those returns. These are time series applications, but it is nonetheless likely that heteroskedasticity is an issue. Even a cursory look at financial data suggests that some time periods are riskier than others; that is, the expected value of the magnitude of error terms at some times is greater than at others. Moreover, these risky times are not scattered randomly across quarterly or annual data. Instead, there is a degree of autocorrelation in the riskiness of financial returns. Financial analysts, looking at plots of daily returns such as in Figure 1, notice that the amplitude of the returns varies over time and describe this as "volatility clustering." The ARCH and GARCH models, ..., are designed to deal with just this set of issues. They have become widespread tools for dealing with time series heteroskedastic models. The goal of such models is to provide a volatility measure—like a standard deviation—that can be used in financial decisions concerning risk analysis, portfolio selection and derivative pricing.

¹¹Engle, R., "GARCH 101: The use of ARCH/GARCH Models in Applied Econometrics," *Journal of Economic Perspectives*, **15**(4), Fall 2001, 157-168.

8.2 The ARCH/GARCH Models

Let $\{\epsilon_t, t \geq 0\}$ be a white noise process with variance 1, thus $E(\epsilon_t|\mathcal{F}_{t-1}) = 0$ and $Var(\epsilon_t|\mathcal{F}_{t-1}) = 1$. We often strengthen this assumption by assuming the ϵ_t process is independent with constant mean 0, constant variance 1 and $E(\epsilon_t^4) < \infty, \forall t$.

We want to model the noise process to create conditional heteroskedasticity, and then use it as the noise process for a time series model like an AR or ARMA model. The simplest model, created by Engle (1982), is the ARCH(1) process where ARCH means *AutoRegressive Conditional Heteroscedastic*.

The process $\{a_t, t \geq 0\}$ is an ARCH(1) process if

$$a_t = \epsilon_t \sigma_t = \epsilon_t \sqrt{\omega + \alpha_1 a_{t-1}^2},$$

for $\omega > 0, 0 \leq \alpha_1 < 1 \Rightarrow \omega + \alpha_1 a_{t-1}^2 > 0, \forall t$. The condition $\alpha_1 < 1$ ensures stationarity of $\{a_t, t \geq 0\}$. This process is built upon the underlying ϵ_t noise process. One can also rewrite this as:

$$a_t^2 = (\omega + \alpha_1 a_{t-1}^2) \epsilon_t^2.$$

The $\{a_t^2, t \geq 0\}$ process is very much like an AR(1) process with multiplicative noise rather than additive noise.

Let $\sigma_\epsilon = 1$. We first compute the conditional mean and variance of the a_t process under the assumption that the $\{\epsilon_t\}$ process is i.i.d. with $E(\epsilon_t) = 0, Var(\epsilon_t) = 1$ and $E(\epsilon_t^4) < \infty, \forall t$. Let $\mathcal{F}_t$ be the σ -algebra generated by $\{a_s, s \leq t\}$.

$$E(a_t|\mathcal{F}_{t-1}) = E(\epsilon_t \sqrt{\omega + \alpha_1 a_{t-1}^2}|\mathcal{F}_{t-1}) = \sqrt{\omega + \alpha_1 a_{t-1}^2} E(\epsilon_t|\mathcal{F}_{t-1}) = 0.$$

$$E(a_t|\mathcal{F}_{t-h}) = E(E(a_t|\mathcal{F}_{t-1})|\mathcal{F}_{t-h}) = E(0|\mathcal{F}_{t-h}) = 0.$$

Let $\sigma_t^2 = Var(a_t|\mathcal{F}_{t-1})$, thus:

$$\begin{aligned} \sigma_t^2 &= Var(a_t|\mathcal{F}_{t-1}) \\ &= E(a_t^2|\mathcal{F}_{t-1}) - (E(a_t|\mathcal{F}_{t-1}))^2 \\ &= E((\omega + \alpha_1 a_{t-1}^2) \epsilon_t^2|\mathcal{F}_{t-1}) - 0 \\ &= (\omega + \alpha_1 a_{t-1}^2) E(\epsilon_t^2|\mathcal{F}_{t-1}) \\ &= (\omega + \alpha_1 a_{t-1}^2) * 1 \\ &= \omega + \alpha_1 a_{t-1}^2 \end{aligned}$$

Taking expectations we get $\gamma_a(0) = E(\sigma_t^2) = E(\omega + \alpha_1 a_{t-1}^2) = \omega + \alpha_1 \gamma_a(0)$. Assuming $0 \leq \alpha_1 < 1$ and solving for $\gamma_a(0)$, we find

$$\gamma_a(0) = \frac{\omega}{1 - \alpha_1}.$$

We can find the covariance function for a_t using similar conditioning methods. For $h > 0$:

$$\begin{aligned}
\gamma_a(h) &= E(a_t a_{t-h}) \\
&= E(E(a_t a_{t-h} | \mathcal{F}_{t-h})) \\
&= E(a_{t-h} E(a_t | \mathcal{F}_{t-h})) \\
&= E(a_{t-h} * 0) \\
&= 0.
\end{aligned}$$

It follows that the ACF of the ARCH(1) process $\{a_t\}$ is $\rho_a(h) = 1$ if $h = 0$ and 0 if $h \neq 0$, a white noise process.

The conditional variance expression $\sigma_t^2 = \omega + \alpha_1 a_{t-1}^2$ gives insight into the behavior of an ARCH(1) process and its generalization to AR(p) and GARCH models. If a_{t-1}^2 takes on an unusually large value, then $\sigma_t^2 = \omega + \alpha_1 a_{t-1}^2$ will be unusually large. Recall that $a_t = \epsilon_t \sigma_t$, thus there is an increased probability that a_t will also be large, as $P(|a_t| > c) = P(|\epsilon_t \sigma_t| > c) = P(|\epsilon_t| > \frac{c}{|\sigma_t|})$ increases with increasing $|\sigma_t|$. Note that $\alpha_1 < 1$, so there is a damping effect; however, the initial large value of a_{t-1}^2 is likely to create a cluster of large values of a_t^2 . This will, of course, die out over time, the length of the cluster being directly related to the size of α_1 . The same argument holds in reverse if a_{t-1}^2 takes on an unusually small value. There will tend to be a cluster of small volatility values. In both cases there is mean reversion to the mean value with α_1 governing the speed of mean reversion.

We can combine the ARCH noise process with the AR time series model to achieve a process both of whose conditional mean and variance depend on the past. Suppose $\{a_t, t \geq 0\}$ is an ARCH(1) noise process

$$a_t = \sqrt{\omega + \alpha_1 a_{t-1}^2} \epsilon_t,$$

where $\{\epsilon_t, t \geq 0\}$ consists of i.i.d. $N(0,1)$ random variables, $\omega > 0$, $0 \leq \alpha_1 < 1$, $|\phi_1| < 1$, and define

$$X_t = \phi_0 + \phi_1(X_{t-1} - \phi_0) + a_t.$$

One can compute $\rho_X(h) = \phi_1^{|h|}$ and $\rho_{a^2} = \alpha_1^{|h|}$. This model has non-constant conditional mean and variance.

The order of the ARCH process can easily be increased from 1 to $p > 1$. Let

$$a_t = \sigma_t \epsilon_t,$$

where

$$\sigma_t = \sqrt{\omega + \sum_{i=1}^p \alpha_i a_{t-i}^2}.$$

The ARCH(p) process is uncorrelated and has constant conditional and unconditional mean. Its unconditional variance is also constant, but the conditional variance is nonconstant. The ACF of a_t^2 has the same form as that of an AR(p) process.

The original ARCH model was created by R. Engle in 1982¹². It was generalized to GARCH (Generalized AutoRegressive Conditional Heteroskedasticity) by Tim Bollerslev in 1986¹³. The GARCH model allows the values of the X process to impact the evolution of the conditional variance, σ_t . The GARCH(p,q) model is given by $\omega > 0$, $\beta_j > 0$:

$$\begin{aligned} a_t &= \sigma_t \epsilon_t, \\ \sigma_t^2 &= \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2, \end{aligned}$$

where we also assume $\{\epsilon_t, t \geq 0\}$ is independent with $E(\epsilon_t) = 0$, $Var(\epsilon_t) = 1$ and $E(\epsilon^4) < \infty$, $\forall t$.

Consider a GARCH(1,1) model with $a_t = \sigma_t \epsilon_t$, and $\{\epsilon_t, t \geq 0\}$

$$\sigma_t^2 = \omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

an ARMA(1,1) process but with a white noise error process.

$$\begin{aligned} E(a_t^2 | \mathcal{F}_{t-1}) &= E(\sigma_t^2 \epsilon_t^2 | \mathcal{F}_{t-1}) \\ &= E(\epsilon_t^2) E(\sigma_t^2 | \mathcal{F}_{t-1}) \\ &= E(\omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 | \mathcal{F}_{t-1}) \cdot 1 \\ &= \omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ &= \sigma_t^2 \end{aligned}$$

Define $\eta_t = a_t^2 - \sigma_t^2$. It follows that

$$E(\eta_t | \mathcal{F}_{t-1}) = E(a_t^2 - \sigma_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2 - \sigma_t^2 = 0,$$

consequently, the $\{\eta_t, t \geq 0\}$ process is a MDS and weak white noise, provided it has constant variance (homoscedastic). We can show that the square of this GARCH(1,1) process is actually an ARMA(1,1) process:

$$\begin{aligned} \sigma_t^2 &= \omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ &= \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ &= \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 (a_{t-1}^2 - \sigma_{t-1}^2) \\ &= \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 \eta_{t-1}, \end{aligned}$$

so

$$a_t^2 = \sigma_t^2 + \eta_t = \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 \eta_{t-1} + \eta_t.$$

Let $\nu = \omega / (1 - (\alpha_1 + \beta_1))$, so $\omega = \nu(1 - (\alpha_1 + \beta_1))$. Taking expectations across the above equation and assuming stationarity, we find $E(a_t^2) = \nu$ and

$$a_t^2 - \nu = (\alpha_1 + \beta_1)(a_{t-1}^2 - \nu) - \beta_1 \eta_{t-1} + \eta_t.$$

¹²R.F. Engle, "Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation," *Econometrica*, **50**, 1982, 987-1007.

¹³T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity, *Journal of Economics*, **31**, 1986, 307-327.

It follows that $\{a_t^2, t \geq 0\}$ is an ARMA(1,1) process with mean ν and AR(1) coefficient $\phi_1 = \alpha_1 + \beta_1$, and MA(1) coefficient $\theta_1 = -\beta_1$. Since this process is ARMA(1,1), we can immediately obtain its ACF from previous results:

$$\rho_{a^2}(h) = (\alpha_1 + \beta_1)^h \frac{\alpha_1(1 - \alpha_1\beta_1 - \beta_1^2)}{1 - 2\alpha_1\beta_1 - \beta_1^2}$$

We have written the square of the GARCH(1,1) as an ARMA(1,1) model with an MDS noise process. This can be generalized to the GARCH(p,q)

$$\begin{aligned} a_t &= \sigma_t \epsilon_t, \\ \sigma_t^2 &= \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2, \end{aligned}$$

where we assume $\{\epsilon_t\}$ are i.i.d. with mean 0 and variance 1. If we define $\alpha_i = 0$ if $i > p$ and $\beta_j = 0$ if $j > q$, then letting $\eta_t = a_t^2 - \sigma_t^2$ or $a_t^2 = \sigma_t^2 + \eta_t$, we find:

$$\begin{aligned} a_t^2 &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_j (a_t^2 - \sigma_t^2) \\ &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_j \eta_t. \end{aligned}$$

Now η_t is a MDS, because $E(a_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2$, so $E(\eta_t | \mathcal{F}_{t-1}) = E(a_t^2 | \mathcal{F}_{t-1}) - \sigma_{t-1}^2 = \sigma_{t-1}^2 - \sigma_{t-1}^2 = 0$. Consequently, $\{a_t^2\}$ is an ARMA(p,q) with AR coefficients $\phi_i = \alpha_i + \beta_i$ and MA coefficients $\theta_i = -\beta_i$. The variance of η_t is constant assuming $E(\epsilon_i^4) < \infty$. Consequently, η_t is a white noise process. Because the error process is a white noise process, all of our previous results hold.

To ensure that the $\{a_t^2\}$ process is weakly stationary, we need the roots of the polynomial

$$1 - \sum_{j=1}^{\max(p,q)} (\alpha_j + \beta_j) z^j,$$

to lie outside the unit circle. We usually assume that all of the α_i and β_j are positive, the weak stationary condition becomes

$$\sum_{i=1}^p \alpha_i + \sum_{j=1}^q \beta_j < 1.$$

If this condition holds, then

$$E(a_t^2) = \omega / (1 - \sum_{i=1}^p \alpha_i - \sum_{j=1}^q \beta_j).$$

8.3 GARCH Forecasting

Suppose we wish to forecast X_{t+k} for some $k \geq 1$ where the $\{X_t\}$ process is ARMA(p_m, q_m) + GARCH(p_v, q_v), that is the mean model is ARMA(p_m, q_m), and the volatility component is GARCH(p_v, q_v). The GARCH(p_v, q_v)

noise process is white noise, and earlier we developed ARMA forecasting assuming white noise errors. So the point forecasts of X_{t+k} will be the same as if we assumed the errors were i.i.d. Gaussian. What is different is that the GARCH noise process will impact the variability of the forecast, specifically the forecast intervals.

We showed that the square of the GARCH(1,1) model can be written as an ARMA process with MDS noise, hence $WN(0, \sigma_a^2)$. Let us review how to forecast ARMA(p,q) processes:

$$X_t = \phi_0 + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t.$$

Recall that for forecasting a future value, say $\hat{X}_{t+k}$, we use the defining model above and forecast future values of the process, $(\hat{X}_{t+1}, \hat{X}_{t+2}, \dots, \hat{X}_{t+k})$, sequentially. At each stage, we plug in the current data $(X_t, X_{t-1}, \dots, X_{t-k})$ as called for by the model, and we plug in the estimates of the unknown parameters.

This leaves the ϵ terms in the equation. For current and previous values of ϵ , we plug in $(\hat{\epsilon}_t, \hat{\epsilon}_{t-1}, \dots, \hat{\epsilon}_{t-q})$ going backward in time, and we plug in 0 going forward, i.e. $\hat{\epsilon}_{t+i} = 0$ for $i \geq 1$.

For a GARCH(1,1) volatility process with parameters $\alpha_1 > 0$ and $\beta_1 > 0$ with $\alpha_1 + \beta_1 < 1$,

$$a_t^2 = \epsilon_t^2(\omega + (\alpha_1 + \beta_1)a_{t-1}^2 - \beta_1\eta_t).$$

Let the constant unconditional variance be σ^2 . Taking expectations across the equation, we find

$$\sigma^2 = \omega / (1 - \alpha_1 - \beta_1).$$

Suppose that the current time is t and we want to forecast $\hat{\sigma}_{t+k}^2$, the k -step ahead forecast of the volatility given $\mathcal{F}_t$. Consider the one-step ahead forecast, $\hat{\sigma}_{t+1}^2$. We have

$$\sigma_{t+1}^2 = \omega + \alpha_1 a_t^2 + \beta_1 \sigma_t^2.$$

The right-side is $\mathcal{F}_t$ measurable, so the above equation provides the one-step-ahead forecast, although estimates must be substituted for the unknown parameters, $\omega, \alpha_1, \beta_1$. For multi-step forecasts, $k \geq 1$, recall that $a_t^2 = \epsilon_t \sigma_t^2$. So the volatility equation can be rewritten as

$$\sigma_{t+1}^2 = \omega + \alpha_1 \epsilon_t^2 \sigma_t^2 + \beta_1 \sigma_t^2.$$

Rearranging

$$\sigma_{t+1}^2 = \omega + (\alpha_1 + \beta_1) \sigma_t^2 + \alpha_1 \sigma_t^2 (\epsilon_t^2 - 1).$$

The forecast at time $t+2$ is $E(\hat{\sigma}_{t+2}^2 | \mathcal{F}_t)$. Changing $t+1$ to $t+2$ in the above equation, taking $E(\cdot | \mathcal{F}_t)$ and noting $E(\epsilon_{t+1}^2 - 1 | \mathcal{F}_t) = 0$, we find

$$\hat{\sigma}_{t+2}^2 = \omega + (\alpha_1 + \beta_1) \hat{\sigma}_{t+1}^2.$$

This generalizes to

$$\hat{\sigma}_{t+k}^2 = \omega + (\alpha_1 + \beta_1) \hat{\sigma}_{t+k-1}^2.$$

We let $\lambda = \alpha_1 + \beta_1$ (a *persistence parameter*) and iterate backward:

$$\begin{aligned}\hat{\sigma}_{t+k}^2 &= \omega + \lambda \hat{\sigma}_{t+k-1}^2 \\ &= \omega(1 + \lambda) + \lambda^2 \hat{\sigma}_{t+k-2}^2 \\ &= \dots \\ &= \omega(1 + \lambda + \dots + \lambda^{k-2}) + \lambda^{k-1} \hat{\sigma}_{t+1}^2.\end{aligned}$$

As $k \rightarrow \infty$ and assuming $\lambda < 1$

$$\hat{\sigma}_{t+k}^2 \rightarrow \sigma^2 = \omega / (1 - \lambda).$$

The **half-life**, T , of the volatility difference is defined by the equation $\lambda^T = \frac{1}{2}$. It follows that $T \approx -\frac{\log 2}{\log \lambda} = -\frac{.693}{\log \lambda}$

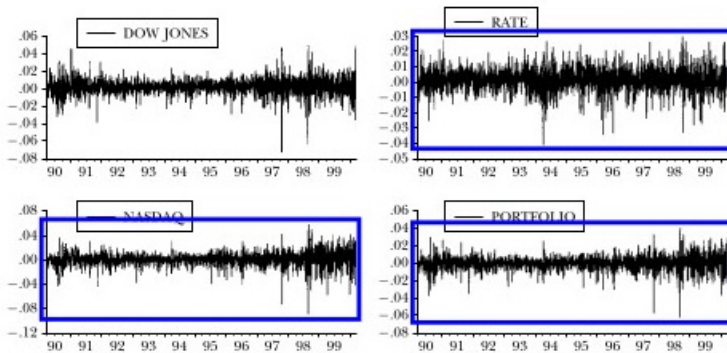
This will be discussed further in the next example,

8.4 Engle Example: Portfolio Risk Management

We now return to the Engle paper. His primary example is the use of GARCH models in the calculation of risk measures such as value-at-risk for a portfolio involving stocks and bonds. He used a GARCH(1,1) model to estimate $\text{VaR}_{.99}$ for a \$1M portfolio on March 23, 2000. The portfolio consists of 50% NASDAQ, 30% Dow Jones and 20% long bonds (10 year constant maturity treasury bond). The proportion was kept constant through rebalancing. This date was chosen to be just before a large market slide and a time of high volatility.

The following figure gives the 10-year returns for the three components (Dow Jones, NASDAQ and Rate) as well as the portfolio itself. The non-constant volatility is obvious in each of the four figures, and the areas of high volatility generally occur at common time periods across the 4 figures.

Figure 1
Nasdaq, Dow Jones and Bond Returns



For the 10-year period, the statistics of the four return processes are given by

	Dow Jones	NASDAQ	Rate	Portfolio
Mean	0.0005	0.0009	0.0001	0.0007
Std Dev	0.0090	0.0115	0.0073	0.0083
Skewness	-0.3593	-0.5310	-0.2031	-0.4738
Kurtosis	8.3288	7.4936	4.9579	7.0026

The $\text{VaR}_{.99}$ values for the portfolio can be calculated from historical data drawn from a time window. This is done by taking the values in the window being considered, sorting them in increasing order and noting the start of the 1% largest values. The value at which the ordered dataset splits into 99% below and 1% above is the historical estimate of $\text{VaR}_{.99}$ for that window. This value will vary as the size of the window being considered varies. Specifically $\text{VaR}_{.99}$ is given by

- \$22,477 using the full 10-year sample
- \$24,653 using the most recent year
- \$35,159 using 1/1/2000 to 3/23/2000

Clearly the last year was more volatile compared with the full 10-year period, while the last 3 months are much more volatile than either the full 10-year period or the final one-year period.

Changing Engle's notation slightly, beginning on page 158, he writes

Let the dependent variable be labeled r_t , which could be the return on an asset or portfolio. The mean value m_t and the variance σ_t^2 are defined relative to a the past information set ($\mathcal{F}_{t-1}$). Then, the return r_t in the present will be equal to the mean predicted for this period, and the new information in this period that is captured by the most recent squared residual. Such an updating rule is a simple description of adaptive or learning behavior and can be thought of as Bayesian updating.

Consider a trader who knows that the long-run average daily standard deviation of the S&P 500 is 1 percent, that the forecast made yesterday was 2 percent and the unexpected return observed today is 3 percent. Obviously, this is a high volatility period, and today is especially volatile, which suggests that the forecast for tomorrow could be even higher. However, the fact that the long-term average is only 1 percent might lead the forecaster to lower the forecast. The best strategy depends upon the dependence between days. If these 3 numbers are each squared and weighted equally, then the new forecast would be $\sqrt{(1 + 4 + 9)/3} = 2.16$. However, rather than weighting these equally, it is generally found for daily data that weights such as

those in the empirical example below (.02, .9, .08) are much more accurate. Hence the forecast is $\sqrt{.02 * 1 + .9 * 4 + .08 * 9} = 2.08$.

To be precise, we can use σ_t^2 to define the variance of the residuals of a regression

$$r_t = m_t + \sigma_t \epsilon_t.$$

In this definition, the variance of ϵ is one. The GARCH(1,1) model for variance looks like:

$$\sigma_{t+1}^2 = \omega + \alpha_1(r_t - m_t)^2 + \beta_1\sigma_t^2 = \omega + \alpha_1\sigma_t^2\epsilon_t^2 + \beta_1\sigma_t^2.$$

The econometrician must estimate the constants $\omega, \alpha_1, \beta_1$; updating simply requires knowing the previous forecast σ_t^2 and residual. The weights as $(1 - \alpha_1 - \beta_1, \beta_1, \alpha_1)$, and the long-run average variance is $\sqrt{\omega/(1 - \alpha_1 - \beta_1)}$. It should be noted that this only works if $\alpha_1 + \beta_1 < 1$, and it only really makes sense if the weights are positive.

...

I turn now to the question of how the econometrician can possibly estimate an equations like the GARCH(1,1) equation above when the only variable on which there are data is r_t . The simple answer is to use maximum likelihood by substituting σ_t^2 for σ^2 in the normal likelihood and then maximizing with respect to the parameters. Many software packages will facilitate this automatically.

... For any set of parameters $\omega, \alpha_1, \beta_1$ and a starting estimate for the variance of the first observation, which is often taken to be the observed variance of the residuals, it is easy to calculate the variance forecast for the second observation. The GARCH(1,1) updating formula takes the weighted average of the unconditional variance, the squared residual for the first observation and the starting variance and estimates the the variance of the second observation. This is input into the forecast of the third variance, and so forth. Eventually an entire time series of variance forecasts is constructed. Ideally, this series is large when the residuals are large and small when they are small. The likelihood function provides a systematic way to adjust the parameters $\omega, \alpha_1, \beta_1$ to give the best fit.

Of course it is entirely possible that the true variance process is different from the one specified by the econometrician. In order to detect this, a variety of diagnostic tests are available. The simplest is to construct the series of $\{\epsilon_t\}$, which are supposed to have constant mean and variance if the model is correctly specified. Various tests such as tests for autocorrelation in the squares are able to detect model failures. Often a “Ljung Box test” with 15 lagged autocorrelations is used.

We continue fitting a GARCH(1,1) model and parameters estimates, and we find p-values for them. These values were used in the standard deviation calculations in the above quotation.

- $\omega = 1.4E - 06(.0018)$
- $\alpha_1 = .0772(.0000)$
- $\phi_1 = .9046(.0000)$

The estimated coefficients are positive and sum to less than 1, but are close to 1, so there is high persistence and slow mean reversion. Engle notes that the forecast next day standard deviation is 0.0146, which is much larger than .0083, the 10 year average standard deviation. So this day is in a period of high volatility. The residuals are relatively heavy tailed, so multiply by 2.844 (instead of 2.327 for a standard normal distribution) to calculate a $\text{Var}_{.99}$ value. Here $\text{VaR}_{.99}$ is given by \$39,996, far higher than the values in any of the historical calculations.

We close this example showing two figures. Figure 2 superposes the $\text{Var}_{.99}$ curve on the actual losses over the 10-year in sample period. Figure 3 is constructed in the same way but for a one-year out-of-sample period starting in March 2000. Engle does not reestimate the parameters, but he does provide a series of 1 day rolling VaR values. Recall, that over a long period, for $\text{Var}_{.99}$ one would expect 1% of the losses to exceed the VaR value. For the 250 trading days, on average there would be 2 to 3 days at which this occurs, and that seems to be the case in Figure 3.

Figure 2

Value at Risk and Portfolio Losses In-Sample

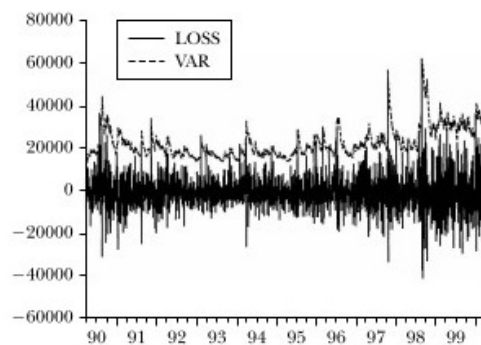
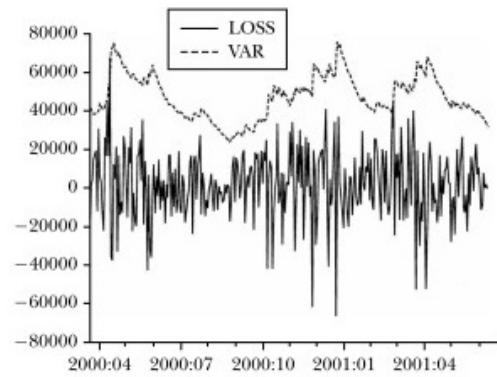


Figure 3

Value at Risk and Portfolio Losses Out of Sample



9 Multivariate Time Series

Many of the ideas we have studied for univariate time series carry over to the multivariate case. This is important, because we would like to conduct an analysis that not only deals with the individual aspects of some asset but its relationship with other assets. Unfortunately, as we go to the multivariate case, the number of parameters can skyrocket - parameters for the individual components and parameters that deal with dependency relationships. We begin with a d -dimensional time series $\mathbf{X}_t = (X_{1t}, \dots, X_{dt})$.

As each component is a time series, each will have a mean function μ_{it} and an autocovariance function, $\Gamma_{ii}(t+h, t) = \text{Cov}(X_{i(t+h)}, X_{it}) = E(X_{i(t+h)}X_{it}) - \mu_{it}^2$.

The covariance function (now a matrix) is given by

$$\mathbf{\Gamma}(t+h, t) = E((\mathbf{X}_{t+h} - \boldsymbol{\mu}_{t+h})(\mathbf{X}_t - \boldsymbol{\mu}_t)^T).$$

We can define a weakly stationary multivariate time series by requiring that the first two moments, the mean vector, $\boldsymbol{\mu}_t$ and the autocovariance matrix, $\mathbf{\Gamma}(t+h, t)$, are independent of t . If $\mathbf{\Gamma}(t, t+h)$ is independent of t , i.e. if it is given by $\mathbf{\Gamma}(h)$, then the diagonal elements of the $\mathbf{\Gamma}(h)$, $\gamma_{ii}(h)$, are the autocovariance functions of the individual component time series.

We can also create the autocorrelation matrix, $\mathbf{R}(h) = \rho_{ij}(h)$, and a typical entry can be computed by

$$\rho_{ij}(h) = \frac{\gamma_{ij}(h)}{(\gamma_{ii}(0)\gamma_{jj}(0))^{1/2}}.$$

From these definitions, we can define a vector white noise process.

Definition

$\{\epsilon_t, t \geq 0\}$ is $\text{WN}(\mathbf{0}, \boldsymbol{\Sigma})$ if and only if the ϵ_t process is weakly stationary, $\boldsymbol{\mu} = \mathbf{0}$, and

$$\mathbf{\Gamma}(h) = \begin{cases} \boldsymbol{\Sigma} & \text{if } h = 0, \\ \mathbf{0} & \text{if } h \neq 0. \end{cases}$$

Note that $\mathbf{R} = (R_{ij})$. $R_{ii}(h)$ is the autocorrelation function of X_{it} . $R_{ij}(h) = \rho(X_{it}(t), X_{jt}(t-h))$ is called the **cross-correlation function (CCF)** of X_{it} and X_{jt} . Note that $R_{ij}(h) = \rho(X_{it}, X_{j(t-h)}) \neq \rho(X_{it}, X_{jt}(t+h))$. Rather $R_{ij}(-h) = \rho(X_{it}(t), X_{jt}(t+h)) = \rho(X_{jt}(t+h), X_{it}(t)) = R_{ji}(h)$. In matrix form, $\mathbf{R}(h) = \mathbf{R}(-h)^T$. The sample estimates for the first and second moments are given next.

The usual estimate of the common mean of $\mathbf{X}$, $E(\mathbf{X}) = \boldsymbol{\mu} = (\mu_1, \dots, \mu_d)$ is the sample mean,

$$\bar{\mathbf{X}}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i.$$

Under the assumption that the multivariate time series is weakly stationary, then the sample mean, $\bar{\mathbf{X}}_n$, converges to $\boldsymbol{\mu}$ and is asymptotically normal. That is, $\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ converges weakly to a $N_d(\mathbf{0}, \boldsymbol{\Sigma}^*)$ where $\boldsymbol{\Sigma}^*$ depends upon the properties of the underlying white noise process.

We next turn to the autocovariance matrix:

$$\Gamma(h) = E((\mathbf{X}_{t+h} - \boldsymbol{\mu})(\mathbf{X}_t - \boldsymbol{\mu})^T),$$

The sample estimate is

$$\hat{\Gamma}(h) = \frac{1}{n} \sum_{i=1}^{n-h} (\mathbf{X}_{t+h} - \bar{\mathbf{X}}_n)(\mathbf{X}_t - \bar{\mathbf{X}}_n)^T, \quad 0 \leq h \leq n-1.$$

Letting $\hat{\Gamma}(h) = (\hat{\gamma}_{ij})$ and $\hat{\mathbf{R}}(h) = (\hat{\rho}_{ij})$, we have

$$\hat{\rho}_{ij}(h) = \frac{\hat{\gamma}_{ij}(h)}{(\hat{\gamma}_{ii}(0)\hat{\gamma}_{jj}(0))^{1/2}}.$$

9.1 Vector Autoregressive Processes

We want to deal with the simultaneous behavior of financial time series in pairs trading and portfolio construction. The generalization of univariate AR processes to multivariate AR processes is straightforward. These are called **Vector Autoregressive Processes** or VAR(p). Let $\mathbf{X}_t = (X_{1t}, \dots, X_{dt})$ be an d-dimensional vector time series satisfying

$$\mathbf{X}_t = \mathbf{a}_0 + \sum_{i=1}^p \mathbf{A}_i \mathbf{X}_{t-i} + \boldsymbol{\epsilon}_t,$$

where $\mathbf{a}_0$ is a d-dimensional vector, $\mathbf{A}_1, \dots, \mathbf{A}_p$ are $d \times d$ matrices and $\{\boldsymbol{\epsilon}_t\}$ are i.i.d. d-dimensional random vectors with mean $\mathbf{0}$ and some covariance matrix $\boldsymbol{\Gamma}$. For stationarity, we would want each of the individual series to be stationary as well as the vector process.

9.2 Stationarity of VAR(1) Processes

Suppose we consider the VAR(1) process

$$\mathbf{X}_t = \mathbf{a} + \mathbf{A}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t.$$

Assuming that the process is stationary and the $\boldsymbol{\epsilon}_t$ process is $\text{WN}(0, \boldsymbol{\Sigma})$ and $E(\mathbf{X}_t) = \boldsymbol{\mu}$, one may take expectations of both sides to find $\boldsymbol{\mu} = \mathbf{a} + \mathbf{A}\boldsymbol{\mu}$. Thus $((\mathbf{I} - \mathbf{A})\boldsymbol{\mu} = \mathbf{a}$. Assuming that $\mathbf{I} - \mathbf{A}$ is invertible, we find

$$\boldsymbol{\mu} = \mathbf{I} - \mathbf{A}^{-1}\mathbf{a}.$$

When the dimension $d = 1$, this reduces to the mean for the univariate AR(1) process, $\phi_0/(1 - \phi_1)$ assuming $|\phi_1| < 1$.

Again assuming stationarity, the covariance matrix of $\mathbf{X}_t$, $\boldsymbol{\Sigma}_X$, can be found by taking the covariance of both sides. Again the constant $\mathbf{a}$ does not change the covariance matrix, so it can be set to $\mathbf{0}$. Recall that the covariance matrix of $\mathbf{A}\mathbf{X}_t$ is given by $\mathbf{A}\boldsymbol{\Sigma}_X\mathbf{A}^T$. Finally, $\mathbf{A}\mathbf{X}_{t-1}$ and $\boldsymbol{\epsilon}_t$ are uncorrelated. We find

$$\boldsymbol{\Sigma}_X = \mathbf{A}\boldsymbol{\Sigma}_X\mathbf{A}^T + \boldsymbol{\Sigma}_\epsilon.$$

To simplify further, we assume that $\Sigma_{\epsilon} = \sigma^2 \mathbf{I}$ where $\mathbf{I}$ is a $d \times d$ identity matrix, so the error components are uncorrelated with constant variance.

One can iterate the above equation as we did with the mean, assuming stationarity to find

$$\Sigma_X = \sigma^2 \sum_{i=0}^{\infty} (\mathbf{A}\mathbf{A}^T)^i = \sigma^2 (\mathbf{I} - \mathbf{A}\mathbf{A}^T)^{-1}.$$

This is in analogy with the result for the univariate case in which the variance was given by $\sigma/(1 - \phi_1^2)$.

To understand the conditions on $\mathbf{A}$ for stationarity, we consider iteration backward in time, just as we did in the univariate case. For simplicity we let $\mathbf{a} = 0$.

$$\begin{aligned} \mathbf{X}_t &= \mathbf{A}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t \\ &= \mathbf{A}^2\mathbf{X}_{t-2} + \mathbf{A}\boldsymbol{\epsilon}_{t-1} + \mathbf{A}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t \\ &= \mathbf{A}^2\mathbf{X}_{t-2} + \sum_{i=0}^2 \mathbf{A}^i \boldsymbol{\epsilon}_{t-i} \\ &\dots \\ &= \mathbf{A}^{t-1}\mathbf{X}_1 + \sum_{i=0}^{t-1} \mathbf{A}^i \boldsymbol{\epsilon}_{t-i} \end{aligned}$$

We need $\mathbf{A}^t \rightarrow \mathbf{0}$. As we have seen before, we need the eigenvalues of $\mathbf{A}$ to be inside the unit circle for stationarity. Now, as we showed in the univariate discussion earlier in these notes, this can be reframed in terms of the autoregressive polynomial,

$$\det(\mathbf{I} - \mathbf{A}x) = 0,$$

having roots entirely outside the unit interval.

Note, in these notes the only multivariate time series we consider are VAR(p), vector autoregressive processes of order p . These will arise in the next section when we discuss the Johansen test for cointegration of time series. In this case, the condition for stationarity is generalized to the roots of the polynomial equation

$$\det(\mathbf{I} - \sum_{i=1}^p \mathbf{A}_i x^i) = 0$$

all lying outside the unit circle.

10 Cointegration

10.1 Definition and Examples

The paper by Engle and Granger¹⁴ lays the basic foundations for this topic. As they state in their introduction:

An individual economic variable, viewed as a time series, can wander extensively and yet some pairs of series may be expected to move so that they do not drift too far apart. Typically economic theory will propose forces which tend to keep such series together. Examples might be short and long term interest rates, capital appropriations and expenditures, household income and expenditures, and prices of the same commodity in different markets or close substitutes in the same market.

For statistical arbitrage trading strategies, the variables in question are often stock prices, although volatility arbitrage, which involves options, or capital structure arbitrage, which involves debt instruments offer additional examples. These price processes cannot be expected to be stationary but will vary, potentially widely. The crucial idea is to find a subset of these processes that “move together” in the sense that some linear combination is a stationary process. If this is the case, then over the long run the processes are “tied together” (i.e. mean reverting), and make good candidates for a pairs trading strategy.

In their paper (p253), Engle and Granger introduce the following definition of cointegration.

Definition:

“The components of the vector X_t are said to be *co-integrated of order d, b* , denoted $X_t \sim CI(d, b)$, if (i) all components of X_t are $I(d)$; (ii) there exists a vector $\alpha \neq 0$ so that $Z_t = \alpha' X_t \sim I(d - b), b > 0$. The vector α is called the *co-integrating vector*.”

Note that arbitrary constants and scale factors can be added to the individual time series without altering the cointegration relationship, since linear transformations won't alter the weakly stationary relationship. We next show that the concepts of cointegration and correlation are different. This is illustrated by examples in which 1) two series are cointegrated but not necessarily highly correlated and 2) two series are highly correlated but not cointegrated.

The first example is from Alexander, **Market Models**, p351¹⁵. Consider three time series, X_t, Y_t, W_t and three independent sequences of i.i.d. random variables, $\{\epsilon_{X,t}\}, \{\epsilon_{Y,t}\}$, and $\{\epsilon_t\}$. Define

$$W_t = W_{t-1} + \epsilon_t,$$

$$X_t = W_t + \epsilon_{X,t},$$

¹⁴Engle, R.F. and Granger, C.W.J., “Co-Integration and Error Correction: Representation, Estimation, and Testing.” *Econometrica*, **55**, 1987, 251-276.

¹⁵Alexander, C., *Market Models*, John Wiley, 2001.

$$Y_t = W_t + \epsilon_{Y,t},$$

The X_t and Y_t are simply independently noisy versions of W_t , and W_t is a random walk. Note that $\nabla X_t = W_t + \epsilon_{X,t} - W_{t-1} - \epsilon_{X,t-1} = \epsilon_{X,t} - \epsilon_{X,t-1} + \epsilon_t$ which is stationary (a linear combination of stationary processes). It follows that X_t is $\mathcal{I}(1)$. Similarly, $\nabla Y_t = W_t + \epsilon_{Y,t} - W_{t-1} - \epsilon_{Y,t-1} = \epsilon_{Y,t} - \epsilon_{Y,t-1} + \epsilon_t$ which is stationary. It follows that Y_t is $\mathcal{I}(1)$. X_t and Y_t are cointegrated since both are $\mathcal{I}(1)$ and $X_t - Y_t = \epsilon_{X,t} - \epsilon_{Y,t}$, which is a stationary noise process with cointegrating vector (1,-1).

Even though these two processes are cointegrated, they may have very low correlation when $\text{Var}(\epsilon_{X,t}) = \sigma_X^2$ and $\text{Var}(\epsilon_{Y,t}) = \sigma_Y^2$ are large compared with $\text{Var}(\epsilon_t) = \sigma^2$, i.e. when the random walk has small variance, but the measurements of the location of the walk are independent and very noisy.

$$\text{Var}(X_t) = \text{Var}(W_t + \epsilon_{X,t}) = t\sigma^2 + \sigma_X^2,$$

$$\text{Var}(Y_t) = \text{Var}(W_t + \epsilon_{Y,t}) = t\sigma^2 + \sigma_Y^2.$$

$$\begin{aligned} \text{Cov}(X_t, Y_t) &= \text{Cov}(W_t + \epsilon_{X,t}, W_t + \epsilon_{Y,t}) \\ &= \text{Cov}(W_t, W_t) \\ &= t\sigma^2 \end{aligned}$$

The correlation between X_t and Y_t is given by

$$t\sigma^2 / \sqrt{(t\sigma^2 + \sigma_X^2)(t\sigma^2 + \sigma_Y^2)},$$

which gives

$$1 / \sqrt{(1 + \frac{\sigma_X^2}{t\sigma^2})(1 + \frac{\sigma_Y^2}{t\sigma^2})},$$

If one or both of σ_X^2 and σ_Y^2 is large compared with $t\sigma^2$, then the denominator can be large compared with the numerator (1). Thus, even though the X_t and Y_t are cointegrated, their correlation can be small. Nevertheless, as t increases, this ratio will decrease and the correlation between X_t and Y_t will increase to 1. We can, however, modify this example

$$W_t = W_0 + \epsilon_t,$$

$$X_t = W_t + \epsilon_{X,t},$$

$$Y_t = W_t + \epsilon_{Y,t},$$

Here, rather than performing a random walk, W_t randomly moves about a fixed point W_0 .

With this modification, the quantity $t\sigma^2$ becomes σ^2 , and if either or both of σ_X and σ_Y is large compared with σ^2 , the correlation between X_t and Y_t will be near 0. Thus cointegration does not imply high correlation.

The reverse is also true. Series that are highly correlated are not necessarily cointegrated. Consider three time series, $\{X_t\}$, $\{Y_t\}$ and $\{Z_t\}$ with

$$X_t = X_{t-1} + \epsilon_{X,t},$$

$$Y_t = Y_{t-1} + \epsilon_{Y,t},$$

$$Z_t = X_t + Y_t,$$

where $\{\epsilon_{X,t}\}$ and $\{\epsilon_{Y,t}\}$ are i.i.d. mean 0 processes independent of each other. $\{X_t\}$ and $\{Y_t\}$ are independent random walks whose innovations have variance σ_X^2 and σ_Y^2 , while $\{Z_t\}$ is the sum of these two random walks.

X_t and Z_t are not cointegrated. Although each is $\mathcal{I}(1)$, any linear combination of the two will still involve both X_t and Y_t , neither of which is stationary. $Cov(X_t, Z_t) = Cov(X_t, X_t + Y_t) = Var(X_t) = t\sigma_X^2$, $Var(X_t) = t\sigma_X^2$ and $Var(Z_t) = t\sigma_X^2 + t\sigma_Y^2$. Hence $\rho_{X_t, Z_t} = 1/\sqrt{1 + (\sigma_Y^2/\sigma_X^2)}$. If σ_Y/σ_X is small, then the correlation will be large. Hence this is an example of two series that are not cointegrated but can have very high correlation.

When two time series, $X_{1,t}$ and $X_{2,t}$, are cointegrated, there is a unique cointegration vector of the form $(1, \alpha)$. The uniqueness follows, since if there were two such vectors, $(1, \alpha_1)$ and $(1, \alpha_2)$ then $(X_1 + \alpha_1 X_2) - (X_1 + \alpha_2 X_2) = (\alpha_1 - \alpha_2)X_2$ would be stationary in contradiction to the assumption that X_2 was not stationary. This uniqueness no longer holds when $m > 2$. There may be as many as $m - 1$ linearly independent cointegration vectors. Those vectors will span the cointegration space; however, the choice of vectors representing the cointegration space is not unique.

10.2 Tests for Cointegration

There are three important approaches to testing for cointegration, one proposed by Engle and Granger (1987) cited earlier, another proposed by Johansen (1988, 1991)^{16,17}, and a third by Phillips and Ouliaris (1990)¹⁸.

There are two parts to the definition of cointegration. First, the individual components must be $I(d)$. Second, there must be a linear combination of those components that is $I(d - b)$ for some integer $b > 0$. Testing of the first requirement that each component is $I(d)$ was considered earlier in these notes, say using a Dickey-Fuller test. Thus we turn to the second issue, finding a linear combination of variables that is $I(d - b)$.

The Engle and Granger method has two steps. Since cointegration refers to having a linear combination of the components that is stationary, the Engle and Granger approach uses linear regression to find the best fitting linear relationship. One of the m component variables is selected as the dependent variable, say X_j , and the model

$$X_{t,j} = \beta_0 + \sum_{i \neq j} \beta_i X_{t,i} + \epsilon_t,$$

is fit using ordinary least squares. One would then test whether the residuals were stationary using a unit root test like the Dickey-Fuller test. If so, then we would have a cointegration vector estimated using

¹⁶Johansen, S., "Statistical analysis of cointegration vectors," *Journal of Economic Dynamics and Control*, **12**, 1988, 231-254.

¹⁷Johansen, S., "Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models," *Econometrica*, **58**(6), 1991, 1551-1580.

¹⁸Phillips, P.C.B. and S. Ouliaris, S. "Asymptotic properties of residual based tests for cointegration," *Econometrica*, **58**(3), 1990, 165-193.

least squares. This process could be repeated with other variables taken to be the dependent variable. The difficulty that arises is that the results from the different regressions can be inconsistent. Consider, for example, the case of two variables, $m = 2$. Assuming that both $X_{1,t}$ and $X_{2,t}$ are $I(1)$, one could use the regression

$$X_{2,t} = \alpha_2 + \beta_1 X_{1,t} + \epsilon_t.$$

or the regression

$$X_{1,t} = \alpha_1 + \beta_2 X_{2,t} + \epsilon_t,$$

One can fit either regression model. Before doing so, to simplify the situation we first standardize the X_1 and X_2 data so that $\sum_{i=1}^T X_{1,t} = \sum_{i=1}^T X_{2,t} = 0$ and $\sum_{i=1}^T X_{1,t}^2 = \sum_{i=1}^T X_{2,t}^2 = 1$. In this case, $\hat{\beta}_1 = \hat{\beta}_2 = \sum_{i=1}^T X_{1,t} X_{2,t}$. Furthermore, the residual sum of squares is given by $1 - \hat{\beta}_1^2 = 1 - \hat{\beta}_2^2$ for the two regressions. Consequently, the R^2 value will be identical for the two regressions, and both regressions will give identical statistical inferences concerning whether cointegration is present or not. Nevertheless, the two different approaches would give $(\hat{\beta}_1, 1)$ and $(1, \hat{\beta}_2)$ respectively as cointegrating vectors. This would lead to the formation of two different pairs in a pairs trading strategy, only one of which would be market neutral except in the relatively rare case in which $\hat{\beta}_1 = \hat{\beta}_2 = 1$. Consequently, the reader should exercise some caution when reading the following quote from Alexander on page 355 as it might pertain to pairs trading: “When $n = 2$ it does not matter which variable is taken as the dependent variable. There is only one cointegration vector, which is the same when estimated by a regression of x on y as when estimated by a regression of y on x .” Alexander goes on to say “But when there are more than two $I(1)$ series the Engle-Granger method can suffer from a serious bias. That is, different estimates of a cointegration vector are obtained depending on the choice of dependent variable, and only one estimate is possible even though there can be up to $n - 1$ cointegration vectors.”

In general, the regression analysis approach will identify linear combinations that minimize least squares deviations; however, minimization of variance is not, in general, associated with finding stationarity in the residual process. For this reason, the approach of Johansen, which is designed to find the linear combination of variables that gives stationarity has gained favor. The Johansen methodology will be discussed next.

10.3 The Johansen Methodology

We present the Johansen methodology taken from the 1988 paper by Johansen cited earlier. We begin with a vector autoregressive process of order 1, i.e. a VAR(1) process given by,

$$\mathbf{X}_t = \mathbf{a} + \mathbf{A}\mathbf{X}_{t-1} + \epsilon_t.$$

Now, subtracting $\mathbf{X}_{t-1}$ from both sides, we find

$$\nabla \mathbf{X}_t = \mathbf{a} + \mathbf{A}\mathbf{X}_{t-1} - \mathbf{X}_{t-1} + \epsilon_t = \mathbf{a} + (\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1} + \epsilon_t.$$

If each component of $\mathbf{X}$ is $\mathcal{I}(1)$, then the left side of the above equation, $\nabla \mathbf{X}$, is stationary. Thus the right side of the above equation must also be stationary. But the term $(\mathbf{A} - \mathbf{I})\mathbf{X}_t$ is stationary only if the rows of the matrix $(\mathbf{A} - \mathbf{I})$ are cointegration vectors. That is, the term $(\mathbf{A} - \mathbf{I})\mathbf{X}_t$ must consist of linear combinations

of $\mathbf{X}_t$ that are stationary. If $(\mathbf{A} - \mathbf{I})$ has rank 0 (and is the $\mathbf{0}$ matrix) there are no cointegrating vectors to be found. If $(\mathbf{A} - \mathbf{I})$ has rank R (for some $1 \leq R \leq n - 1$), then there are R linearly independent combinations of the components of $\mathbf{X}$ that are stationary. That is, there are R linearly independent cointegration vectors. This formulation leads us to the question of determining the rank of the matrix $(\mathbf{A} - \mathbf{I})$, or put equivalently, determining the number of eigenvalues that are 0. Note, $R < n$, because if the rank of $R = n$, then $\mathbf{A}$ and $\mathbf{A} - \mathbf{I}$ would be invertible. Thus, if $(\mathbf{A} - \mathbf{I})\mathbf{X}_t$ were stationary, then $(\mathbf{A} - \mathbf{I})^{-1}(\mathbf{A} - \mathbf{I})\mathbf{X}_t = \mathbf{X}_t$ would also be stationary, in contradiction that $\mathbf{X}_t$ is $\mathcal{I}(1)$. Thus, the first part of the Johansen methodology involves making a statistical inference about the rank of the $(\mathbf{A} - \mathbf{I})$ matrix.

The idea described for the VAR(1) process easily carries over to the VAR(p) case. We begin with the model:

$$\mathbf{X}_t = \mathbf{a} + \sum_{i=1}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t.$$

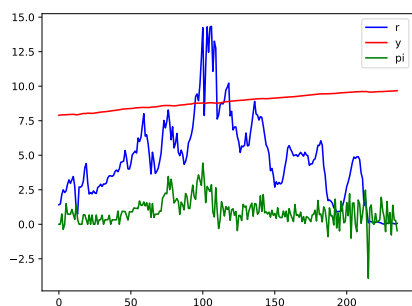
In this model, p is the order of the vector autoregression, n is the number of components of $\mathbf{X}$, and T is the data sample size. Subtracting $\mathbf{X}_{t-1}$ from both sides, then sequentially subtracting and adding $(\sum_{j=1}^k \mathbf{A}_j - \mathbf{I})\mathbf{X}_{t-k}$, $k = 2, \dots, p - 1$, from the right side, we find:

$$\begin{aligned} \nabla \mathbf{X}_t &= \mathbf{a} + (\mathbf{A}_1 - \mathbf{I})\nabla \mathbf{X}_{t-1} + (\mathbf{A}_1 - \mathbf{I})\mathbf{X}_{t-2} + \sum_{i=2}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t \\ &= \mathbf{a} + \mathbf{B}_1 \nabla \mathbf{X}_{t-1} + (\mathbf{A}_1 - \mathbf{I})\mathbf{X}_{t-2} + \sum_{i=2}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t \\ &= \mathbf{a} + \sum_{j=1}^2 \mathbf{B}_j \nabla \mathbf{X}_{t-j} + (\mathbf{A}_1 + \mathbf{A}_2 - \mathbf{I})\mathbf{X}_{t-2} + \sum_{i=3}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t \\ &\dots \\ &= \mathbf{a} + \sum_{j=1}^{p-1} \mathbf{B}_j \nabla \mathbf{X}_{t-j} + \mathbf{B}_p \mathbf{X}_{t-p} + \epsilon_t, \end{aligned}$$

where $\mathbf{B}_i = (\mathbf{A}_1 + \dots + \mathbf{A}_i - \mathbf{I})$. We again want the left side to be stationary, but the term $\mathbf{B}_p \mathbf{X}_{t-p}$ must be consistent with stationarity. If the rank of $\mathbf{B}_p$ is R , $1 \leq R \leq n - 1$, then this implies there must be R linearly independent combinations of the components that are stationary, i.e. the $\mathbf{X}$ must be cointegrated with R linearly independent cointegration vectors.

The Johansen procedure estimates the eigenvalues of the matrix $\mathbf{B}_p$ (ordered by size, call them $|\hat{\lambda}_{(1)}| \leq |\hat{\lambda}_{(2)}|, \dots, \leq |\hat{\lambda}_{(n)}|$), and then makes a series of one-sided hypothesis tests: $H_0 : R = 0$ vs. $H_1 : R > 0$. If this is rejected, we next test $H_0 : R = 1$ vs $H_1 : R > 1$. If that is rejected, we next test $H_0 : R = 2$ vs $H_1 : R > 2$, and so forth. Determining the number of non-zero eigenvalues identifies the rank of the matrix, R . The test statistic that is used is $-T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_{(i)})$. The null-hypothesis (of R or fewer cointegration relations) is rejected if the test statistic is large, indicating that at least one of the larger eigenvalues is not very near 0. Note that this series of hypothesis tests are done individually, so this procedure does not take into account simultaneous hypothesis testing.

There is a second approach used in multivariate statistical analysis to determine the rank of a matrix based on the trace of the matrix. The trace is the sum of the eigenvalues, and testing whether the trace is 0 leads



to determine if the rank is 0. If a test shows that the trace is not 0, then the largest eigenvalue is eliminated and a second test is conducted if the sum of the other eigenvalues is 0. this is continued until the number of non-zero eigenvalues is determined.

Once the number of cointegration relations has been established, the Johansen methodology goes on to estimate the specific cointegration vectors. Of course, almost all will rely on an appropriate statistical package to automate the calculation, for example the arch python package. It is, however, appropriate to caution that Johansen's work and the common statistical package implementations of it will assume that the errors in the statistical model have a multivariate normal distribution. Thus the user should be a bit cautious as many financial time series are known to have tails that are heavier than the normal distribution would imply.

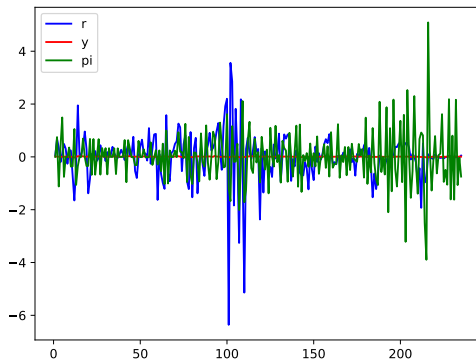
10.4 Examples of Cointegration Testing

The data set contains three time series, r (the 91-day T-bill rate), y (the log of real GDP), and pi (the inflation rate). The goal is to perform an analysis to determine if any of these series are cointegrated.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from arch.unitroot import engle_granger
from arch.unitroot.cointegration import phillips_ouliaris
from statsmodels.tsa.vector_ar.vecm import coint_johansen

data = pd.read.csv('TbGdpPi.csv')
```



```
plt.plot(data['r'], 'b-', label = 'r')
plt.plot(data['y'], 'r-', label = 'y')
plt.plot(data['pi'], 'g-', label = 'pi')
plt.legend(loc = 'best')
plt.show()
```

- `sm.tsa.stattools.adfuller(data['y'])`
 (-1.45, 0.56, 2, 233, '1%': -3.46, '5%': -2.87, '10%': -2.57,)
`sm.tsa.stattools.adfuller(data['r'])`
 (-1.58, 0.49, 12, 223, '1%': -3.46, '5%': -2.87, '10%': -2.57,)
`sm.tsa.stattools.adfuller(data['pi'])`
 (-1.78, 0.39, 15, 220, '1%': -3.46, '5%': -2.87, '10%': -2.57,)

- All are nonstationary and H_0 (unit root) is clearly not rejected.

- `diffdata = data.diff()[1:]`

```
plt.plot(data_diff['r'], 'b-', label = 'r')
plt.plot(data_diff['y'], 'r-', label = 'y')
plt.plot(data_diff['pi'], 'g-', label = 'pi')
plt.legend(loc = 'best')
plt.show()
```

```
sm.tsa.stattools.adfuller(diffdata['y'])
(-7.87, 4.89e-12, 1, 233, '1%': -3.46, '5%': -2.87, '10%': -2.57,)
sm.tsa.stattools.adfuller(diffdata['r'])
(-5.35, 4.25e-06, 11, 223, '1%': -3.46, '5%': -2.87, '10%': -2.57)
sm.tsa.stattools.adfuller(diffdata['pi'])
(-5.98, 1.87e-07, 14, 220, '1%': -3.46, '5%': -2.87, '10%': -2.57)
All three  $H_0$  are rejected, so all are  $I(1)$ 
```

```
eg_test = engle_granger(data['r'], data['pi'], trend = 'n')
eg_test
```

```
Engle-Granger Cointegration Test (r, pi)
Statistic: -2.377
P-value: 0.119
Null: No Cointegration, Alternative: Cointegration
ADF Lag length: 3
Trend: c
Estimated Root: 0.856
eg_testry = engle_granger(data['r'], data['y'])
```

```
eg_testry
Engle-Granger Cointegration Test (r,y)
Statistic: -2.29
P-value: 0.378
Null: No Cointegration, Alternative: Cointegration
ADF Lag length: 3
Trend: c
```

```
po_zt_testrpi = phillips_ouliaris(data['r'], data['pi'], trend = 'n', test_type = 'Zt')
po_zt_testrpi.summary()
```

```
Phillips-Ouliaris zt Cointegration Test
=====
Test Statistic      -11.755
P-value              0.000
Kernel              Bartlett
Bandwidth            12.323
=====
Trend: Constant
Critical Values: -2.48 (10%), -2.79 (5%), -3.39 (1%)
Null Hypothesis: No Cointegration
Alternative Hypothesis: Cointegration
```

Distribution Order: 2

```
po_zt.testry = phillips_ouliaris(data['r'],data['y'],trend = 'n',test_type = 'Zt')
```

```
po_zt.testry.summary()
```

Phillips-Ouliaris zt Cointegration Test

```
=====
Test Statistic      -2.051
P-value              40.217
Kernel              Bartlett
Bandwidth            4.313
=====
```

Trend: Constant

Critical Values: -2.48 (10%), -2.79 (5%), -3.39 (1%)

Null Hypothesis: No Cointegration

Alternative Hypothesis: Cointegration

Distribution Order: 2

```
po_zt.testpiy = phillips_ouliaris(data['pi'],data['y'], trend = 'n',test_type = 'zt')
```

```
po_zt.testpiy.summary()
```

Phillips-Ouliaris zt Cointegration Test

```
=====
Test Statistic      -11.880
P-value              0.000
Kernel              Bartlett
Bandwidth            11.959
=====
```

Trend: Constant

Critical Values: -2.48 (10%), -2.79 (5%), -3.39 (1%)

Null Hypothesis: No Cointegration

Alternative Hypothesis: Cointegration

Distribution Order: 2

We next turn to the Johansen methodology. Recall that there are two different statistics that can be used to assess the rank of the matrix based on the number of non-zero eigenvalues in the matrix $(\mathbf{A}_1 - \mathbf{I})$ or $\mathbf{B}_p$, the trace statistics and the maximum eigenvalue statistics. The first set of three numbers represent the values of the trace statistic for the three hypotheses: $H_0 = 0$ vs $H_A > 0$, $H_0 = 1$ vs $H_A > 1$, and $H_0 = 2$ vs $H_A > 2$. Note that since all components of the time series are $\mathcal{I}(1)$, we will accept the final hypothesis. The .90, .95 and .99 critical values for of these tests is given in the table below. Similarly, the maximum eigenvalue statistics for the three hypothesis are given as are the .90, .95 and .99 critical values.

```
johansen_rst=coint_johansen(data,det_order=0, k_ar_diff=0)
```

```
johansen_rst.lr1    # trace statistic
```

```
([131.27584764, 10.78357386, 4.18281327])
```

```
johansen_rst.lr2    #max eigenvalue statistic
```

```
([120.49227378, 6.60076059, 4.18281327])
```

```
johansen_rst.Results()
johansen_rst.cvt, critical values (.90,.95,.99) trace
((27.0669, 29.7961, 35.4628),
(13.4294, 15.4943, 19.9349),
( 2.7055, 3.8415, 6.6349))
johansen_rst.cvm critical values (.90.95,.99) max eigenvalue
((18.8928, 21.1314, 25.865 ),
(12.2971, 14.2639, 18.52 ),
( 2.7055, 3.8415, 6.6349))
```

One can see that the hypothesis test of no cointegrating vector (rank = 0) is clearly rejected by both tests (131.28 > 35.46 for the trace statistics and 123.49 versus 25.865 for the maximum eigenvalue test). However, when we move to the second test ($H_0 : R = 1$), we find that the null hypothesis is accepted ($10.78 < 13.43$ for the trace statistic and $6.60 < 12.30$ for the maximum eigenvalue test). Thus there is a single cointegrating relationship,

```
trace_rst = 'Reject': list(johansen_rst.lr1 > johansen_rst.cvt[:,1])
trace_rst = pd.DataFrame(trace_rst)
trace_rst.index = ['r', 'y', 'pi']
trace_rst
```

```
max_eigv_rst
= 'Reject': list(johansen_rst.lr2 > johansen_rst.cvm[:,1])
max_eigv_rst = pd.DataFrame(max_eigv_rst)
max_eigv_rst.index = ['r', 'y', 'pi']
max_eigv_rst
```


11 State space Modeling

11.1 State Space Model Formulation

State space modeling has a long history and is connected to important technology problems (e.g. Hidden Markov Models, which have been central to problems such as speech and image recognition, image reconstruction, and autonomous mobility). The model is very flexible in its ability to apply to many of the time series models being discussed in this Financial Times Series course.

The idea is that there is a stochastic process that describes the true state of a system (the *state process*), $\{\mathbf{X}_t, t \geq 1\}$. There is a second time series that represents the observation of this state process (the *observation process*), $\{\mathbf{Y}_t, t \geq 1\}$. The observation process can be thought of as a noisy version of the true state process. The models can be univariate or multivariate, and are parameterized as follows in the multivariate context:

$$\mathbf{X}_{t+1} = \mathbf{F}_t \mathbf{X}_t + \mathbf{V}_t,$$

$$\mathbf{Y}_t = \mathbf{G}_t \mathbf{X}_t + \mathbf{W}_t,$$

The state process is *unobservable*. The above model indicates that the state process progresses according to a linear model and is impacted by a noise process. The $\mathbf{Y}_t$ process is *observable*. It, too, behaves as a linear process governed by the current unobservable state, $\mathbf{X}_t$, and is impacted by a separate noise term. The two noise processes, $\mathbf{V}_t$ and $\mathbf{W}_t$ are assumed to be $\text{WN}(\mathbf{0}, \Sigma_{X,t})$ and $\text{WN}(\mathbf{0}, \Sigma_{Y,t})$ respectively, and the two noise processes are uncorrelated, i.e. $E(\mathbf{V}_s \mathbf{W}_t) = \mathbf{0}, \forall s, t$. In much of what follows, we assume that the matrices $\mathbf{F}_t$ and $\mathbf{G}_t$ are known. In reality, they will need to be estimated from data, and this can really only be done when $\mathbf{F}_t = \mathbf{F}$ and $\mathbf{G}_t = \mathbf{G}$.

Now the general state space model is easiest to understand in the context of target tracking. A target is moving according to a linear model subject to noise, and its trajectory is the state process giving the true location of the target at time t . The location of the target is measured by the observation process, which is a linear function of the state plus noise. We use the observation process to make inferences about the state. While the step-by-step transitions are described as a linear function, the coefficients can change at each time point, which does allow for non-linear behavior.

Now, this model is very flexible and can be applied in many contexts including the classical time series models discussed in this course. Consider, for example, a simple AR(1) process:

$$\begin{aligned} X_t &= \phi_1 X_{t-1} + \epsilon_t, \\ X_{t-1} &= X_{t-1}. \end{aligned}$$

Define $\mathbf{X}_t = (X_t, X_{t-1})^T$, $Y_t = X_t$, $G_t = G = 1$, $\mathbf{V}_t = (\epsilon_t, 0)^T$, $W_t = 0$, and

$$F_t = F = \begin{pmatrix} \phi_1 & 0 \\ 1 & 0 \end{pmatrix}.$$

It follows that

$$\begin{aligned}\mathbf{X}_t &= \mathbf{F}\mathbf{X}_{t-1} + \mathbf{V}_t \\ Y_t &= X_t\end{aligned}$$

Note that X_t is said to be unobservable, but $Y_t = X_t$, so in this framework X_t is actually observable, because it is the same as the observed Y_t . Note that this example can be easily extended to an AR(p) process.

A second familiar example is the standard linear regression model. Here we need to change the notation from the ordinary regression notation. Assume that we have data arriving in time, $\{(Z_{1t}, Z_{2t}, \dots, Z_{kt}, Y_t), t \geq 1\}$. Here, $\mathbf{Z}_t = (Z_{1t}, \dots, Z_{kt})$ represents the independent (explanatory) variables, while Y_t is the dependent variable. We let $\mathbf{X}_t = (\beta_1, \dots, \beta_k)$ be the unknown parameter set, and $\mathbf{X}_t = \mathbf{X}_{t-1} = \dots = \mathbf{X}_1$, so the parameters do not change over time. This can be easily put into state space form:

$$\mathbf{X}_t = \mathbf{1}\mathbf{X}_{t-1} + \mathbf{0},$$

and

$$Y_t = \mathbf{G}_t\mathbf{X}_t + \epsilon_t.$$

where ϵ_t are i.i.d. with $E(\epsilon_t) = 0$ and $Var(\epsilon_t) = \sigma^2$. Clearly $\mathbf{F}_t = \mathbf{1}$, and

$$\mathbf{G}_t = (Z_{ij}), 1 \leq i \leq t, 1 \leq j \leq k.$$

Here, the Z_{ij} are observed (and in some cases are actually chosen), the dependent variable Y_t is observed and is a linear function of the unknown, unobserved parameters $\mathbf{X}_t$ plus noise.

11.2 Kalman Recursions

State space modeling is flexible and covers many important statistical models, but what is the benefit of writing models in this form? One of the great benefits is that state space models allow for algorithms, the *Kalman recursions*, to solve three important statistical problems: prediction, filtering and smoothing, and in the first two cases does so recursively. For example, with respect to prediction, the current predictions and the new data together are sufficient for computing the new predictions.

There are three distinct types of problems that arise in this context, each solved by Kalman Recursions: prediction/forecasting, filtering, and smoothing. Suppose there is an unknown, latent state vector process $\mathbf{X}_t$. We have noisy observations of this state vector, $\mathbf{Y}_t$, and some initial state $\mathbf{Y}_1$. We want to use the observations, $\mathbf{Y}_t$, to make some inference about $\mathbf{X}_t$. For notational purposes, it is convenient to define $\mathcal{F}_s$ as the σ -algebra generated by $(\mathbf{Y}_1, \dots, \mathbf{Y}_s)$

- **Prediction or Forecasting**

Estimate $\mathbf{X}_{t+1}$ or $\mathbf{X}_{t+k}$, $k \geq 1$ using $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_t$. We denote this estimate by $\hat{\mathbf{X}}_{(t+k)|t}$ indicating the forecast of X_{t+k} given $\mathcal{F}_t$.

- **Filtering**

Estimate $\mathbf{X}_t$ using $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_t$. We denote this by $\hat{\mathbf{X}}_{t|t}$ indicating the estimate of the current state $\mathbf{X}_t$ given $\mathcal{F}_t$.

- **Smoothing**

Estimate $\mathbf{X}_t$ using $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_T$, where $1 \leq t \leq T$. This is denoted by $\hat{\mathbf{X}}_{t|T}$ indicating the estimate of $\mathbf{X}_t$ conditional on $\mathcal{F}_T$.

Consider Kalman filtering. Suppose we have the information $\mathcal{F}_{t-1}$ and an estimate $\hat{\mathbf{X}}_{t-1|t-1}$ along with the mean square error of this estimate, $\Omega_{t-1} = E((\hat{\mathbf{X}}_{t-1|t-1} - \mathbf{X}_{t-1})(\hat{\mathbf{X}}_{t-1|t-1} - \mathbf{X}_{t-1})^T)$. New values $(\mathbf{X}_t, \mathbf{Y}_t)$ are generated, where the new state $\mathbf{X}_t$ is not observed but is to be estimated, while $\mathbf{Y}_t$ is observed. Now one can ignore the new observed value $\mathbf{Y}_t$ and directly estimate $\mathbf{X}_t$ using the minimum mean square error estimator, $E(\mathbf{X}_t | \mathcal{F}_{t-1})$ with its associated standard error. We can also use $\mathbf{Y}_t$. We know that $\mathbf{Y}_t = \mathbf{G}_t \mathbf{X}_t + \text{noise}$. We don't observe $\mathbf{X}_t$; however, we can estimate (predict) $\mathbf{Y}_t$ by $\mathbf{G}_t \hat{\mathbf{X}}_{t-1|t-1}$. The method is the same as treating $\mathbf{Y}_t$ as if it were a control variable. Specifically, we use it to adjust the standard estimate. Our estimates are unbiased, so the quantity $\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_{t-1|t-1}$ is an unbiased estimate of 0. We need to combine these two, taking into account the variability (in this case, the covariance matrix) for both. The explicit estimate, $\hat{\mathbf{X}}_{t|t}$, and the associated standard error, $\Omega_{t|t}$ is given below. The expressions are complex, especially since they are given in the most general matrix case.

In Kalman prediction, the goal is to predict/forecast $\mathbf{X}_{t+k}$, $k \geq 1$, given $(\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_t)$, a future value of $\mathbf{X}$ given the information from $\mathbf{Y}$ up to time t . It is generally convenient to define $\mathcal{F}_t$ to be the σ -algebra generated by $(\mathbf{Y}_1, \dots, \mathbf{Y}_t)$. We let the mean squared error of the prediction be denoted by

$$\Omega_{t+k} = E((\hat{\mathbf{X}}_{t+k|t} - \mathbf{X}_{t+k})(\hat{\mathbf{X}}_{t+k|t} - \mathbf{X}_{t+k})^T).$$

The Kalman prediction recursion begins with $\hat{\mathbf{X}}_{1|1}$ and proceeds one-step at a time to $\hat{\mathbf{X}}_{t+1|t}$, as follows:

$$\begin{aligned} \hat{\mathbf{X}}_{1|1} &= E(\mathbf{X}_1 | \mathbf{Y}_1), \\ \hat{\mathbf{X}}_{t+1|t} &= \mathbf{F}_t \hat{\mathbf{X}}_{t|t} + \mathbf{\Theta} \Delta_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_{t|t-1}), \end{aligned}$$

where

$$\begin{aligned} \Delta_t &= \mathbf{G}_t \Omega_t \mathbf{G}_t^T + \mathbf{R}_t, \\ \mathbf{\Theta}_t &= \mathbf{F}_t \Omega_t \mathbf{G}_t^T. \end{aligned}$$

Consider a simple linear model

$$Y_t = \beta Z_t + \epsilon_t,$$

where ϵ_t satisfies $E(\epsilon_t) = 0$ and $\text{Var}(\epsilon_t) = \sigma^2$, and the explanatory (independent) variable Z_t is constant and known.

We know the least squares estimator of β is given by

$$\hat{\beta}_t = \sum_{i=1}^t Y_i Z_i / \sum_{i=1}^t Z_i^2.$$

We next consider that the data in this situation arrive as a time series, although with the same distributional assumptions as before. This would be the case for financial data when we have data $(Z_1, Y_1), \dots, (Z_{t-1}, Y_{t-1})$ and estimate β based on those data. We denote that estimate by $\hat{\beta}_{t-1}$. Now, more data arrives at time t , (Z_t, Y_t) , and we wish to update our estimate $\hat{\beta}_{t-1}$ to $\hat{\beta}_t$. Ideally, we would like to do this on-line, that is based only on the current estimate $\hat{\beta}_{t-1}$ and the new data (Z_t, Y_t) without knowing the detailed earlier data. This is derived as follows:

$$\begin{aligned}
\hat{\beta}_t &= \frac{\sum_{i=1}^t Y_i Z_i}{\sum_{i=1}^t Z_i^2} \\
&= \frac{\sum_{i=1}^{t-1} Z_i^2}{\sum_{i=1}^t Z_i^2} \frac{\sum_{i=1}^{t-1} Y_i Z_i + Y_t Z_t}{\sum_{i=1}^{t-1} Z_i^2} \\
&= \frac{\sum_{i=1}^{t-1} Z_i^2}{\sum_{i=1}^t Z_i^2} (\hat{\beta}_{t-1} + \frac{Y_t Z_t}{\sum_{i=1}^{t-1} Z_i^2}) \\
&= \hat{\beta}_{t-1} - \frac{Z_t^2}{\sum_{i=1}^t Z_i^2} \hat{\beta}_{t-1} + \frac{Y_t Z_t}{\sum_{i=1}^t Z_i^2} \\
&= \hat{\beta}_{t-1} + \frac{Z_t}{\sum_{i=1}^t Z_i^2} (Y_t - Z_t \hat{\beta}_{t-1})
\end{aligned}$$

The expression above shows how to update a current estimate of β , $\hat{\beta}_{t-1}$, using just the new data, $\hat{\beta}_{t-1}$ and $\sum_{i=1}^{t-1} Z_i^2$. Now the quantity $\hat{\beta}_{t-1} Z_t$ is a prediction of Y_t , using the linear model and the new value of Z_t . The quantity $(Y_t - \hat{\beta}_{t-1} Z_t)$ is the error of that prediction. If the error is positive and $Z_t \geq 0$, then we conclude that $\hat{\beta}_{t-1}$ should be increased. Conversely, if $(Y_t - \hat{\beta}_{t-1} Z_t) \leq 0$ and $Z_t \leq 0$, then we believe that $\hat{\beta}_{t-1}$ should be decreased. The coefficient applied to this prediction error, $K_t = Z_t / \sum_{i=1}^t Z_i^2$ is called the *Kalman gain*.

Moreover, one can also compute the variance of the new estimate given the current variance and K_t

$$Var(\hat{\beta}_t) = (1 - K_t Z_t) Var(\hat{\beta}_{t-1}).$$

We now rewrite this model $Y_t = \beta Z_t + \epsilon_t$, in state space form where ϵ_t are i.i.d. $N(0, \sigma^2)$ and the independent variable Z_t is known.

We first express this as a state-space model with the goal being to estimate the unknown parameter β :

$$X_t = \beta,$$

$$Y_t = G_t X_t + W_t,$$

where $G_t = Z_t$, $R_t = \sigma^2$, $F_t = 1$.

In Kalman filtering we want to estimate X_t given Y_i , $i \leq t$.

$$\Omega_t = E((\hat{\beta}_t - \beta)^2) = Var(\hat{\beta}_t) = \sigma^2 / \sum_{i=1}^t Z_i^2,$$

where the last equality follows from a standard result for this one parameter linear model. Also

$$\Delta_t = G_t \Omega_t G_t^T + R_t = \frac{Z_t^2 \sigma^2}{\sum_{j=1}^{t-1} Z_j^2} + \sigma^2.$$

For Kalman filtering, we want to predict/estimate the current state, $\mathbf{X}_t$ given $(\mathbf{Y}_0, \dots, \mathbf{Y}_t)$. This is denoted by $\hat{\mathbf{X}}_{t|t}$.

The estimate is a combination of the one-step-ahead prediction from time $t-1$, $\hat{\mathbf{X}}_{t|t-1}$, and the difference between the observed $\mathbf{Y}_t$ and its predicted value using the observation equation, $\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_{t|t-1}$

$$\mathbf{X}_{t|t} = \hat{\mathbf{X}}_t + \mathbf{\Omega}_t \mathbf{G}_t^T \Delta_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_{t|t-1}),$$

where the relevant terms were defined earlier.

$$\begin{aligned} \hat{X}_{t|t} &= \hat{X}_{t|(t-1)} + \frac{\sigma^2}{\sum_{i=1}^{t-1} Z_i^2} \left(\frac{Z_t^2 \sigma^2}{\sum_{i=1}^{t-1} Z_i^2} + \sigma^2 \right)^{-1} (Y_t - Z_t \hat{X}_{t|(t-1)}) \\ &= \hat{X}_{t|(t-1)} + \frac{Z_t}{\sum_{i=1}^t Z_i^2} (Y_t - Z_t \hat{X}_{t|(t-1)}). \end{aligned}$$

Recognizing that $\hat{X}_{t|t} = \hat{\beta}_{t|t}$, and $\hat{X}_{t|(t-1)} = \hat{\beta}_{t-1}$, we find that the Kalman recursion equations are the same as those derived earlier using standard least squares results.

Recall that previously we have found

$$\hat{\beta}_t = \hat{\beta}_{t-1} + \frac{Z_t}{\sum_{i=1}^t Z_i^2} (Y_t - Z_t \hat{\beta}_{t-1}),$$

and $K_t = Z_t / \sum_{i=1}^t Z_i^2$, the *Kalman Gain*. Note assuming the ϵ_t are independent, Y_t is independent of $\hat{\beta}_{t-1}$, we take variances across the above equation:

$$\begin{aligned} \text{Var}(\hat{\beta}_t) &= (1 - K_t Z_t)^2 \text{Var}(\hat{\beta}_{t-1}) + K_t^2 \text{Var}(Y_t) \\ &= \frac{\sum_{i=1}^{t-1} Z_i^2}{\sum_{i=1}^t Z_i^2} (1 - K_t Z_t) \text{Var}(\hat{\beta}_{t-1}) + \frac{Z_t^2}{(\sum_{i=1}^t Z_i^2)^2} \sigma^2 \\ &= \frac{\sum_{i=1}^{t-1} Z_i^2}{\sum_{i=1}^t Z_i^2} (1 - K_t Z_t) \text{Var}(\hat{\beta}_{t-1}) + \frac{Z_t^2 \sum_{i=1}^{t-1} Z_i^2}{(\sum_{i=1}^t Z_i^2)^2} \text{Var}(\hat{\beta}_{t-1}) \\ &= (1 - K_t Z_t) \text{Var}(\hat{\beta}_{t-1}) \left(\frac{\sum_{i=1}^{t-1} Z_i^2}{\sum_{i=1}^t Z_i^2} + \frac{Z_t^2}{\sum_{i=1}^t Z_i^2} \right) \\ &= (1 - K_t Z_t) \text{Var}(\hat{\beta}_{t-1}). \end{aligned}$$

This result concurs with the standard result from simple linear regression.

Consider Kalman prediction, $\hat{\mathbf{X}}_{(t+1)|t}$, the prediction of $\mathbf{X}_{t+1}$ given $\mathcal{F}_t$. Suppose we begin with information $\mathcal{F}_{t-1}$ and then move to time t with unobserved $\mathbf{X}_t$ and observed $\mathbf{Y}_t$. We could use $\mathbf{Y}_t$ to obtain $\hat{\mathbf{X}}_{t|t}$ and then predict $\mathbf{X}_{t+1}$ using $\mathbf{G}\mathbf{F}_t\hat{\mathbf{X}}_{t|t}$, or we could forecast $\mathbf{X}_{t+1}$ using $\mathbf{F}_t\hat{\mathbf{X}}_{t|t-1}$ and adjust this by predicting $\mathbf{Y}_t$ and using the prediction error, $\mathbf{Y}_t - \mathbf{G}_t\hat{\mathbf{X}}_{t|t-1}$ to adjust the forecast $\mathbf{F}_t\hat{\mathbf{X}}_{t|t-1}$ as we did in the Kalman filter case.

The Kalman prediction recursion algorithm is given by the initialization step:

$$\begin{aligned}\hat{\mathbf{X}}_{1|1} &= E(\mathbf{X}_{1|1}|\mathcal{F}_1), \\ \Omega_1 &= E((\hat{\mathbf{X}}_{1|1} - \mathbf{X}_1)(\hat{\mathbf{X}}_{1|1} - \mathbf{X}_1)^T).\end{aligned}$$

Define

$$\begin{aligned}\Delta_t &= \mathbf{G}_t\Omega_t\mathbf{G}_t^T + \mathbf{R}_t, \\ \Theta_t &= \mathbf{F}_t\Omega_t\mathbf{G}_t^T,\end{aligned}$$

then the recursion is given by

$$\begin{aligned}\hat{\mathbf{X}}_{t+1|t} &= \mathbf{F}_t\hat{\mathbf{X}}_{t|t-1} + \Theta_t\Delta_t^{-1}(\mathbf{Y}_t - \mathbf{G}_t\hat{\mathbf{X}}_{t|t-1}), \\ \Omega_{t+1} &= \mathbf{F}_t\Omega_t\mathbf{F}_t^T + \mathbf{Q}_t - \Theta_t\Delta_t^{-1}\Theta_t^T.\end{aligned}$$

11.3 Stochastic Volatility Models

Recall we introduced ARCH/GARCH modeling to create models that better conform to the stylized facts of financial markets: heteroscedasticity, volatility clustering, heavy tail behavior. The GARCH model created this behavior “within itself,” not based on external events/shocks. Yet we have seen that many stocks (particularly those impacted by the same factors, such as the market, industry sectors, etc.) exhibit similar behavior with respect to volatility clustering.

An alternative model would allow for exogenous processes to impact the volatility behavior of a financial asset. Such models are called stochastic volatility models, and a state space formulation of these models allows us to employ “Kalman methodology” to filter and predict.

In state space models, there is an unobserved or latent variable, which could be multivariate, modeled and an observation/measurement process. We often want to make inferences about the latent process. A very good example of this situation is volatility. Volatility is unobserved, yet crucial to the valuation of assets (e.g. an option price). We want to model volatility as the unobserved state and then have an observation process that can be used to estimate (filter) and predict the latent process. An example using the Heston model is presented in the paper “Filtering in Finance”¹⁹. Another good article with many relevant references is by Date and Ponomareva²⁰.

¹⁹Javaheri, A., Lautier, D, and . Galli, “Filtering in Finance,” **Wilmott Magazine**, **3.**, 2003, 67-83

²⁰Date, P. and K. Ponomareva, “Linear and nonlinear filtering in mathematical finance: a review,” *IMA Journal of Management Mathematics*, **21(3)**, June, 2010, 1-18.

The Heston model involves the state process, v_t , and the observation process, S_t , defined by the SDE:

$$\begin{aligned} dv(t) &= (\omega - \theta v_t)dt + \xi \sqrt{v_t} dW_t^v, \\ dS_t &= r S_t dt + \sqrt{v(t)} S_t dW_t^S. \end{aligned}$$

The volatility process is a CIR process, whereas the stock price process is GBM with stochastic volatility where:

- r is the risk free rate,
- θ is the speed of mean reversion,
- ω/θ is the long-run equilibrium variance
- ξ is the volatility of the variance process.
- ρ is the correlation between W_t^v and W_t^S

Conditional on v_t , the price process is a GBM.

The Heston model is defined in continuous time, and we need to convert it to discrete time, and put it into state space form. The parameter set $(\omega, \theta, \xi, \rho)$ must be calibrated with market data to have the model in risk-neutral form. We can rewrite the stock price equation in risk neutral form

$$\log S_{t+\Delta} = \log S_t + (r - .5v_t)\Delta + \sqrt{v_t\Delta} Z_t^S.$$

We next use an Euler discretization for the two SDEs with time step Δ :

$$\begin{aligned} \log S_{(i+1)\Delta} &= \log S_{i\Delta} + (r_{i\Delta} - .5v_{i\Delta})\Delta + \sqrt{v_{i\Delta}\Delta} Z_{(i+1)\Delta}^S, \\ v_{(i+1)\Delta} &= v_{i\Delta} + (\omega - \theta v_{i\Delta})\Delta + \xi \sqrt{v_{i\Delta}\Delta} Z_{(i+1)\Delta}^v, \end{aligned}$$

where $\{(Z_i^S, Z_i^v), 1 \leq i \leq N\}$, $\Delta = T/N$, are i.i.d. standard bivariate normal random vectors with correlation ρ .

It is possible to eliminate the correlation parameter ρ by introducing the independent pair $(\tilde{Z}_i^v, Z_i^S)$ where

$$Z_i^v = \rho Z_i^S + \sqrt{1 - \rho^2} \tilde{Z}_i^v$$

and substituting $\tilde{Z}_{i\Delta}^v = \frac{1}{\sqrt{1-\rho^2}}(Z_{i\Delta}^v - \rho Z_{i\Delta}^S)$ into the volatility equation.

Substituting $\sqrt{v_k\Delta} Z_i^S = \log(S_{(i+1)\Delta}/S_{i\Delta}) - (r_{i\Delta} - .5v_{i\Delta})\Delta$ into the volatility equation, we arrive at the state space version of the Euler discretized Heston model with independent noise terms:

$$\begin{aligned} v_{(i+1)\Delta} &= v_{i\Delta} + [\omega - \rho\xi r_{i\Delta} - (\theta - .5\rho\xi)v_{i\Delta}]\Delta \\ &+ \rho\xi \log(S_{(i+1)\Delta}/S_{i\Delta}) + \xi \sqrt{1 - \rho^2} \sqrt{v_{i\Delta}\Delta} \tilde{Z}_k^v. \end{aligned}$$

for the state equation and

$$\log S_{(i+1)\Delta} = \log S_{i\Delta} + (r_{i\Delta} - .5v_{i\Delta})\Delta + \sqrt{v_{i\Delta}\Delta}Z_k^S,$$

for the observation equation.

Once the stochastic volatility model is in state space form, we can utilize the Kalman procedures for statistical inference. In the particular case, the volatility is of great interest, and inferring the current volatility using Kalman filtering has significant implications (e.g. in valuation of options, comparing current volatility estimates with implied volatility)

There are a number of stochastic volatility models, e.g. the Hull-White model we used in the simulation course. Estimation of these stochastic volatility models is known to be more complicated than GARCH models.

12 Time Series Frequency Domain Analysis

These notes have focused exclusively on the time-domain analysis of time series. This means we focused our analysis on the time evolution of a stochastic process and its autocovariance or autocorrelation function. Generally, we do not control a time series, like a stock price or economic series; however, in many engineering examples, a major goal is to understand and then modify or control the process. For example, if a time series is a signal, like the output of a radio, it is useful to recognize if there is some sort of noise structure superposed on the actual signal, and then build a device that will filter out that noise from the signal. In such a case, there is a different way to view the time series called frequency domain analysis. This type of analysis is well-known to MSCF students whose undergraduate major was electrical engineering and who have taken a signal processing course. The methodology is much less commonly used to study financial economic time series where time domain analysis prevails. In addition, frequency domain analysis requires some new (to some) mathematical tools that arise in complex analysis. As a result, this chapter is to introduce frequency analysis; however, it is nearly tangential to the most important material that MSCF students must master.

12.1 Some Technical Details

MSCF students are familiar with Laplace transforms and know them as moment generating functions (m.g.f.s). Specifically, if X is a random variable with cdf F , the $h(t) = E(e^{tX}) = \int_{-\infty}^{+\infty} e^{tx} dF(x)$ defines the mgf. The mgf is an equivalent way to describe a probability distribution, and mgfs are especially useful in dealing with sums of independent random variables. If $\{X_i, 1 \leq i \leq n\}$ is a sequence of independent random variables, and X_i has mgf $h_i(t)$, then $S_n = \sum_{i=1}^n X_i$ has mgf $h(t) = \prod_{i=1}^n h_i(t)$. Unfortunately, the mgf $h(t)$ may not exist for all values of t , a fact that limits its universality and hence its use.

A more general transform is the *Fourier transform*, or *characteristic function* when applied to probability measures. It is defined by $\phi(t) = E(e^{itX}) = E(\cos(tX) + i \sin(tX))$ where $i = \sqrt{-1}$. Complex numbers are plotted in two dimensions with the real part, $E(\cos(tX))$, associated with the X-axis and the imaginary part, $iE(\sin(tX))$, associated with the Y-axis. Note that $|\cos(tx)| \leq 1$ and $|\sin(tx)| \leq 1$, so $\phi(t)$ always exists and lies on or inside the unit circle. Finally, $\phi(0) = 1$ for all probability distributions F .

The interesting connection between characteristic functions and time series analysis is that the autocorrelation function, $\rho(h)$, associated with a weakly stationary time series $\{X_t, -\infty < t < +\infty\}$ is a characteristic function. Similarly, the autocovariance function, $\gamma(h)$, is a characteristic function (Fourier transform) differing from $\rho(h)$ only by the scale factor σ^2 . Since $\rho(h)$ is a characteristic function (Fourier transform), it must correspond to a probability distribution, and that distribution is known as the *spectral distribution* of the weakly stationary process. Up to this point, we have used t as the dummy variable for the characteristic function $\phi(t)$; however, we are now changing this to λ , $\phi(\lambda)$ where λ corresponds to “frequency.” We define the spectral distribution function, $F(\lambda)$, on $[-\pi, \pi]$, the weight put on various frequencies, similar to an ordinary cdf on $(-\infty, +\infty)$ corresponds to the weight the distribution puts over the real line. If $F(\lambda)$ is absolutely continuous (has a density), then $f(\lambda) = dF(\lambda)/d\lambda$ is called the *spectral density function*. In an engineering context, this function is often referred to as the *power spectrum*. The total power in the signal is σ^2 and the power spectrum shows the power at each frequency.

It is important to find the relationship between $\gamma(k)$ and $F(\lambda)$. This brings us to the topic of Fourier inversion formulas. We say that the time series analysis based on the spectral distribution function is a “frequency-domain analysis” as opposed to “time-domain analysis” when we are focused on the autocovariance or autocorrelation function. We next present two theorems relating the autocorrelation function to the spectral distribution and how one can be computed from the other.

The first theorem indicates that $\gamma(k)$ is the Fourier transform of a spectral distribution and shows how the autocovariance function can be computed from the spectral distribution function.

Theorem

$\gamma(k)$ is the ACF of a weakly stationary process $\{X_t\}$ if and only if

$$\gamma(k) = \int_{-\pi}^{\pi} e^{i\lambda k} dF(\lambda),$$

where F is a right continuous, nondecreasing function on $[-\pi, \pi]$ with $F(-\pi) = 0$ and $F(\pi) = \gamma(0) = \sigma^2$.

F is the spectral distribution function corresponding to $X(t)$ in cdf form, although F is a probability measure only if $\sigma = 1$. If $F(\lambda) = \int_{-\pi}^{\lambda} f(x)dx$, then $f(\lambda)$ called the spectral density function, in analogy with the usual pdf.

The second theorem is a converse result, the representation of the spectral distribution from the autocovariance function²¹

Theorem

An absolutely summable function $\gamma(k)$ is the ACF of a stationary process $\{X_t\}$ if and only if it is even and the function f is defined as

$$f(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma(k) e^{-ik\lambda} \geq 0,$$

$\forall \lambda \in [-\pi, \pi]$. f is called the spectral density function of $\gamma(k)$.

Note, a function g is even if $g(x) = g(-x)$. Direct observation shows that $f(\lambda)$ is even and

$$\gamma(k) = \int_{-\pi}^{\pi} \cos(k\lambda) f(\lambda) d\lambda.$$

Because the spectral density is even and repeats after one period of length 2π , we only need to consider the function over a half-period, $[0, \pi]$.

We now consider three simple examples:

²¹Note, throughout this section the definition of the spectral distribution, spectral density and periodogram differs in different texts by a constant factor of π or 2π . This can cause confusion; however, is not of great importance as it is only a simple scale factor.

Example 1: X_t is $\text{WN}(0, \sigma^2)$.

For white noise $\gamma(k) = \sigma^2$ for $k = 0$ and is 0 otherwise. In this case, the infinite sum has only one non-zero term corresponding to $k = 0$, while the others are all 0. Hence $\forall \lambda, f(\lambda) = \frac{1}{2\pi} \sigma^2 e^0 = \frac{1}{2\pi} \sigma^2$, a constant for all λ , hence a uniform distribution over $(-\pi, \pi)$ with total integral σ^2 .

Example 2: X_t is $\text{AR}(1)$, $X_t = \phi X_{t-1} + \epsilon_t, |\phi| < 1$.

Recall that $\gamma(k) = \frac{\sigma^2}{1-\phi^2} \phi^k$. Again using the formula above to find the spectral density function, we have

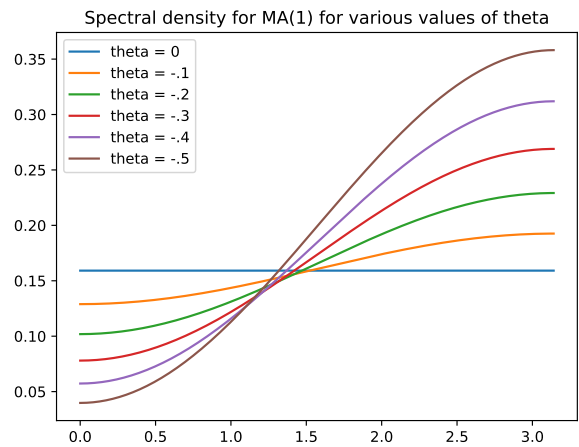
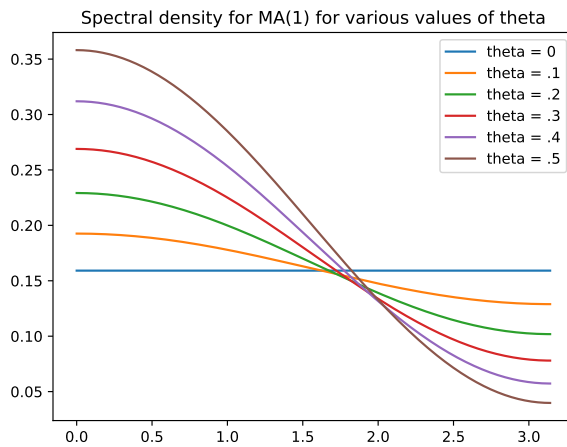
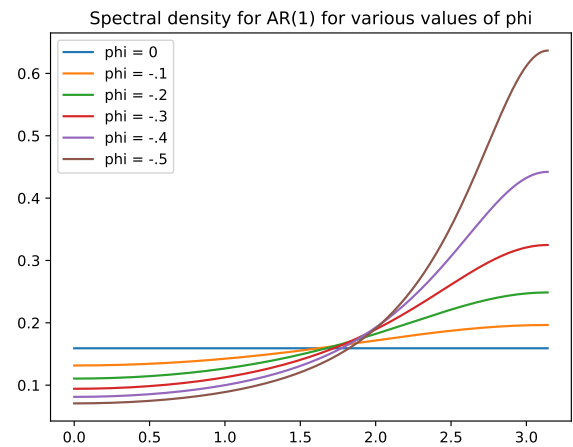
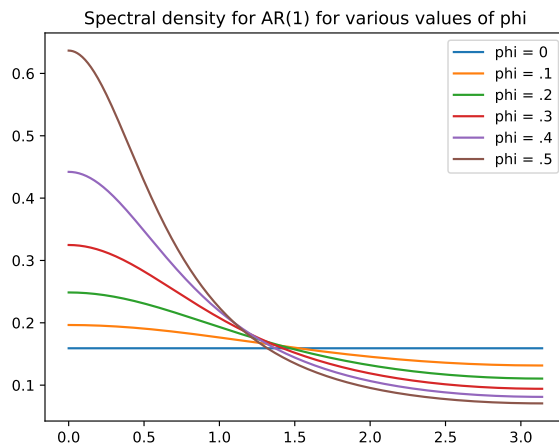
$$\begin{aligned}
 f(\lambda) &= \frac{\sigma^2}{2\pi(1-\phi^2)} \sum_{k=-\infty}^{\infty} \phi^{|k|} e^{-ik\lambda} \\
 &= \frac{\sigma^2}{2\pi(1-\phi^2)} \left(1 + \sum_{k=1}^{\infty} \phi^k e^{-ik\lambda} + \sum_{k=-\infty}^{\infty} \phi^{-k} e^{-ik\lambda} \right) \\
 &= \frac{\sigma^2}{2\pi(1-\phi^2)} \left(1 + \sum_{k=1}^{\infty} \phi^k e^{-ik\lambda} + \sum_{k=1}^{\infty} \phi^k e^{ik\lambda} \right) \\
 &= \frac{\sigma^2}{2\pi(1-\phi^2)} \left(1 + \frac{\phi e^{-i\lambda}}{1-\phi e^{-i\lambda}} + \frac{\phi e^{i\lambda}}{1-\phi e^{i\lambda}} \right) \\
 &= \frac{\sigma^2}{2\pi(1-\phi^2)} \frac{1-\phi^2}{1-2\phi \cos(\lambda) + \phi^2} \\
 &= \frac{\sigma^2}{2\pi(1-2\phi \cos(\lambda) + \phi^2)}
 \end{aligned}$$

Example 3: X_t is $\text{MA}(1)$, $X_t = \theta \epsilon_{t-1} + \epsilon_t$.

$\gamma(k) = \sigma^2 c_k$ where $c_0 = 1 + \theta^2$, $c_1 = \theta$ and $c_k = 0, k \geq 2$.

$$\begin{aligned}
 f(\lambda) &= \frac{\sigma^2}{2\pi} (1 + \theta^2 + \theta e^{-i\lambda} + \theta e^{i\lambda}) \\
 &= \frac{\sigma^2}{2\pi} (1 + \theta^2 + 2\theta \cos(\lambda))
 \end{aligned}$$

We show graphs of theoretical spectral density functions for $\text{AR}(1)$ and $\text{MA}(1)$ processes in the next pair of figures. The area beneath each of the curves is $1/2$, since the symmetric portion of the spectrum over $[-\pi, 0]$ is excluded. The flat curves corresponding to $\phi = 0$ and $\theta = 0$ are all white noise processes.



12.2 Periodogram and Empirical Spectral Densities

We have seen nice smooth graphs of the spectral densities for simple time series models. These would lead one to believe that the time series model could be identified through its spectral density. While this is in principle correct, we have graphed the spectral density using its known mathematical form. In reality, one will have a noisy data series and will need to estimate the spectral density function using that series, possibly using the empirical autocovariance or autocorrelation function.

Suppose we have data $(X_1, \dots, X_n)$ from a weakly stationary process with unknown autocovariance function $\gamma(k)$. We want to use these data to estimate the spectral density. A basic tool is the *periodogram*,

$$I(\lambda) = \frac{1}{2\pi n} \left| \sum_{j=1}^n X_j e^{ij\lambda} \right|^2.$$

Recall that a complex number, z , can be decomposed into a real and complex part, $x+iy$, and $|z|^2 = x^2 + y^2$. Thus, one can express $I(\lambda)$ by

$$I(\lambda) = \frac{1}{2\pi n} \left(\left(\sum_{j=1}^n X_j \cos(ij\lambda) \right)^2 + \left(\sum_{j=1}^n X_j \sin(ij\lambda) \right)^2 \right).$$

There are only n data points, so we cannot expect to estimate $I(\lambda)$ for all $\lambda \in [0, \phi]$. Instead, we focus on the so-called Fourier frequencies, $\lambda_j = \frac{2\pi j}{n}$, $1 \leq j \leq n$. These are equally spaced angles over $[0, 2\pi]$. Consequently,

$$\sum_{j=1}^n e^{it\lambda_j} = \sum_{j=1}^n e^{i2\pi t \frac{j}{n}} = 0.$$

Define

$$\bar{X} = \frac{1}{n} \sum_{j=1}^n X_j,$$

from which we have

$$nI(\lambda_j) = \sum_{j=1}^n (X_j - \bar{X})^2 + 2 \sum_{k=1}^n \sum_{l=k+1}^n (X_l - \bar{X})(X_{k-l} - \bar{X}) \cos(\lambda_j k).$$

Now

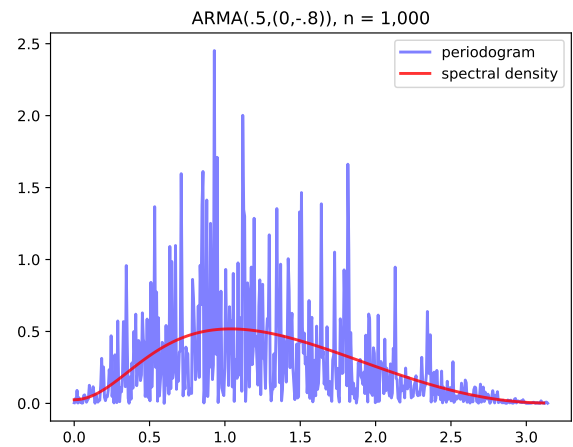
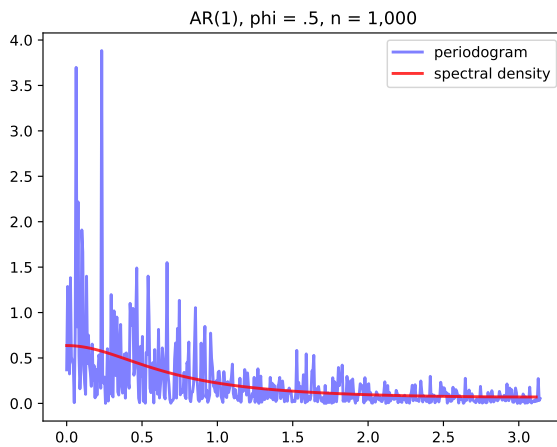
$$\hat{\gamma}(k) = \sum_{k=1}^n \sum_{l=k+1}^n (X_l - \bar{X})(X_{k-l} - \bar{X}),$$

hence

$$I(\lambda_j) = \frac{1}{2\pi} (\hat{\gamma}(0) + 2 \sum_{k=1}^{\infty} \hat{\gamma}(k) \cos(\lambda_j k)),$$

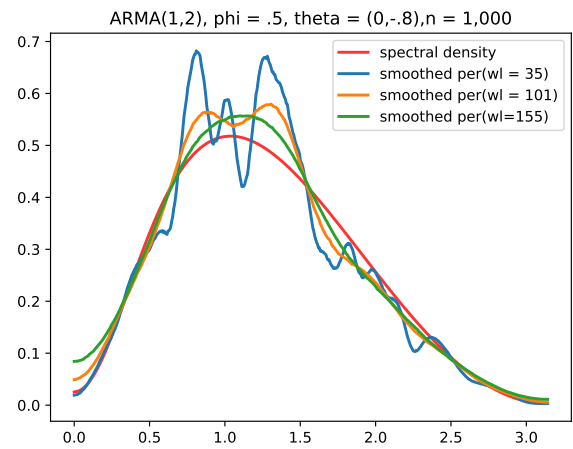
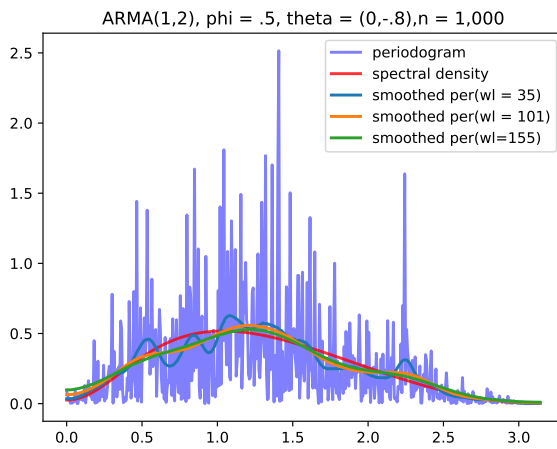
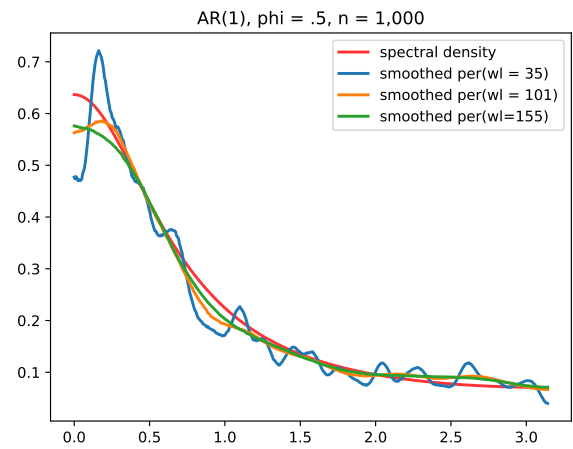
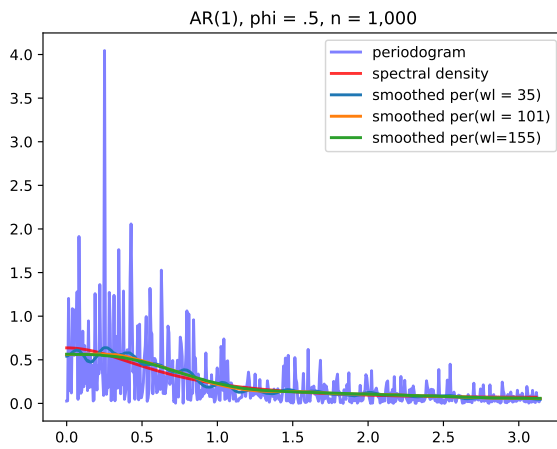
which is the natural sample estimate of $f(\lambda_j)$.

Consequently, the periodogram gives the usual sample estimates of the spectral density at the Fourier frequencies. It is not clear that the periodogram itself gives good estimates at other frequencies. We first show two plots showing the periodogram and the true spectral density for the models: AR(1) with $\phi_1 = .5$ and ARMA(1,2) with $\phi_1 = .5, \theta_1 = 0, \theta_2 = -.8$. The simulations in this section use the ARMA, periodogram and smoothing methods from `quantecon.py` package, but many of these are available through `scipy` and other packages. Specific code is given later.



One can see that the periodogram, even with a large sample size of $n = 1,000$ is a very poor estimate of the spectral density. The estimate is non-parametric, and smoothing is required to obtain a reasonable estimate. Smoothing is done through local averaging. Suppose we let $I(\lambda_j)$ be the periodogram estimate at the frequency λ_j . We consider adjacent values $I(\lambda_{j-k}), I(\lambda_{j-k+1}), \dots, I(\lambda_j), \dots, I(\lambda_{j+k})$. Each is assigned a weight, w_l , $-k \leq l \leq k$, where $\sum_{l=-k}^k w_l = 1$. The estimate becomes $\hat{I}(\lambda_j) = \sum_{l=-k}^k w_l I(\lambda_{j+l})$.

Generally, a window function defines a set of weights that sum to 1, and this is equivalent to a discrete probability distribution centered at the location of interest. There are many, many such window functions. The uniform and triangular have obvious shapes. Many others have more curvature with more mass around the middle and then tapering off more like the normal distribution. In the figures below, the Hamming window is used, a window more like the normal distribution shape. In addition to the choice of window functions, a second important parameter is the *window length*, which determines the extent to which distant values enter into the local averages.



The code for the above plot follows:

```
import numpy as np
from quantecon import periodogram, ARMA
import matplotlib.pyplot as plt

c = 1/(2*np.pi)
```

```

n = 1000
lp = ARMA(.5,(0,-.80))
X = lp.simulation(ts.length = n)

fig, ax = plt.subplots()
x, y = periodogram(X)
ax.plot(x,c*y, 'b-', lw=2,alpha = 0.5, label = 'periodogram' )
x_sd, y_sd = lp.spectral_density(two_pi = False, res = 120)
ax.plot(x_sd, c*y_sd, 'r-', lw = 2, alpha = 0.8, label = 'spectral density')
x, y_sm = periodogram(X>window = 'hamming', window_len = 35)
ax.plot(x,c*y_sm, lw = 2, label = 'smoothed per(wl = 35)')
x, y_sm = periodogram(X>window = 'hamming', window_len = 101)
ax.plot(x,c*y_sm, lw = 2, label = 'smoothed per(wl = 101)')
x, y_sm = periodogram(X>window = 'hamming', window_len = 155)
ax.plot(x,c*y_sm, lw = 2, label = 'smoothed per(wl=155)')
plt.title('ARMA(1,2), phi = .5, theta = (0,-.8),n = 1,000')
plt.legend()
plt.show()

```